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The ECB-Multi Country Model. A semi-structural model for forecasting and policy analysis for the largest euro area countries

--Manuscript Draft--

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Full Title:	The ECB-Multi Country Model. A semi-structural model for forecasting and policy analysis for the largest euro area countries
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Keywords:	Semi-structural model; Forecasting; euro area countries; Monetary policy
Abstract:	This paper introduces the European Central Bank's Multi Country model (ECB-MC), a coherent macroeconomic framework designed to support economic forecasting and policy analysis within the Eurosystem. The ECB-MC captures the economic dynamics of the five major economies in the euro area – Germany, France, Italy, Spain, and the Netherlands – which account for more than 80 percent of the euro area total GDP. By incorporating detailed structural features and data-driven insights, the model provides the main reference for the ECB's staff macroeconomic projections, acting as a disciplined tool for forecasting, enabling scenario, risk and sensitivity analyses, and giving a framework to understand the transmission channels of various economic shocks. The paper offers a detailed account of the structure, the estimation and the model properties, and provides a primer on the potential uses of the ECB-MC in the Eurosystem macroeconomic projections.

Replies to comments by the Associate Editor (AE)

We thank the AE for the comments and suggestions which helped improve greatly the paper. We hope our revision replies carefully to your queries and meets your expectations for the publication.

We answer to the various comments in bundles. The questions/comments are reported in black while the authors' replies are in blue.

Questions:

1. The manuscript reads like an internal technical report. For publication, it must be positioned for a scientific audience by surveying the landscape of macroeconomic models and explaining where this approach fits, justifying the modelling choices, data selection, and identifying assumptions, and discussing the trade-offs faced and why specific decisions were made. It would be extremely useful to add a section on caveats, with an honest assessment of the weaknesses of the model and directions for further work.

4. The model follows the Cowles Commission tradition of structural econometric modelling. Over the past decades, economists and policy institutions have largely moved toward VAR models and DSGE frameworks. The revision must explain how this approach relates to this methodological evolution, compare the model—both theoretically and empirically—with these alternative frameworks, and discuss models maintained at other institutions and at the ECB itself. Some alternative ECB models include authors of this paper among their developers. The absence of any reference to this related work is surprising and must be corrected.

Answer:

We have included a new section (Section 2) which, as suggested by the AE and by some of the referees, surveys the landscape and explains how the model fits this landscape. The section includes a comparison with other models and honestly admits caveats and weaknesses of the approach. We also add references to other models that are used in the projection framework, with more detailed references being added in Section 4 on the role of the model in the projections (see also the answer below). Most of this material has been published elsewhere and therefore we refer to those publications for fully fledged explanations.

Questions:

2. An important dimension is the fact that the model is linear and does not provide any meaningful assessment of risks and nonlinearities. For risk and density forecasting, other approaches have been developed at the ECB, such as the quantile regression forest model for inflation (Lenza, Moutachaker, and Paredes, 2024) and the Growth-at-Risk model (Figueres and Jarociński, 2020). How does the model of the paper connect with and reconcile forecasts of risks?

3. It is also important that the authors describe how the current model is used together with other tools such as the New Area Wide Model, the ECB's DSGE model (Christoffel, Coenen, and Warne, 2011; Christoffel et al., 2010), multi-country and global VARs (Angelini et al., 2019; Altavilla, Giannone, and Lenza, 2016; Dees et al., 2007), and satellite models for inflation

forecasting (Banbura, Lenza, and Paredes, 2023; Giannone et al., 2014), monetary policy measurement (Altavilla et al., 2019), and other ad hoc and targeted models.

5. The paper should clearly articulate what problems this model is designed to solve. Is the goal forecasting, scenario analysis, understanding transmission mechanisms, or a combination? How are the model's outputs used in the policy process? Answering these questions will provide a coherent narrative and help justify the modelling choices.

Answer:

We thank the Associate Editor for highlighting the importance of clarifying the role of the ECB-MC within the broader ECB modelling framework, particularly with respect to risk and scenario analysis. In the revised version, we clarify that the ECB-MC is a non-linear semi-structural model primarily designed to support the construction and discipline of the baseline macroeconomic projections and the associated sensitivity analyses. Within the ECB's suite-of-models approach, different models are designed to serve complementary purposes in order to avoid unnecessary overlap. While the ECB-MC plays a central role in the projection process and in the analysis of alternative assumptions, the assessment of risks around the baseline and density forecasting is typically complemented by additional tools, including DSGE models and dedicated empirical models such as inflation-at-risk and growth-at-risk frameworks.

To address these points, we substantially expanded the section on the use of the ECB-MC in the projection exercises (new Section 4) where we explain the whole projection process and the various models used, to arrive to a final projection narrative. These models include various DSGE models (the NAWM in particular) and several 'satellite' time series tools used for the short-term forecast of the main variables (GDP growth and inflation), as well as for scenario, sensitivity and risk analysis. The latter includes also the inflation and growth at risk models used by the ECB. We have tried to streamline the description of the process. First because many aspects of the process have been published in other papers (referenced in this version), and second (but not less important) because the full projection process is cumbersome and includes the involvement of several models and several ECB divisions. In Section 4 we limit our discussion to what is strictly functional to the projection process and complementary to the use of ECB-MC. Therefore, for instance we exclude from this discussion all ECB models that have been designed for pure monetary policy analysis (such as the papers that the AE cites, e.g. Altavilla et al, 2019) or those additional models that have indeed been developed at the ECB but that are not regularly (or not any longer) used in the projection process as such (such as Angelini et al., 2019; Altavilla, Giannone, and Lenza, 2016; Dees et al., 2007; Giannone et al., 2014).

The new Section 4 should therefore clarify all the problems this model is designed to solve in the context of the ECB projections. Its main goal is to discipline the baseline forecast of the large euro area countries, which implies to (i) provide a starting point for the derivation of the baseline projections; (i) accommodate additional and judgmental information obtained outside the model (the ECB-MC is especially designed to include systematic judgment into macroeconomic forecasts, based on the add-factor adjustment that is quite common across central banks and institutions), and (iii) perform sensitivity analysis to evaluate the potential impact of uncertain assumptions on the baseline. We hope this section now provides a coherent narrative and helps justify the modelling choice as requested by the AE.

Replies to comments by Referee 1

We thank the referee for the comments and suggestions which helped improve greatly the paper. We hope our revision replies carefully to your queries and meets your expectations for the publication.

We answer to the various comments in a point-by point manner. The questions/comments are reported in black while the authors' replies are in blue.

Comments/Questions:

The paper introduces a large-scale, semi-structural, multi-country model for forecasting and policy analysis and would be a very useful reference for those who are developing or enhancing their models in research/policy institutions. I see two major contributions of this paper assuming that this paper is subsequent to Angelini et al (2019) [2]. First, this paper presents how to expand a one-country model (ECB-BASE) to a multi-country model. Second, this paper demonstrates forecasting process of ECB with a realistic example.

I have some suggestions on editing and clarifying questions mostly to enhance readability and highlight the two contributions.

1. I find the descriptions on the relationship between ECB-BASE and ECB-MC a little bit unclear or confusing in the second paragraph in Section 1 and the rest of the draft. I read two papers cited in the draft to understand this better. Angelini et al (2019) [2] confused me more by saying that 'Since the structure of the models are almost identical, we will refer to the model interchangeably as ECB-BASE and ECB-MC unless stated otherwise.' in footnote 1. Ciccarelli et al (2024) [1] didn't help much either just by saying 'For multi-country analysis, the ECB-multi-country model (ECB-MC) is the main semi-structural model and is available for the five largest euro area countries' in page 14 and 'Two variants of the model are available, one relating to the euro area (ECB-BASE) and another, the ECB multi-country (ECB-MC) model, which is a single-country version for the big five euro area countries' in page 16. My understanding that ECB-BASE is a single-country or single-economy model for the overall euro area while ECB-MC is a 5-country model where each model for one country has the similar structure as ECB-BASE and the 5 countries interact each other through trade. If that is right, ECB-BASE would forecast the overall EU variables while ECB-MC would project those of 5 countries (as well as the overall EU variables?). ECB-MC would be especially useful to examine the effect of one country on the euro area. Since the 5 countries consist of more than 80% of the euro area GDP, the results from both models should be comparable. It would be very helpful to readers if the authors would clarify such points in the introduction. In addition, it would be much easier for the authors to compare this paper and Angelini et al (2019) [2]. They can refer Angelini et al (2019) [2] when they describe the common structures and assign more space to explain what is not in ECB-BASE and what is added to ECB-MC. I don't think that the current draft presents the differences between the two models clearly. For example, 'a granular financial block' and 'an extensive fiscal block' are improvements of ECB-BASE, not of ECB-MC, as long as I understand (I might be wrong.). But the paper describes them as if they are improvements of ECB-MC.

Answer:

We agree with the referee that the previous version could generate confusion. In the new version we have improved this aspect and the references to the ECB-BASE model have been made only

to the extent that they were functional to the description and illustration of the ECB-MC. We clarify that both ECB-BASE and ECB-MC are constructed using methodologies like those employed in the FRB-US model of the Federal Reserve Board. The primary objective of both models is to support central bank forecasting and policy analysis. This requires that the models: (a) capture the interdependencies among key macroeconomic variables; (b) incorporate realistic transmission channels for monetary policy and other economic shocks; and (c) provide a coherent accounting framework consistent with the needs of the forecasting process.

However, the two models differ in their scope and intended application. ECB-BASE is designed as an aggregate euro area model for area-wide policy simulations and complementary scenarios analyses around the baseline projections, whereas the ECB-MC model extends the same semi-structural framework to a multi-country environment – that explicitly captures heterogeneity across euro area economies – and is mostly used in the construction of the country baseline projections and for sensitivity analyses.

This is explained in the new Section 2 and in the Subsection 3.2 (ECB-MC-Specific Features), where the following text has been added: “Despite sharing the same structural framework as the ECB-BASE, the ECB-MC model incorporates distinct features within its country-specific blocks, particularly in the trade and financial sectors. Moreover, the core-VAR component of the model integrates both national and euro area variables in the expectations formation, enhancing its analytical depth.”

Notice also that the version of the ECB-MC provided in the paper is the one used in the projection process in a country-by-country mode and not in a linked mode. Therefore, with this version, we do not examine spillovers from one country to the euro area. We have also removed the reference to the fact that the 5 countries represent 80% of the euro area as it is not really relevant for the purpose of this paper.

2. (continued) Even though the authors cite Ciccarelli et al (2024) [1] for a comprehensive overview of ECB’s models, I suggest that the authors briefly present the overview in the introduction, such as Table 2 in the cited paper. One suggestion is as follows. The models consists of three categories: structural, semi-structural, and empirical time series models. The main structural model is NAWM, which is a DSGE model. The main semi-structural models are ECB-BASE and ECB-MC. (It would be helpful to explain briefly what ‘semi-structural’ means.) This kind of background would help readers understand the usefulness and scope of ECB-MC.

This suggestion has been fully taken onboard. The new Section 2 and the new Section 4.1 respectively survey the landscape of models and describe the projection process and the role that the various models play in it. An overview of the suite-of models has been reproduced in Section 2 (Figure 1) which provides a schematic representation of most ECB models used in projections and scenario analyses.

We have also better explained what semi-structural means already in the introduction, where we say: *These models are typically categorized as “semi-structural”, because they exist in a middle ground between purely empirical models (data-driven/time-series) and fully structural models (theory-driven/DSGE). Therefore, they balance data consistency and modelling adaptability (the ‘Semi’ part), with fundamental theoretical principles (the ‘structural’ part).*

3. I think the most interesting part of this paper is the section on the forecasting process with examples (Section 3). I hope the authors take another look at the section and revise it to maximize readability. Especially, I suggest to move Figure 7 and the corresponding paragraphs to the end of 3.3 so that Figure 7 illustrates all the steps. I would present the process by the order as following. First, impact of assumption and new data as blue dots. The short-term forecast is primarily shaped by satellite models and 'can' be significantly different from the previous forecast (red dotted line). For example, the first quarter projection of the quarterly real GDP growth of Germany that was 0.3% in the previous forecast is now lowered to be 0.1% based on new assumptions and data. In this stage, the financial, fiscal, and external variables are exogenous. Second, projection update as green dots. In this stage, all the variables are determined endogenously, which means that the financial, fiscal, and external variables are interact with the other variables. (Is it correct?) This forecasts starts with the 3rd quarter. (I don't see green dots in the blue shade.) Third, final forecast with judgment as a blue line. One thing I want to make sure that I understand correctly here is this. The blue dots are forecasts conditioned on the 'exogenous' forecast of financial, fiscal, and external variables. The green dots are forecasts where all variables are endogenous. I would appreciate it if the authors clarify this in the revised version. Here are some related suggestions or questions.
 - a) I think the authors use the term 'conditional' as two different meaning in 3.1 and 3.3. 3.3 is about off-model adjustments, and I don't understand why the authors call it conditional forecast.
 - b) While Figure 3 shows the forecasting horizons, it would be helpful to specify 2Q horizon for the short-term forecast and the rest of 3 years for the mid-term forecast in the text as well.

The section has been revised (and renamed to Section 4) to increase readability, in line with the very useful suggestions. In particular, we have followed the suggestion to move Figure 7 to the end of the section and amend the flow of arguments to explain the steps of constructing the final baseline projections. The description of the figure has been improved to clarify that the blue dots are the results of the conditioning on the exogenous assumptions, corresponding to the 'Impact of assumptions' described in the Subsection 4.2.1. In turn, the green dots depict 'Projection update' indicator, also described in the Subsection 4.2.2, that is conditioned on the same exogenous assumptions and the short-term outlook.

As regards point a), in general, whenever the exogenous information is added to the model (being either assumptions, short-term outlook, or add-factored residuals) we refer to the resulting model simulation as the "conditional" forecast. We have improved the description to make sure that each time the conditioning factors are spelled out explicitly in the paper.

As regards point b), this has been also incorporated into the description of Figure 7.

4. Here are minor comments.

- (a) The last paragraph of Section 1 seems to need to be rewritten. The details of model components and equations are in Appendix, not in Section 3. Section 3.5 could be a separate section such as Section 4 as in Angelini et al (2019) [2].

The last paragraph of the Introduction has been rewritten to more accurately reflect the content of the various sections and to better reflect the structure of the paper. Section 3.5 has now been

moved to an online appendix because it generated confusion with the typical IRF and because, following referees' suggestions, in the new version we have preferred to emphasize the forecasting role of the ECB-MC in the broad context of the projection process. A small section to evaluate the forecasting performance was also added.

(b) I don't see any descriptions or an example on policy analysis using ECB-MC in the paper while it says that the model is used for both forecasting and policy evaluation. It would be helpful to add those or just to clarify in the introduction that this paper demonstrates only how to forecast using the model.

The paper now clarifies that the primary objective of the paper is to present the model in the context of ECB forecasting process. The introduction has been amended accordingly and references to other policy analyses have been removed or toned down.

(c) Can you explain what $\oplus$ or D means in the following?

$$D = (T \oplus \infty)$$

D denotes the optimisation horizon. It points out that the optimisation problem in principle can be either finite (D takes the value of T denoting T-th time period) or infinite. The previous version used $\oplus$ sign which represents the logical OR quantifier which is not used anymore. Instead, we have defined D to be a set of two elements representing T or infinity as an "end" of the optimisation horizon. The text is adjusted accordingly.

(d) Can you check the following revised equation?

$$x^*t = \arg\max_{x^*t} E_t X D_{s=t} \beta t - s f(x_s, u_s)$$

The equation stands for a generic optimisation problem which is applicable to many blocks in the model.

(e) Can you check the following revised Equation (1) in the paper?

$$\Delta x_t = a_0(x^*t-1 - x_{t-1}) + m \sum_{i=1}^{X-1} a_i \Delta x_{t-i} + E_{t-1} \sum_{j=0}^{\infty} d_j \Delta x^*t+j$$

Can you explain why you use ∞ in the last term while you use a finite term in the lagged term? If not a typo, can you explain how to calculate the last term? Or if it is a typo and you use a finite term, can you share how many quarters you use in general?

This is the consequence of the optimisation problem. In short, the infinite summation is replaced by the complex expression which is derived by assuming the conformable VAR model can be used to determine the expectations. The details related to ECB BASE model can be found in Angelini et al (2019) and more detailed derivation exposition can be found in Tinsley, P. (2002), 'Rational error correction', Computational Economics 19 or Brayton, F., Davis, M. & Tulip, P. (2000), Polynomial Adjustment Costs in the FRB-US, mimeo, Federal Reserve Bank.

(f) I think one term is missing in Equation (4).

$$IH_t = PU_t - IOA_t - SCT$$

The equation has been adjusted as suggested.

(g) There is no text on Figure 8.

The text has been added and the paragraph accordingly adjusted.

(h) Can you explain more about what the 10 by 10 matrix represents in B.1? It would be helpful if you could compare the matrix with Equation A.14 in Angelini et al (2019) [2]. since the former seems an extension of the latter. For example, does each row represent one VAR? Does the first row represent a VAR with variables with 1s in the first row? A reference would be very appreciated if any.

We thank the referee for this suggestion. We agree that the previous presentation of the VAR structure was not sufficiently transparent, in particular regarding the interpretation of the block-inclusion matrix and the role of the block exogeneity restrictions. In the revised version, we rewrote this subsection substantially. The VAR structure is now presented using a clearer matrix representation where rows denote the dependent variable in each VAR equation and columns denote the variables included in the corresponding information set. We also added a more detailed explanation of the economic interpretation of the restrictions, clarified the role of euro area versus country-specific variables, and explicitly discussed how the ECB-MC specification extends the ECB-BASE VAR setup in Angelini et al. (2019). We believe the revised presentation substantially improves readability and makes the structure of the expectations system considerably easier to interpret.

Replies to comments by Referee 2

We thank the referee for the comments and suggestions which helped improve greatly the paper. We hope our revision replies carefully to your queries and meets your expectations for the publication.

We answer to the various comments in a point-by point manner. The questions/comments are reported in black while the authors' replies are in blue.

This paper introduces the ECB's Multi-Country Model (ECB-MC), a new large-scale semi-structural macroeconomic model covering the five largest euro-area economies (Germany, France, Italy, Spain, and the Netherlands). The ECB-MC is presented as a core tool for the Eurosystem's MPE and BMPE, serving as core model underpinning staff projections. The paper provides a detailed description of the model's structure, neatly organized in multiple behavioral blocks for demand, supply, fiscal, etc., in the Appendix a brief discussion of the estimation and (selected) calibration procedures drawing on existing frameworks like the ECB-BASE model, and the German block in detail. The paper documents how the ECB-MC is used in forecasts and scenario analyses, including illustrations of model responses to various shocks, rather scenarios, referred to as "Basic Model Elasticities" (BMEs). The main contribution of the paper is its documentation of the use of this new multi-country forecasting model and showing how it can be used for scenario analysis and policy evaluation.

This is a timely and relevant contribution given the importance of forecasting models in policy institutions, and it will interest practitioners and researchers looking to understand the ECB's forecasting apparatus.

Overall, the manuscript is ambitious and policy-relevant. It describes an impressive modeling effort: a comprehensive macroeconomic model that accounts for the big five euro-area countries. The paper is exceptionally well-written and structured logically, guiding the reader through the model's integration in the staff projections, construction, and usage. The authors are to be commended for the scope of the work: bringing together a multi-country model of this scale is non-trivial. Furthermore the paper should be praised for explaining effectively at a high level the model's theoretical underpinnings and equation structure. The policy relevance of the ECB-MC is its main merit: it replaces the NMCM of Dieppe et al. (2011) and is now one of the ECB's main forecasting tools. The paper successfully conveys that the model is a useful operational tool, capable of producing baseline projections, scenario analyses, risk assessments, and narrative decompositions of forecasts. These features make the model a valuable asset for policy analysis, and documenting them for an external audience is a positive step. In summary, my overall impression is that the paper addresses an important topic and provides a rich description of a model that could be of great interest to forecasting experts and policymakers alike.

Despite its merits, the paper in its current form falls short in several key areas that are critical for a publication in an academic forecasting journal such as IJF. I outline below my main concerns, along with suggestions for improvement:

The manuscript does not evaluate how the ECB-MC improves upon the ECB-BASE model or the NMCM it aims to replace. Without a structured comparison, either in terms of model structure, forecast performance, or use in scenario analysis, it is difficult to assess what modeling trade-

offs were made. For instance, ECB-BASE is a detailed aggregate euro area model that has been used for years in policy forecasting, while NMCM provided a country-level decomposition, albeit with more rigid structure. The ECB-MC claims to build on the strengths of both, but no explicit discussion is provided on what was gained or lost in the transition. Did forecast accuracy improve at the country level? Were certain blocks restructured (e.g., expectations or fiscal rules)? Were there tractability or stability issues that ECB-MC resolved? Without such comparisons, the justification for ECB-MC's structure and scale remains largely implicit. I think addressing this concern would also address my other

Answer:

In the new version of the paper we now make it clearer what the main differences with the ECB-BASE model are and we contextualise the choice of the model with a new section on the model landscape (Section 2 offers a comparison with other models used at the ECB and elsewhere) and with an expanded section on forecasting (Section 4 offers the description of the role of the ECB-MC in the projections and its evaluation).

We have also added a small paragraph on the main differences with the previous models, the NMCM, which was replaced for several reasons. A protracted underinvestment in the model had reduced its effectiveness in contributing to the projection process, and this was happening precisely at the time when multi-country analysis was needed the most. From a modelling perspective, a too stringent adherence to an optimization framework sometimes reduced the model's ability to provide empirical responses that matched closely with the data. At the same time, from a user viewpoint, the NMCM had a number of shortcomings that detracted from its effectiveness in aiding the development of the baseline forecast and in the scenario analyses. In particular, the only rudimentary incorporation of a financial block undermined the role of macro-financial linkages of significant importance in recent years. Along the same lines, the model not only displayed lower responses to standard monetary policy shocks than usual benchmarks but also lacked the channels through which non-standard monetary policy can affect euro area economies. The model also lacked a separate treatment of business investment. And more generally, many users view the model's complexity (especially the expectations formation based on learning and Kalman filtering) as often preventing an intuitive interpretation of its output, thus making it less helpful in providing consistent interpretations of the economic developments underlying the forecast.

Some of this explanation has been added in the text and a footnote in the first paragraph of Section 3.

Comments:

1. The paper provides no out-of-sample forecast evaluation or benchmark comparison: A major concern is the lack of any formal forecast performance evaluation, that I think is central for any forecast model. The manuscript does not present out-of-sample forecasting tests, accuracy metrics, or comparisons of ECB-MC's forecasts against simple benchmarks or alternative models. This omission is problematic, as academic research expects evidence of how well the model forecasts relative to naive models or existing approaches. Without such evidence, it is difficult to judge the practical forecasting value-added of the ECB-MC. Even though the paper's focus is on model design, I think for IJF it would significantly strengthen the contribution to demonstrate that the model yields competitive forecasts. I strongly encourage the authors to include an out-of-sample forecast evaluation

exercise and to report standard accuracy metrics. As the abstract hints, the model is intended as a "disciplined tool for forecasting", but the paper needs to show this by reporting how well it forecasts in practice.

In the new Section 4, the paper now describes the role of the ECB-MC in the projections and hopefully clarifies what 'forecasting' means for the ECB-MC. As now stated in the paper: "It is important to understand in the first place that, while the ECB/Eurosystem staff projection exercises are model-based, like in most central banks and policy institutions they are not purely defined by the bounds of traditional statistical time series analysis. They are instead nuanced exercises, continuously calibrated via expert judgement to assimilate a plethora of information streams and inputs, mirroring the dynamic economic landscape. The process itself therefore relies less on purely mechanical usage of the economic-models, and rather uses economic models as a starting point, as an organizing and constraining device and finally as a cross-checking device. Therefore, the final projections incorporate a fair amount of judgement as a result of additional information included in the model and iterative discussions that happen during the projection exercise". The forecasting process at the ECB is therefore distinct in two fundamental ways from standard statistical exercise. Firstly, it is not a purely statistical endeavour but a blend of data, models, and insights from diverse sectors, ensuring a reflection of the everchanging economic context. Secondly, the forecasting is inherently continuous, building on a rich repository of insights from past iterations, and is primarily conducted as a sequence of updates.

Given these features, we are clearer now that the ECB-MC model has been especially designed to include a systematic judgment into macroeconomic forecasts, based on practices (add-factor adjustment) that are quite common across central banks and institutions. Therefore, we claim, it is important that the model have some desirable characteristics which make it conformable to economic theory and consistent with the data. However, its purpose in the whole 'assembly line' of the projection exercises is not to produce accurate point forecasts, and hence a standard evaluation of its forecasting properties based e.g. on RMSE, or similar metrics, is almost meaningless and must be performed with accurate caveats.

Notwithstanding, we have added a new Subsection (4.3) on forecast evaluation, where the 'evaluation metric' is not the same used in standard time series practices and has been tailored to the main purposes of the model. Concretely, to evaluate its forecasting ability we ask the question: What would the (conditional) forecast of the model be if we fed it with the 'right' assumptions, namely with the financial, fiscal, and external data that occurred over the evaluation period (as opposed to the real time assumptions). The forecast performance is then evaluated relative to a simple random walk benchmark -- which provides a transparent and model-free reference point. We also illustrate what the model would predict for HICP inflation at 2022Q1, i.e. at the beginning of the inflation surge period, had it been fed with the realised assumptions as compared to a projection based on the real-time assumptions.

2. Impulse response analysis presented in the paper, the BMEs, is not comparable to standard IRFs, and thus are hard to assess. The paper has a section presenting "Basic Model Elasticities" as a way to illustrate the model's properties. While informative, these BMEs are essentially conditional forecasts or scenario responses under the assumption of no "policy reaction", rather than traditional IRFs used in academic literature. The distinction needs to be made much clearer. Currently, the narrative could confuse readers, at times it refers to these responses in language similar to IRFs, mentioning an "unchanged policy rule," whereas what is actually being held fixed are the policy instruments, e.g. interest rates and

exchange rates, in the simulations. The BME framework's "partialization" approach means policy variables (! and not policy rules!!!- see page 26.) are held constant so that only the "direct effects" of shocks are seen. This is fine for internal consistency across Eurosystem models, but it is not a standard IRF. The authors themselves note that in this forecasting setup, fully endogenous IRFs "would not make much sense". Thus, it should be emphasized that the BMEs are conditional model elasticities under specific scenarios, and they are not directly comparable to IRFs from a VAR or DSGE model that include monetary policy reactions. I recommend the authors clarify this in the text to avoid confusion. For an academic audience, the IRFs are more informative about the model properties. As it stands, the presentation of results in Section 3.5 could be misleading to readers unfamiliar with the ECB's internal (B)MPE conventions.

We further clarify that BMEs are not directly comparable to IRFs. BMEs are derived under model "partialization" assumptions that reflect the ECB forecasting framework. As such, they illustrate how forecasts respond to standard changes in conditioning assumptions (e.g. fiscal, external, and financial variables), rather than providing fully structural impulse responses. We acknowledge the referee's concern that this distinction may not be clear to a broader audience less familiar with the ECB framework. To avoid potential misunderstanding, we have therefore removed this section, and we might place it in an online appendix.

3. Insufficient documentation of parameter estimates and use of ECB-BASE priors: The model's estimation and calibration strategy relies heavily on the earlier ECB-BASE model, but the paper does not fully document the parameters adopted or estimated. For example, the manuscript indicates that many coefficient priors were calibrated to ECB-BASE estimates. However, it provides few details on what those prior values are or how close the posterior estimates of ECB-MC came to them. Key parameters are not tabulated in the paper. This gap makes it difficult for readers to verify and reproduce the model. An academic paper should ideally either provide full estimated parameter tables, e.g. as an Appendix or online file, or at least summarize the main parameter values and compare them to those from ECB-BASE. Currently, one has to take on trust that the ECB-BASE calibration was used not opportunistically but systematically, since the actual numbers are not shown. For transparency and verifiability, the authors should include a table of the main estimated coefficients for each economy and clearly state which parameters were calibrated vs estimated. Without this, the model remains somewhat of a black box and this brings me to another comment:

The key parameters are tabulated in the Appendix – which now also contains the estimates of the financial block– and the estimation strategy has been entirely added in the new version (see new Section 3.3 and Appendix C).

Given that we deal with five countries, the total number of parameters (as can be seen from the full model equation appendix) is very large. There are two reasons why we report only a (large) subset of the full parameter space. First, some parameters are estimated using restricted (non-publicly) available data and therefore we do not report them. Second, the full model is used for official forecasting within ECB and therefore it is restricted to the public in its latest version. In addition, we have significantly extended the material by adding detailed descriptions on the estimation strategies employed for different model block. This is now placed in Section 3. We

will have an online appendix with the full model and the data used to estimate it. Most of the data are publicly available with the exception of some data that are property of the ECB or National Central Banks and cannot be distributed. In the online appendix we will also publish the replication code for simulating the basic model elasticities to a monetary policy shock.

4. Lack of reproducibility - data and code: The paper in its current form does not enable independent replication of the results. Crucial details are missing about the data and estimation. While an appendix lists the variable definitions and equations, the data series and sample used for estimation are not fully provided. While the authors mention that an interactive webpage will list data codes in the future, without it it is hard to assess the model's completeness. Moreover, only a selected sample of coefficient estimates is reported in the Appendix, rather than the full set of estimated parameters. Ensuring reproducibility would greatly enhance the paper's credibility. If confidentiality is a concern for certain data series, one solution might be to release a smaller version of the model using publicly available data for academic purposes.

Please see the comment above. In addition, we have added the detailed series keys of the publicly available data used for the estimation. We believe that the interested reader by combining the details provided in this paper, together with details in Angelini et al (2019) and examples provided in Dynare manual can tackle the replication. Obviously, the model is updated as the new crucial data arise and therefore the parameters values reported should be taken as a snapshot in time.

5. Broader context and comparisons: Lastly, the paper could situate ECB-MC in the context of other policy models to highlight its innovations. The authors note that both ECB-BASE and ECB-MC follow a methodology similar to the Fed's FRB/US model, but they stop short of comparing outcomes or design choices. It would improve the paper to briefly discuss how ECB-MC differs from other central-bank forecasting models (for example, in handling expectations or fiscal policy) and why those differences matter. Such context would help readers understand the unique contribution of ECB-MC. In particular, explicitly contrasting the new model's approach to expectation formation and model complexity with that of a fully micro-founded DSGE (like the NMCM or the ECB's New Area-Wide Model) or a reduced-form VAR model could illustrate the middle ground that ECB-MC occupies. Providing this broader comparison would underline the trade-offs the authors chose in balancing theoretical rigor with empirical usability, thereby giving a deeper insight into the model's strengths and intended role.

We have now included a new section (Section 2) which, as suggested by the referee, surveys the landscape and explains how the model fits this landscape. The section includes a comparison with other models and honestly admits caveats and weaknesses of the approach. We also add references to other models that are used in the ECB projection framework, with more detailed references being added in the new Section 4 on the role of the model in the projections. Most of this material has been published elsewhere and therefore we refer to those publications for fully fledged explanations.

Regarding the comparison with other central banking models, we refer to two papers published by the ECB in occasion of the strategy review in 2021 (Darracq et al, 2021) and of its assessment in 2025 (Angelini et al, 2025). From those papers it clearly emerges that most central banks of the world use a suite-of-models approach (to balance the need for robustness across different uses provided by model diversity against flexible deployment of selected types of models for specific purposes) and large-scale semi-structural models for projections, while structural (typically dynamic stochastic general equilibrium, or DSGE) models are preferred for scenario and policy analysis (for their forward looking expectation formation for instance).

Suggestions for Improvements:

1. Provide a proper forecast evaluation: To reiterate, including an out-of-sample forecasting experiment would greatly strengthen the paper. For example, the authors could simulate how the model would have forecast GDP and inflation for each country over a past period (say, using recursive or rolling estimates), and report standard accuracy measures (RMSE, etc.), comparing them to a simple benchmark (like a random walk or a VAR). This would directly address the academic audience's expectation and demonstrate the model's practical utility.

We have added a new Subsection 4.3 on forecast evaluation, where the 'evaluation metric' is not the same used in standard time series practices and has been tailored to the main purposes of the model. Concretely, to evaluate its forecasting ability we ask the question: What would the (conditional) forecast of the model be if we fed it with the 'right' assumptions, namely with the financial, fiscal, and external data that occurred over the evaluation period (as opposed to the real time assumptions). The forecast performance is then evaluated relative to a simple random walk benchmark -- which provides a transparent and model-free reference point. We also illustrate what the model would predict for HICP inflation at 2022Q1, i.e. at the beginning of the inflation surge period, had it been fed with the realised assumptions as compared to a projection based on the real-time assumptions.

2. Clarify and possibly supplement the impulse-response analysis: The paper should make clear that the presented BMEs are conditional partial responses. The authors might consider moving some of the technical explanation (currently in the footnote and text) into a more prominent description so that readers understand the limitations of these results. If possible, showing standard IRFs, perhaps in an online appendix or a brief subsection, would be ideal for completeness, but if not, a clear statement of why they are omitted (as the authors note, because in this forecasting version policy is exogenous) will suffice. The goal is to prevent confusion and ensure readers do not over-interpret the BME charts or that the readers can reproduce these charts in their model of choice.

This is a very good comment. However, given also the comments from other referee, we have decided to remove the section on BMEs from the main paper and move it to the online appendix. In this new version, we have further enhanced the description of this exercise to make it clear that BMEs are the model responses under substantial model "partialization" which is required by the BMPE forecasting process (e.g. no monetary policy rule, exogenous fiscal block). We do not show standard IRFs given that we focus on this forecast use of the ECB-MC. We have also added this clarification in the Introduction.

3. Enhance transparency of the model's inputs and outputs: It would be very useful if the authors can publish additional materials alongside the paper. One remedy could be to release a simplified public version of the model. For instance, a smaller-scale version or even just the core equations with baseline data could be made available for researchers to test. The authors mention plans for an interactive web page with data codes - that is a great initiative. Building on that, even providing the baseline forecast dataset or the code used to generate the elasticity figures would allow external academics to validate and experiment with the model. The operational full model might be complex or proprietary, but a public toolkit version (with perhaps fewer variables or shortened history) would go a long way in allowing external scrutiny and learning. I strongly encourage the ECB to consider this, as it aligns with best practices in model development (for example, the Federal Reserve often publishes simplified versions or detailed documentation of its FERPA/FRB-US models for researchers). At minimum, the paper should include a comprehensive technical appendix with the full list of equations and all estimated coefficients for each country, as well as details on data sources and transformations.

We have prepared an online appendix which contains an interactive HTML file listing all of the model variables per country, and the corresponding data series keys that are publicly available via ECB SDW portal. We also provide MATLAB replication codes for all 5 countries that allow user to compute Basic Model Elasticities (BMEs) to a “monetary policy shock”. The codes also include the “Enterprise” version of Dynare developed specifically for the purposes of estimating and simulating ECB-BASE/MC models. README.html file contains detailed guide to setup and run the simulation codes. We need to point out that, for confidentiality reasons, a subset of parameters (like price and wage Phillips curves) are not the official ones used in the context of ECB (B)MPE forecasting process and represent averages across countries. Also, this version of the model does not represent the latest version of the model used for the ECB forecasting process, nor do the submitted datasets represent the latest revisions used in the same process.

Reframe the paper toward an academic audience: In revising, the authors should keep in mind that IJF readers will be looking for evidence of forecast performance and clear comparisons, rather than only a model description. The revision should highlight any empirical validation of the model. For example, include a section on forecast performance (even if it's just within-sample fit or a pseudo real-time exercise), and a subsection comparing results with the single-country approach versus an aggregate approach. The writing can also be tightened to focus on insights that generalize beyond the ECB context (what can other forecasters learn from this model's design?). By adding these elements, the paper will speak more directly to academic forecasters, not just central bank insiders.

See above. We have added a new Subsection (4.3) on forecast evaluation, where we also explain that the ‘evaluation metric’ is not the same used in standard time series practices and has been tailored to the main purposes of the model. Concretely, to evaluate its forecasting ability we ask the question: What would the (conditional) forecast of the model be if we fed it with the ‘right’ assumptions, namely with the financial, fiscal, and external data that occurred over the evaluation period (as opposed to the real time assumptions). The forecast performance is then evaluated relative to a simple random walk benchmark -- which provides a transparent and model-free reference point. We also illustrate what the model would predict for HICP inflation at 2022Q1, i.e. at the beginning of the inflation surge period, had it been fed with the realised assumptions as compared to a projection based on the real-time assumptions.

In summary, I see considerable potential in this paper. The ECB-MC is clearly an important tool, and documenting it for a wider audience is commendable. However, to meet the standards of an academic journal, the authors need to address the above points, especially regarding forecast evaluation, clarity of model outputs (IRFs vs BMEs), and reproducibility. I encourage the authors to undertake a thorough revision. If the suggested improvements are incorporated - in particular, adding forecast accuracy results, cross-model comparisons, and providing replication materials - this paper could become a valuable contribution to the forecasting literature. It would then not only present a new model but also convincingly demonstrate its effectiveness and allow others to build on the work. I appreciate the effort that has gone into developing the ECB-MC and I hope the authors will refine the manuscript accordingly for a stronger impact.

Final note to the authors: Thank you for your significant work on this model. I offer these critiques in the spirit of constructive feedback. I believe that addressing (some of) them will greatly enhance the paper's quality. I encourage you to embrace transparency and take on the challenge of turning this policy focused operational model description into a more scholarly piece that stands on its own in terms of methodological rigor and empirical insight. I look forward to seeing a revised version.

We thank for the insightful comments and the appreciation of this exercise of transparency. We sincerely hope that the amendments of the paper meet the referee's expectations. The paper now contextualizes the model in a better way than the previous version, describes the forecast process and the role of the model in a structured manner, and includes a forecast evaluation tailored to the purposes and the characteristics of the model. We have removed the section on model elasticities to avoid misleading interpretations. And we have added the full model code and the estimation strategy. We believe this goes a long way towards more transparency around ECB models and processes, in an unprecedented manner.

Replies to comments by Referee 3

We thank the referee for the comments and suggestions which helped improve greatly the paper. We hope our revision replies carefully to your queries and meets your expectations for the publication.

We answer to the various comments in a point-by point manner. The questions/comments are reported in black while the authors' replies are in blue.

This paper presents the ECB's semi-structural multi-country model (ECB-MC), a newly developed tool for building the ECB's staff macroeconomic projections. The model itself is quite standard, e.g. it is very similar to the FRB-US model used by the Fed since many years. The most interesting aspect of the paper is the description of the model's use in the projection, with details of the forecast-generation process. What is sorely missing from it at the moment is a discussion of the motivation for choosing to build a semistructural model now, almost 50 years since Chris Sims criticized Cowles commission type models in the 1980 seminal paper "Macroeconomics and Reality."

Comments and suggestions

1. Add a discussion of the role of different models. Many central banks have grappled with the question of what model to use for its projections. Most central banks follow a suite-of-models approach, where many models are used to inform the forecast, each with their relative strengths and weaknesses. The suite comprises fully fledged DSGE models for policy like COMPASS and the NAWM, which are ideally suited causal analysis and studying optimal policy, but can be more difficult to use within a forecast process that merges judgment and models. Most central banks also use (structural) VARs (e.g. Angelini et al., 2019, Domit et al., 2019, and Crump et al. 2025) for forecasting, but also scenario analysis. While these models are viewed as somewhat complementary – DSGEs are more grounded in theory and best for policy analysis, BVARs more data-driven and best for forecasting – more recent work shows that you can forecast quite well with DSGEs (see e.g. NYFed's DSGE) and you can do useful policy analysis with VARs (e.g. Banbura et al., 2015, Antolin-Diaz et al., 2021, and Cimadono et al. 2022, among many others). Semi-structural models, like the ECB-MC, are neither strongly grounded in theory nor driven by the data. They are nonetheless an often integral part of the suite of models for most central banks. The reason, I believe, is that these models have the advantage of allowing to incorporate judgement more straightforwardly. This can be very important in a policy setting and the paper should put more emphasis on this. It would be very important for the authors to reflect in the paper this debate about the strengths and weaknesses of the various modelling approaches, so that the reader can appreciate the context in which this model was developed. The paper skirts this discussion completely, even claiming that, since the 2020 ECB strategy review "confirmed that the majority of the forecasting models employed are semi- structural," this affirms "their adequacy for policy scenario analysis and economic forecasting."

Answer:

The referee is fully right in the above considerations. We have now included a new section (Section 2) which surveys the landscape and explains how the model fits this landscape. The section includes a comparison with other models and honestly admits caveats and weaknesses of the approach. We also add references to other models that are used in the ECB projection

framework, with more detailed references being added in the new section 4 on the role of the model in the projections (see below). Most of this material has been published elsewhere and therefore we refer to those publications for fully fledged explanations.

Regarding the comparison with other central banking models, for instance, we refer to two papers published by the ECB in occasion of the strategy review in 2021 (Darracq et al, 2021) and of its assessment in 2025 (Angelini et al, 2025). From those papers it clearly emerges that most central banks of the world use a suite-of-models approach (to balance the need for robustness across different uses provided by model diversity against flexible deployment of selected types of models for specific purposes) and large-scale semi-structural models for projections, while structural (typically dynamic stochastic general equilibrium, or DSGE) models are preferred for scenario and policy analysis (for their forward looking expectation formation for instance).

In the new Section 4, the paper now describes the role of the ECB-MC in the projections and hopefully clarifies what ‘forecasting’ means for the ECB-MC. As now stated in the paper: “It is important to understand in the first place that, while the ECB/Eurosystem staff projection exercises are model-based, like in most central banks and policy institutions they are not purely defined by the bounds of traditional statistical time series analysis. They are instead nuanced exercises, continuously calibrated via expert judgement to assimilate a plethora of information streams and inputs, mirroring the dynamic economic landscape. The process itself therefore relies less on purely mechanical usage of the economic-models, and rather uses economic models as a starting point, as an organizing and constraining device and finally as a cross-checking device. Therefore, the final projections incorporate a fair amount of judgement as a result of additional information included in the model and iterative discussions that happen during the projection exercise”. The forecasting process at the ECB is therefore distinct in two fundamental ways from standard statistical exercise. Firstly, it is not a purely statistical endeavour but a blend of data, models, and insights from diverse sectors, ensuring a reflection of the everchanging economic context. Secondly, the forecasting is inherently continuous, building on a rich repository of insights from past iterations, and is primarily conducted as a sequence of updates.

Given these features, we are clearer now that the ECB-MC model has been especially designed to include a systematic judgment into macroeconomic forecasts, based on practices (add-factor adjustment) that are quite common across central banks and institutions. Therefore, we claim, it is important that the model have some desirable characteristics which make it conformable to economic theory and consistent with the data. However, its purpose in the whole ‘assembly line’ of the projection exercises is not to produce accurate point forecasts, and hence a standard evaluation of its forecasting properties based e.g. on RMSE, or similar metrics, is almost meaningless and must be performed with accurate caveats.

Notwithstanding, we have added a new Subsection 4.3 on forecast evaluation, where the ‘evaluation metric’ is not the same used in standard time series practices and has been tailored to the main purposes of the model. Concretely, to evaluate its forecasting ability we ask the question: What would the (conditional) forecast of the model be if we fed it with the ‘right’ assumptions, namely with the financial, fiscal, and external data that occurred over the evaluation period (as opposed to the real time assumptions). The forecast performance is then evaluated relative to a simple random walk benchmark -- which provides a transparent and model-free reference point. We also illustrate what the model would predict for HICP inflation at 2022Q1, i.e. at the beginning of the inflation surge period, had it been fed with the realised assumptions as compared to a projection based on the real-time assumptions.

2. Implementation of the assumptions analysis. The breakdown of the forecast process in various steps is for me the most relevant point of the paper, as it helps to understand what types of constraints central banks' staff often face when producing a forecast, especially the "main" projection. The first step here is to assess the impact of the "assumptions" on the exogenous variables –e.g. fiscal, financial, world demand,.... These are treated as exogenous, but one would expect that they comove and are interrelated: for example, world demand and fiscal behaviours might change together in the face of a geopolitical shock. Are these changes to the assumptions implemented one at a time or jointly? What models guide them?

We have revised the section (now Section 4) to provide a clearer description of the forecasting process and the role of the model. In particular, we clarify how technical assumptions are incorporated and explain the iterative interaction between the model and these assumptions. This interaction does not take place within the ECB-MC model itself, but rather within the broader ECB forecasting framework. The assumptions are either generated with other models (e.g. Fiscal) or taken from the market (interest rates and oil futures). An ideal general equilibrium convergence process (use assumptions, generate macro forecast, check again assumptions, generate new forecast, etc. until convergence is reached) is never possible.

Replies to comments by Referee 4

We thank the referee for the comments and suggestions which helped improve greatly the paper. We hope our revision replies carefully to your queries and meets your expectations for the publication.

We answer to the various comments in a point-by point manner. The questions/comments are reported in black while the authors' replies are in blue.

The author(s) of this paper describe a multi-country semi-structural economic model (based on the FRB-US) designed to support forecasting and policy analysis at the European Central Bank (ECB). It is interesting to learn about the approaches used by leading central banks, and the author(s) seem to know a great deal on the ECB forecasting process.

However, as it stands, the paper does not meet the standard to publish in the International Journal of Forecasting. I would expect that the author(s) address the following comments.

1. I completely miss key elements in the introduction:

- FRB-US is an old model with many shortcomings; other approaches have become prevalent since FRB-US was first released, taking into account rigorous micro-foundations, heterogeneity, clear economic frictions and so on. Why FRB-US style models are appropriate for the ECB? The author(s) should spend more time arguing in favor of the approach.
- One reason why the model in this paper may be empirically appropriate is the fact that the "multi-country" micro-founded models are not as empirically successful as their closed economy counterparts. In other words, there is not a Christiano-Eichenbaum-Evans or Smets-Wouters model for cross-country analysis. The literature on micro-founded multi-country models should be referenced and discussed in the paper.
- At the same time, there are many successful statistical approaches which have been used for multi-country analysis (the GVAR of Pesaran and co-authors - including the application of Burriel and Galesi to the EU, the Panel VAR of Canova and Ciccarelli, the Bayesian VAR of Altavilla, Giannone and Lenza - applied to the euro area policy analysis in Angelini, Lalik, Lenza and Paredes, the spatial VAR of Gefang, Hall, Tavlás and Wang). This literature should be referenced and discussed in the paper, highlighting the complementarities and the differences with the semi-structural model.

Answer:

The referee is right in the above considerations. We have now included two new sections (Section 2 and Section 4) which survey the landscape, explain how the model fits this landscape, and describe the precise role that the ECB-MC has in the suite-of-model approach of the ECB. The sections include a general comparison with other models (e.g. micro-founded and multi-country) and give details on the projection process of the ECB but discussing only the most important aspects that make the use of a semi-structural model like the ECB-MC an appropriate choice in the full process. In this respect, institutional constraints are important and some of the multi-country models cited by the referee are valid references and have been regularly used to complement the analysis, as now discussed in the paper. The sections also honestly admit caveats and weaknesses of the approach, and make it clear that the model is

never used in isolation and is instead complemented in the analysis by several of the models referenced to by the referee

Regarding the comparison with other central banking models, we refer to two papers published by the ECB in occasion of the strategy review in 2021 (Darracq et al, 2021) and of its assessment in 2025 (Angelini et al, 2025). From those papers it clearly emerges that most central banks of the world use a suite-of-models approach (to balance the need for robustness across different uses provided by model diversity against flexible deployment of selected types of models for specific purposes) and large-scale semi-structural models for projections, while structural (typically dynamic stochastic general equilibrium, or DSGE) models are preferred for scenario and policy analysis (for their forward looking expectation formation for instance).

2. This model is supposed to support forecasting, but there are no indications of its ability to predict some key variables. I would like to see some forecasting results from the model (maybe, by using true values for the conditioning variables).

We agree with the referee's comment. In the new Section 4, the paper now describes the role of the ECB-MC in the projections and hopefully clarifies what 'forecasting' means for the ECB-MC. As now stated in the paper: "It is important to understand in the first place that, while the ECB/Eurosystem staff projection exercises are model-based, like in most central banks and policy institutions they are not purely defined by the bounds of traditional statistical time series analysis. They are instead nuanced exercises, continuously calibrated via expert judgement to assimilate a plethora of information streams and inputs, mirroring the dynamic economic landscape. The process itself therefore relies less on purely mechanical usage of the economic-models, and rather uses economic models as a starting point, as an organizing and constraining device and finally as a cross-checking device. Therefore, the final projections incorporate a fair amount of judgement as a result of additional information included in the model and iterative discussions that happen during the projection exercise". The forecasting process at the ECB is therefore distinct in two fundamental ways from standard statistical exercise. Firstly, it is not a purely statistical endeavour but a blend of data, models, and insights from diverse sectors, ensuring a reflection of the everchanging economic context. Secondly, the forecasting is inherently continuous, building on a rich repository of insights from past iterations, and is primarily conducted as a sequence of updates.

Given these features, we are clearer now that the ECB-MC model has been especially designed to include a systematic judgment into macroeconomic forecasts, based on practices (add-factor adjustment) that are quite common across central banks and institutions. Therefore, we claim, it is important that the model have some desirable characteristics which make it conformable to economic theory and consistent with the data. However, its purpose in the whole 'assembly line' of the projection exercises is not to produce accurate point forecasts, and hence a standard evaluation of its forecasting properties based e.g. on RMSE, or similar metrics, is almost meaningless and must be performed with accurate caveats.

We have added a new Subsection 4.3 on forecast evaluation, where we explain that the 'evaluation metric' is not the same used in standard time series practices and has been tailored to the main purposes of the model. Concretely, to evaluate its forecasting ability we ask the question: What would the (conditional) forecast of the model be if we fed it with the 'right' assumptions, namely with the financial, fiscal, and external data that occurred over the evaluation period (as opposed to the real time assumptions). The forecast performance is then evaluated relative to a simple random walk benchmark -- which provides a transparent and

model-free reference point. We also illustrate what the model would predict for HICP inflation at 2022Q1, i.e. at the beginning of the inflation surge period, had it been fed with the realised assumptions as compared to a projection based on the real-time assumptions.

3. The estimation approach has to be explained much better, with all the elements spelled out transparently and precisely.

We have significantly extended the material by adding detailed descriptions on the estimation strategies employed for different model block (this material is in the new Section 3.3 and Appendix C) as well as additional parameter estimates relative to the financial block.

In addition, we have prepared an online appendix which contains an interactive HTML file listing all of the model variables per country, and the corresponding data series keys that are publicly available via ECB SDW portal. We also provide MATLAB replication codes for all 5 countries that allow user to compute Basic Model Elasticities (BMEs) to a “monetary policy shock”. The codes also include the “Enterprise” version of Dynare developed specifically for the purposes of estimating and simulating ECB-BASE/MC models. README.html file contains detailed guide to setup and run the simulation codes. We need to point out that, for confidentiality reasons, a subset of parameters (like price and wage Phillips curves) are not the official ones used in the context of ECB (B)MPE forecasting process and represent averages across countries. Also, this version of the model does not represent the latest version of the model used for the ECB forecasting process, nor do the submitted datasets represent the latest revisions used in the same process.

4. We do not need all the impulse responses at the end. I suggest to report only one set of impulse responses in the text and, for that one, I would ask (to the extent possible) to explain the sources of cross-country heterogeneity (which is a key element of the approach). The other impulse responses can go in the appendix.

This is a very good suggestion. In line with this and also taking into account comments from the other referees, we have decided to remove the section on BMEs from the main paper and move it to the online appendix.

We have also improved the description of this exercise to make it clearer that BMEs represent model responses under a “partialised” setup required by the BMPE forecasting process (e.g. no monetary policy rule, exogenous fiscal block). As the paper focuses on the forecasting use of the ECB-MC, we do not report standard IRFs.

As regards the sources of cross-country heterogeneity in the model, we also added a paragraph that explains that “while originally conceived as a country-level framework, the ECB-MC was also developed as a multi-country counterpart to the ECB-BASE model. As a result, the model structure is deliberately kept relatively homogeneous across countries, closely mirroring the architecture of ECB-BASE. Cross-country heterogeneity in model outcomes therefore arises primarily from differences in calibration and estimation, rather than from structural differences in model specification”. We believe that this design choice facilitates a meaningful comparison between the aggregated ECB-MC results and the ECB-BASE model, thereby supporting the interpretation of euro area forecasts while preserving the country-level dimension.

The ECB-Multi-Country Model.

A semi-structural model for forecasting and policy analysis for
the largest euro area countries

Abstract

This paper introduces the European Central Bank's Multi-Country model (ECB-MC), a coherent macroeconomic framework designed to support economic forecasting analysis within the Eurosystem. The ECB-MC captures the economic dynamics of the five major economies in the euro area – Germany, France, Italy, Spain, and the Netherlands. By incorporating detailed structural features and data-driven insights, the model provides the main reference for the ECB's staff macroeconomic projections, acting as a disciplined tool for forecasting, and enabling scenario, risk and sensitivity analyses. The paper offers a detailed account of the structure, the estimation and the model properties, and provides a primer on the uses of the ECB-MC model in the Eurosystem macroeconomic projections.

Key words: Semi-structural model, forecasting, euro area countries, monetary policy

JEL Classification: C3, C5, E5, E6

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1 Introduction

This paper introduces the European Central Bank’s Multi-Country (ECB-MC) model, a pivotal component of the ECB’s modelling toolbox. The ECB-MC is designed to model the macroeconomic dynamics of the major euro area economies - Germany, France, Italy, Spain, and the Netherlands - and serves as one of the main tools in the Eurosystem’s quarterly (Broad) Macroeconomic Projection Exercises ((B)MPE). More specifically, ECB-MC is the core model used for the projection exercises and provides a framework for the forecast narrative of these countries. The model structure is quite similar across countries with the main parameter estimates providing the necessary differences to capture country-specific elasticities.

The model has been developed in the context of a larger project aiming at enriching the ECB’s collection of analytical tools to bolster forecasting accuracy and policy analysis efficacy. The general modelling strategy of the ECB is grounded on a suite-of-models approach which aims at building core models complemented by ‘satellite’ tools.¹ In this landscape, the ECB-MC stands as one of the core models, together with its euro area counterparts - the ECB-BASE model and the New Area Wide Model (NAWM).² The ECB-BASE model, described in [Angelini et al. \(2019\)](#), has provided critical benchmarks and methodological guidance for the development of a multi-country model that captures individual country specifics within the euro area while delivering a consolidated view at an aggregate level. The iterative refinement of model equations and estimation methodologies during the ECB-BASE development was instrumental in establishing a harmonized modelling environment for the ECB-MC.

The development of both the ECB-MC and ECB-BASE models drew inspiration from a well-established policy model, namely the FRB-US, which is the primary macroeconomic model used by the Federal Reserve Board ([Brayton & Tinsley \(1996\)](#)). This choice means that while adhering to conventional academic frameworks, the models also incorporate a degree of flexibility and adaptability to respond to shifts in the economic environment. These models are typically categorized as “semi-structural”, because they exist in a middle

¹See [Darracq Pariès et al. \(2021\)](#) for a comprehensive review of the Eurosystem modelling framework.

²See [Christoffel et al. \(2008\)](#) and [Coenen et al. \(2018\)](#) for the introduction of the NAWM, and [Ciccarelli et al. \(2024\)](#) for more information and references about the macroeconomic models used at ECB for projections and policy analysis.

1 ground between purely empirical models (data-driven/time-series) and fully structural models
2 (theory-driven/Dynamic Stochastic General Equilibrium (DSGE)). Therefore, they balance
3 data consistency and modelling adaptability (the 'semi' part), with fundamental theoretical
4 principles (the 'structural' part).
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8 To highlight the role of macroeconomic models, a dedicated workstream has been established
9 during the 2020 ECB strategy review, which documented the models and tools available
10 within the Eurosystem for forecast analysis and monetary policy preparation.³ This review
11 confirmed that the majority of the forecasting models employed are semi-structural, affirming
12 their adequacy for policy scenario analysis and economic forecasting. Despite this, the review
13 also pinpointed areas for potential enhancement. In this respect, the ECB-MC model
14 introduces several advancements in line with the recommendations of that strategy review.
15 Notably, it incorporates an expanded analytical framework for examining monetary policy
16 transmission through a granular financial block, and it features an extensive fiscal block that
17 allows for the examination of various fiscal-monetary policy interactions.
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27 This paper focuses especially on the role of the model in the ECB/Eurosystem forecasting
28 exercises. The properties and uses of the model are examined under a “partial” country-
29 by-country setup, namely the setup that matches the assumptions of the ECB forecasting
30 process whereby several variables (e.g. interest rates, exchange rates, international outlook)
31 are treated as exogenous assumptions to the model.
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37 The remainder of this paper is organized as follows: Section 2 surveys the landscape of
38 macroeconomic models and explains where the model fits. Section 3 offers a concise overview
39 of the ECB-MC model, its fundamental structure, the calibration and estimation techniques
40 employed. It also describes the modalities under which the model can be operationalized.
41 Section 4 exemplifies how the model has been (or can be) used in the projections exercises.
42 Section 5 concludes. The paper appendix includes the model's structure with the basic
43 equations, the VAR used to form expectations, and the estimation strategy with selected
44 parameter estimates. In an online appendix, we publish a version of the full model with
45 data key codes (when publicly available) and the necessary variables to simulate the model
46 elasticities of the five economies to a euro area monetary policy shock.
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54 ³See [Darracq Pariès et al. \(2021\)](#) for the complete report.
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2 Surveying the landscape

The aim of this section is to recall how the ECB-MC model fits into the academic and central banking modelling landscape, also with a view to better understand some specification, identification and data choices that will be explained in the next sections. In this respect, this section does not add substantially new elements to what is already contained in other papers (see, for instance: [Darracq Pariès et al. \(2021\)](#), [Ciccarelli et al. \(2024\)](#), [Angelini, Darracq Pariès, Haertel & Lalik \(2025\)](#)) and concludes with a humble assessment of the model weaknesses.

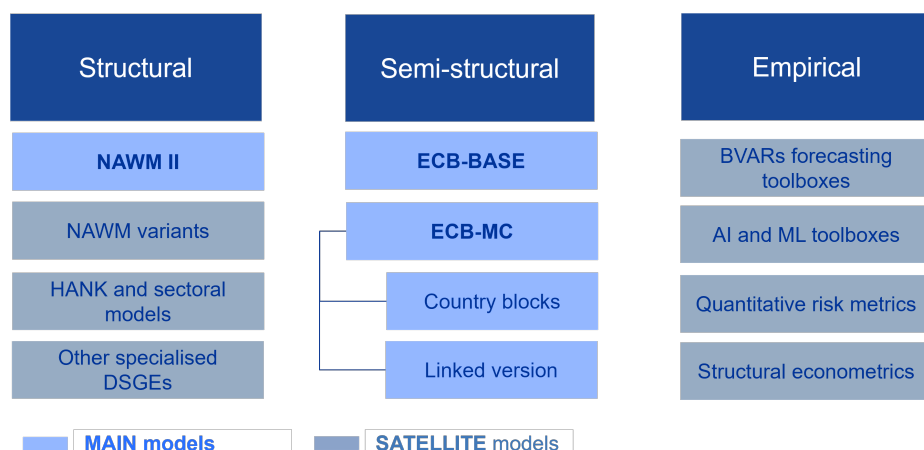
The ECB maintains a suite of models as a general strategy (see Figure 1). The suite contains state-of-the-art economic models that range from general equilibrium micro-founded models, empirically validated (either with calibration or with estimation), to more data-driven linear and non-linear time series models (e.g. VARs, Machine Learning models). Semi-structural models, such as the ECB-MC, are somewhat in the middle of the spectrum. The consensus surrounding this configuration has been shaped by the intersection of constant tensions between three forces – the institutional context, the technological paradigm and the economic environment – and a continuous interaction between academic research and policy analysis. As a result, these modelling choices could be seen as an optimal response to various trade-offs between complexity and simplicity, operational risks and business continuity, and empirical fit and theoretical soundness.⁴

In a sense, the ECB-MC model follows the Cowles Commission tradition of large structural econometric modelling, a tradition that was criticised for its rigid reliance on theoretical assumptions over data reality (e.g. [Sims \(1980\)](#)), the lack of model identifiability, and its limited usefulness in policy evaluation (Lucas’ critique). So why should this class of models still be included in the set of the ECB main models? After all, across the modelling portfolio DSGE and time-series models (including multi-country or non-linear models) have close ties with large streams of the academic literature, while such a relationship is less clear cut in the case of semi-structural models.

The compelling rationale for ECB staff to keep on investing in large semi-structural models

⁴The evolution has been explained in detail in [Ciccarelli et al. \(2024\)](#), which also contains a short history of model development at the ECB in connection with the economic, conceptual, and institutional evolution of the profession.

Figure 1: A schematic overview of the ECB suite of models used for forecasting and policy analyses.



Source: Ciccarelli et al. (2024).

Note: The table reports a schematic representation of some of the ECB models used in projections and scenario analyses.

has been also explained in other publications that the interested reader can consult (see e.g. Ciccarelli et al. (2024)). Here we briefly recall that the rational has to do with their flexible use, their modular estimation regardless of the large scale, and their conceptual and institutional flexibility, which allows to introduce alternative specifications when needed, while accounting for the complexity of the policy process.⁵ This institutional aspect is important because model-based analysis for policy purposes entails the use of judgement in forecasting activities or a need to compare the general equilibrium model perspective with sectoral and/or country-specific expert assessments, that DSGE or time-series (multi-country) models might typically miss. These models are instead used to provide a judgement-free input that can be used to cross check the analysis informed by the semi-structural models.

In fact, the models of the suite are rarely used in isolation, especially in the projection process. The specific use of the ECB-MC in the projections is explained in detail in Section 4. The reader should imagine an ‘assembly line’ where the nowcast and short-term forecasts of inflation and GDP growth at various levels of disaggregation – produced with VAR and other state-of-the art empirical models (see Eraslan et al. (2025)) – are used as starting points for the semi-structural model. This in turn is projected to derive the medium-term forecasts, conditional on several other assumptions. This forecast is then ‘improved’ with

⁵See, for instance, Angelini et al. (2023) who modify the ECB-BASE model to flexibly include an epidemiological module which enabled our teams to assess the interplay between the epidemiological fundamentals of the COVID-19, containment policies and the economy during the difficult years of the pandemics.

expert judgement (see [Ciccarelli et al. \(2023\)](#)) and finally scrutinised through the lens of other models (typically structural DSGE or time series models). The latter add two important elements to the outcome: (1) they deploy a cross-check and a narrative in terms of shock decomposition of the projections that is in line with an internally consistent economic theory; (2) they perform policy simulations and scenarios (including risk analysis and the assessment of optimal policy) around the baseline projections (see e.g. [Carboni et al. \(2026\)](#) for a growth-at-risk model and [Mazelis et al. \(2023\)](#) for optimal monetary policy).

The ECB is not the only institution that uses such a modelling strategy for policy preparation. A recent assessment of the macroeconomic modelling strategies across the Eurosystem (see [Darracq Pariès et al. \(2021\)](#)), for instance, confirms that the typical macroeconomic modelling strategies in use across the Eurosystem feature a suite-of-models approach. “This balances the need for robustness across different uses (provided by model diversity) against flexible deployment of selected types of models for specific purposes. Most NCBs use large-scale semi-structural models for projections, while structural (typically DSGE) models are preferred for scenario and policy analysis. The main macroeconomic models (especially projection models) are usually operated in conjunction with a variety of other satellite tools, notably for short-term forecasts and sectoral or financial analysis.”

Finally notice that, outside the Eurosystem, all major central banks have also maintained a similar suite-of-model approach, with semi-structural models at the core of the framework.⁶

The fact that so many policy institutions use semi-structural models, does not make these models perfect. The weaknesses of these models (and hence of the ECB-MC) are well known and that explains why they are never used alone and are, instead, always complemented by expert judgement and more theoretically founded models. Compared to Dynamic Stochastic General Equilibrium (DSGE) models, for instance, semi-structural models have weaker theoretical foundations, making them more vulnerable to the Lucas’ critique. More precisely, these models often rely on reduced form specifications with expectations either completely absent or not being forward looking.⁷ As many other estimated models (including DSGEs),

⁶See, for example, the CNB blog post https://www.cnb.cz/en/about_cnb/cnblog/Less-theory-more-reality-Semi-structural-models-have-a-lot-to-offer and the article in *Central Banking*, <https://www.centralbanking.com/benchmarking/economics/7715701/semi-structural-models-are-the-forecast-weapon-of-choice>.

⁷Although, as pointed out by e.g. R. Tetlow, “In the original 1976 Critique article, Lucas warned about failing to distinguish between ‘parameters’ that were amenable to policy – such as expectations – and those

they have clear difficulty in handling unprecedented shocks, and they may be slow to adapt to structural changes or new types of economic shocks that break historical correlations. Moreover, these models are sometimes subject to intensive expert judgement in fixing the parameters, which can introduce subjective biases and reduce the model’s transparency or reproducibility. Finally, the absence of full system estimation, might lead to issues with overall consistency and difficulties in capturing two-way feedback mechanisms, among other things.⁸

With this landscape in sight, we will now describe the main features of the model and its centrality in the ECB/Eurosystem projections. We will come back to (and expand) some of the key aspects mentioned in this section also in Section 4.

3 An overview of the model

The ECB-MC model is a large-scale quarterly macroeconomic model for the five-largest euro area economies (Germany, France, Italy, Spain, and the Netherlands). It has been built following the ECB’s modelling strategy to develop a robust suite of models for forecasting and policy analysis. The ECB-MC is one of the main models used in the ECB’s forecasting process. It replaced the previous multi-country model, the NMCM, which was documented in Dieppe et al. (2011). Compared to the NMCM, among other things, the ECB-MC has a less stringent adherence to an optimization framework – which makes it better suited to provide empirical responses that match closely with the data – and a more sophisticated financial block – that captures more realistically the role of macro-financial linkages and the channels through which standard and non-standard monetary policy can affect euro area economies.⁹

that were immutable to policy – ‘structural’ parameters.” The extent to which some simulations are done in a model-consistent expectation approach that the ECB-MC like the FRB/US can have should alleviate the problem. See this blog post: https://noahpinionblog.blogspot.com/2014/04/the-foxy-fed_7.html and the discussion included therein.

⁸This is a not a clear cut drawback while the use of limited information estimation is sometimes justified on the grounds of misspecification prevention whereby potential misspecification of a particular model block/sector does not transfer to the rest of the model.

⁹The NMCM had gone through a protracted underinvestment that reduced its effectiveness in contributing to the projection process, and this had happened precisely at the time when multi-country analysis was needed the most. In addition, some key parameters such as the sensitivity of inflation to economic activity—the coefficient in the Phillips curve—needed to be recalibrated for a correct use of the model in the forecasting process. The model also lacked a separate treatment of business investment. More generally, many users viewed the complexity of the model as embedded in the learning mechanisms of expectations often as preventing an intuitive interpretation of its output, thus making it less helpful in providing consistent

The ECB-MC builds on the methodological framework of the ECB-BASE model. While ECB-BASE is a semi-structural model designed for policy and scenario analysis at the aggregate euro area level, ECB-MC extends this framework to a multi-country setting by modelling the largest euro area economies individually. The ECB-MC therefore preserves the core behavioural structure of ECB-BASE while introducing country-specific trade, financial elements that are essential for forecasting. Both the ECB-BASE and ECB-MC models are constructed using methodologies similar to those employed in the FRB-US model of the Federal Reserve Board. For an in-depth exposition of the methodologies common to both the ECB-MC and ECB-BASE models, readers are directed to consult [Angelini et al. \(2019\)](#). However, the two models differ in their scope and intended application. ECB-BASE is designed as an aggregate euro area model for area-wide policy analysis, whereas ECB-MC extends the same semi-structural framework to a multi-country environment that explicitly captures heterogeneity across euro area economies and the interactions between them. The primary objective of both models is to support central bank forecasting and policy analysis. This requires that the models: (a) capture the interdependencies among key macroeconomic variables; (b) incorporate realistic transmission channels for monetary policy and other economic shocks; and (c) provide a coherent accounting framework consistent with the needs of the forecasting process. The key features distinguishing the two models are described in Subsection 3.2.

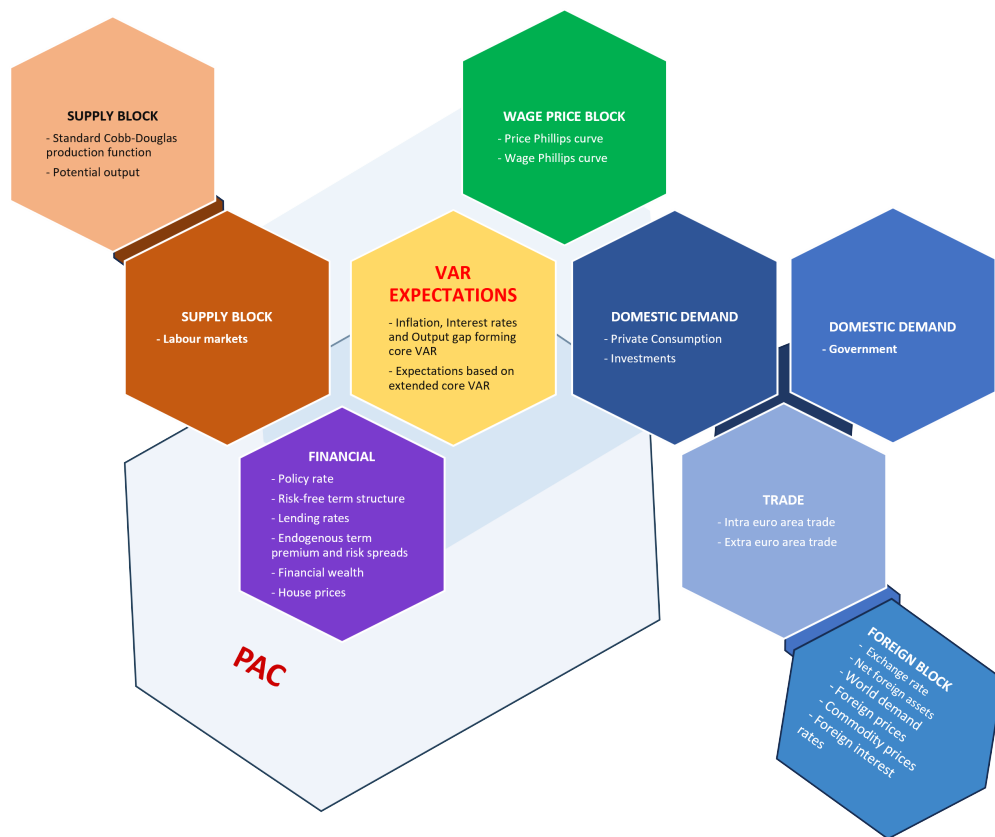
The forecasting and policy analysis requirements necessitate various approaches for solving the model. For instance, the ECB-MC can operate in a “single-country version”, which allows for individual country simulations. Alternatively, it can be executed in a “linked version” that simultaneously solves for all (Big-5) countries, integrating active trade connections, a unified monetary policy, and shared financial variables such as exchange rates and interest rate expectations. Furthermore, the model permits “partialization”, whereby certain policy functions or variables are exogenised. This process eliminates feedback loops within a specific channel. This technique is particularly valuable during forecasting, where numerous assumptions are imposed exogenously. Further elaboration on these methodologies is available in Section 4.

interpretations of the economic developments underlying the forecast.

3.1 Model's structure

To meet the analytical demands of the forecasting process and various policy applications, the ECB-MC model is structured into several interdependent blocks that effectively capture the dynamics of key macroeconomic variables and their interactions. As illustrated in Figure 2, for each of the Big-5 euro area countries, the model is organized into five distinct blocks. We report here the main characteristics of each block and in the appendix a more detailed account of the main equations. The full list of model equations and variables is available in the online appendix.

Figure 2: A schematic overview of the main building blocks of ECB-MC



Source: [Adjemian, Bokan, Darracq Pariès, Müller & Zimic \(2024\)](#).

Notes: Each hexagon corresponds to a particular block which is attracted by the specific long-run anchors that via expectations serve as a model disciplining device. In this framework, the core VAR pins down the main relationship between inflation, output-gap and interest rate expanded with the additional variable-specific determinants block ensuring (at least partially) system wide coherence. In addition, the main model dynamics governed by the PAC equations is influenced by these expectations. The large blue shaded hexagon denotes the former while the framed light-blue hexagon denotes the latter.

3.1.1 Domestic Demand Block

This block encompasses household consumption, business and residential investment, and variables related to fiscal policy. It is designed to provide a comprehensive view of the domestic economic activity driving demand within each country.

For consumption, we distinguish two types of households. Some households are liquidity constrained and follow a rule-of-thumb behaviour. Others have optimizing behaviour and maximize their expected lifetime utility subject to resource constraint. The consumption of liquidity constrained households moves in line with their labour and transfer income.

For investment, we also consider two types of agents (firms): cash-flow constrained and optimizing agents that maximize profits subject to capital accumulation constraint. The output of a firm is given by the Cobb-Douglas production function with constant returns to scale and two production inputs, capital and labour.

Finally, the government block provides a comprehensive representation of government revenues and expenditures and their interactions with the domestic economy. On the revenue side, the model incorporates implicit tax rates aligned with data from government finance statistics, covering taxes on income and wealth, as well as taxes on production and imports. On the expenditure side, it includes government consumption, investment, and social transfers.

3.1.2 Trade Block

This segment accounts for intra- and extra-euro area trade, including the trade of commodities. It covers both volumes and prices, accounts for the role of exchange rates, and models the impact of external economic factors that influence the domestic economies of the euro area.

The modelling of extra-euro area trade volumes and prices is essential to analysing the impact of external shocks that materialize outside the euro area. On the other hand, the intra-euro area variables enable the consistency of the euro-area dynamics as the demand for exports in one euro area country is automatically reflected in imports of other euro area countries, taking into account the respective trade shares. All trade volumes are modelled in real terms.

Trade deflators are also modelled using a ‘top-down’ approach. This means that we model total import and export deflators and the extra-euro area deflators, whereas the intra-euro area prices are re-computed in a consistent manner.

3.1.3 Supply Block

Central to this block is the labour market and the associated factor demands, which are derived from a Cobb-Douglas production function. This framework also allows for the analysis of how inputs such as labour and capital combine to produce output and assess the underlying growth potential of the economy. The labour market is centred around an equation for total employees. Furthermore the block features an equation for hours worked, a measure for unemployment and a separate modelling of self-employed and the labour force participation rate.

3.1.4 Wages and Prices Block

This block models wage dynamics and price adjustments, critical for understanding inflationary pressures and wage-price spirals within the economy. It also integrates the interplay between labour market conditions and inflation, essential for monetary policy considerations. Central to this block, as in the ECB-BASE model, is a small ‘reduced-form’ general equilibrium model called WAPRO (Wage-Price-Output gap). By ‘reduced-form’ we mean that, while some elements, including microfoundations and the explicit modelling of expectations, are borrowed from the dynamic stochastic general equilibrium literature, the economic restrictions are relaxed to improve the data fit. Therefore, even if the structure of the equations is guided by optimizing behaviour of the economic agents, the block is estimated without imposing cross-equation restrictions of the structural parameters. In this sense the loading coefficients are estimated (or calibrated) in a ‘reduced-form’.

The block is centred around a wage-price and output gap specification, that determines jointly output gap, inflation, wages and unemployment. The key inflation indicator in the model is the GDP deflator that follows a New Keynesian Phillips Curve, with inflation depending both on a measure of expected inflation and on past inflation. The long-run values of HICP sub-components are modelled as weighted averages of GDP deflator and import prices, with weights given by the import content in the final use of the respective

sector in the economy. The short-run sub-component dynamics follows the error-correction model. The total HICP is then modelled as a weighted sum of the various components.

Finally, the wage inflation gap is also modelled with a reduced form of the New Keynesian Phillips Curve. It is a function of future wages, of past, present and future domestic price inflation, of present and future euro area price inflation, of the deviation of the labour share of income from its trend, of the deviation of trend output growth from its steady state, and of the deviation of the unemployment rate from its trend (Galì (2011)).

3.1.5 Financial Sector Block

This block models the financial intermediation process, encompassing interest rates, banking lending rates, and broader financial market dynamics. Specifically, in the context of the monetary union, we model a common, area-wide, reference short-term rate and country-specific long-term market rates. The reference short-term interest rate is assumed to follow a simple monetary policy rule while the long-term rates are determined as the sum of the average of expected short-term rates, term premium, and country-specific risk premium. Expectations about the future short-term rates are derived from a VAR.¹⁰ The term premium is modelled separately as a function of expected macroeconomic conditions on euro area level and external market developments. The country risk premium depends on the expected macroeconomic conditions and the fiscal position of the country. Finally, the short-term rate and the long-term market rate, alongside the risk premia associated with particular debt and equity instruments, provide the basis for the construction of lending rates and financing cost measures used in other parts of the ECB-MC model.

3.1.6 Additional building blocks

Together, these blocks form a robust framework that enables a detailed examination of the economic mechanisms at play within the largest economies of the euro area, providing crucial insights for forecasting and policy-making. As in all semi-structural models, each block of the model can be decomposed into behavioural equations, bridging equations, and identities. The equation residuals play a crucial role in aligning the model with empirical data. This adjustment is vital for the forecasting process, where expert judgement is incorporated on

¹⁰For more details see the definition of expectations at the end of this section and the Appendix B.

top of model-based outcomes. This integration of expert analysis is further detailed in Section 4.

Behavioural Equations: These equations are derived from theoretical underpinnings and are primarily estimated to reflect the optimizing behaviours of firms, households, and financial markets, subject to polynomial adjustment costs (PAC) (as in [Tinsley \(2002\)](#) and [Kozicki & Tinsley \(2002\)](#)). This framework encapsulates three core components:

- **Theory-based Target:** This element represents the ideal or goal that agents (firms, households, etc.) strive to achieve under “perfect” conditions.
- **Expectations about Future Targets:** Agents form expectations about future values of their targets based on available information, which guide their current decisions.
- **Dynamic Adjustment Mechanisms:** These are typically modelled using a PAC error-correction approach. The dependent variable in these equations is influenced by its historical values, the deviation of its previous state from the theoretical target (error-correction term), and anticipations regarding future adjustments in the target variable.

The target variables are derived as:

$$\begin{aligned}
 x_t^* &= \underset{x}{\operatorname{argmax}} E_t \sum_{t=0}^D \beta^t f(x_t, u_t) \\
 &\quad \text{s.t} \\
 &\quad u_{t+1} = g(x_t, u_t), \\
 &\quad x_t \in \Phi, \\
 &\quad x_0 \text{ given and } D \in \{T, \infty\}
 \end{aligned}$$

where u_t is the state variable, x_t is the control variable chosen for the optimisation problem, the constraint x_{t+1} specifies the influence that the control exerts on the state and the control variable takes values in a given set $\Phi \subseteq \mathbb{R}$. D denotes the optimisation horizon that can be finite or infinite. Loosely speaking, this is a generic representation for all optimising problems in the model, such as households’ lifetime utility optimisation in consumption

or profit maximisation by firms subject to capital accumulation, to determine investment and the optimal level of employment. However, the empirical success of the model crucially depends on the assumption that agents cannot adapt their behaviour instantaneously in line with optimal/theory-based conditions as they face adjustment costs. The assumption of the existence of adjustment costs implies rich short-run dynamic behaviour that resembles an error-correction formulation for each modelled variable x_t . Formally:

$$\Delta x_t = a_0(x_{t-1}^* - x_{t+1}) + \sum_{i=1}^{m-1} a_i \Delta x_{t-i} + E_{t-1} \sum_{j=0}^{\infty} d_j \Delta x_{t+1}^* \quad (1)$$

where a_0 is the coefficient on the deviation of the variable of interest from its target and a_i gives the weights on the backward-looking terms. The term $E_{t-1} \sum_{j=0}^{\infty} d_j \Delta x_{t+1}^*$ represents the expectations of weighted changes in future targets with d_j being weights on the forward-looking terms.

Polynomial Adjustment Costs (PAC): the main idea behind the PAC approach is that agents cannot change their behaviour to adapt the optimal conditions immediately. There are some adjustment costs, which the agents take into account when forming their expectations. Formally, it requires finding an admissible representation to the last term in Equation (2). Following [Brayton et al. \(2000\)](#) and [Tinsley \(2002\)](#), we do it by building a small-scale auxiliary VAR model for each behavioural equation that employs PAC approach. Those VARs consist of the specific variable for which expectations are being formed and a core set of macro variables: the policy rate (euro area), the GDP deflators (both euro area and country), and the output gaps (both euro area and country).¹¹ The general derivation of the PAC approach and its econometric implementation following [Brayton et al. \(2000\)](#) is described in [Angelini et al. \(2019\)](#).

Expectations are modelled explicitly and can be derived under a VAR framework or model-consistent approach. In the former approach, future values of the lead variables are replaced by forecasts from a VAR model and the model is then simulated. The VAR expectation case assumes only limited knowledge of the joint dynamics of the variables and corresponds to the same restricted information set used in the estimation of the model as detailed in the previous

¹¹In the specification of the country specific "core" VAR we allow the country variable to be affected by the euro area counterpart but eliminate the direct feedback of the country variables to the euro area counterparts. In other words, we utilize so called block-exogenous approach.

paragraph. This design loosely speaking may be interpreted as a limited form of rational expectations. Moreover, PAC expectations can be conformably rewritten so that the model can also be simulated using standard techniques for simulating models with forward-looking terms in a model-consistent manner (rational expectations) or as a combination of the two expectation formation processes.

3.2 ECB-MC specific features

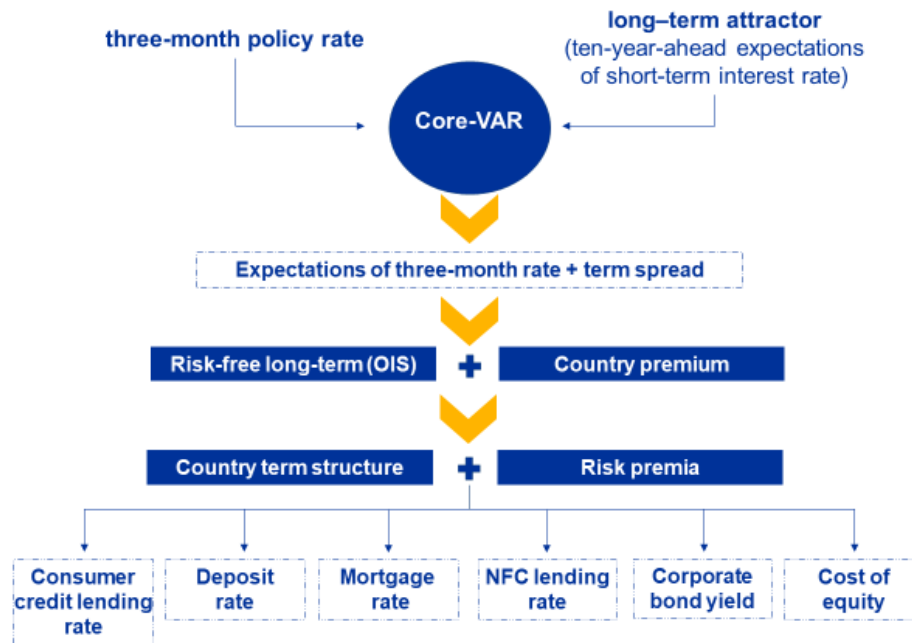
Despite sharing the same structural framework as the ECB-BASE, the ECB-MC model incorporates distinct features within its country-specific blocks, particularly in the trade and financial sectors. Moreover, the core-VAR component of the model integrates both national (country) and euro area variables in the expectations formation, enhancing its analytical depth. The main elements introduced into the ECB-MC structure can be summarized as follows:

The **Trade Block** is split in intra- and extra-euro area components. However, the estimation strategy for the countries is done on total trade directly and on extra-euro area trade, with intra-euro area being a residual. More precisely, the trade block includes error-correction equations for total real exports and imports, and prices. The modelling approach for the trade block is quite traditional. We have chosen a highly aggregated approach with no split between goods and services. Moreover, bilateral trade flows are not directly considered. External indicators for demand and prices are the effective exchange rate based on weighted averages of main trading partners, world demand and competitors prices. The import demand indicator is calculated as a weighted average of final demand components and can be considered as an indicator of import absorption. The weights are determined by the import content derived from the input output tables.

The **Financial Sector** within the ECB-MC closely mirrors the architecture used in the ECB-BASE model, but it has been expanded to include the dynamics of country-specific risk premiums (see Figure 3). The central focus of the financial block is on interest rates, primarily structured around the risk-free term-structure. The euro area's long-term risk-free rate is derived from the short-term interest rate, the anticipated trajectory of this rate as forecasted by the core VAR, and the term premium. Country-specific premiums are then modelled as a function of macroeconomic fundamentals, such as expected output gap and

fiscal positions. The sum of the euro area risk-free long term rates and the country premium determines the country-specific long-term interest rates. This methodology aligns with that used in the ECB-BASE model and serves as the foundation for constructing nominal lending rates, which are calculated as the sum of a weighted average of short-term and long-term rates, augmented by a risk spread.

Figure 3: A schematic overview of the financial sector in ECB-MC



The **Expectation Formation Mechanism** has been enhanced to incorporate both euro area and country-specific information. The VARs utilized for constructing expectations are structured into four blocks:

- **Target Variable:** This block includes the variable for which expectations are being formed, such as expected consumption. It consists solely of this variable and its own historical values, making it equation-specific.
- **Country Core/Base-VAR Variables:** This block comprises key domestic economic indicators like inflation, the output gap, and the interest rate. These variables, often referred to as core-VAR variables, represent the most significant state variables of the national economy and are consistent across all equations.
- **Euro Area Core/Base-VAR Variables:** These variables reflect broader euro area

dynamics, including the common monetary policy and euro area-specific interest rates that are influenced by regional economic developments, rather than country-specific factors.

- **Long-term Attractors:** This final part includes variables that represent the perceived long-term equilibrium states, modelled using data on long-term trends in growth, inflation, and interest rate expectations.

For detailed exposition of the VAR setup and the estimation, see the next subsection and the Appendix [B](#).

3.3 Calibration and estimation

The parameters of the model are derived from a mix of calibration and estimation. The general rule is that the parameters of the target variables are mostly calibrated based on theory and micro evidence, while the coefficients of the dynamic adjustment are data driven and are estimated. Regarding the estimation techniques we follow the approach detailed in [Angelini et al. \(2019\)](#) with few notable exceptions. For various PAC blocks (notably consumption, private and residential investments and house prices) we employ a panel pooled FGLS estimation method to obtain initial conditions (mean) for the parameters of various country specific target variables. They are subsequently used as prior information in a Bayesian estimation framework to robustify the estimates. We illustrate the main estimation techniques used in ECB-MC, by presenting the estimation strategy for two important blocks: the consumption and the financial blocks.

3.3.1 Estimation of the consumption block

The block combines long-run target relationships, VAR-based expectations formation and dynamic PAC adjustment equations, and is representative of the broader estimation framework used throughout the model.¹²

The consumption block follows the PAC framework and combines forward-looking households

¹²An additional estimation step involves the estimation of the permanent income components required for the target consumption equation. Detailed description can be found in [Angelini et al. \(2019\)](#). In ECB-MC, the main extension is the use of pooled FGLS estimation for the income-gap ECMs across countries. The approach is analogous to the pooled FGLS framework used for the estimation of the VAR systems underlying expectations formation explained in this section.

with hand-to-mouth consumers. The estimated equation is:

$$\Delta c_t = \theta \left[a_0(c_{t-1}^* - c_{t-1}) + \sum_{i=1}^{m-1} a_i \Delta c_{t-i} + Z_t \right] + (1 - \theta) \Delta y_t, \quad (2)$$

where c_t^* denotes target consumption, Z_t captures expected future target growth implied by the VAR system and Δy_t denotes current income growth. The parameter θ measures the share of forward-looking households. The estimation proceeds sequentially in three steps:

1. Estimation of the long-run target consumption equation using Bayesian linear regression with priors centered on ECB-BASE estimates.
2. Estimation of the VAR systems underlying expectations formation. The estimation combines pooled FGLS and Bayesian SUR methods in order to exploit cross-country information while allowing for limited parameter heterogeneity across countries.
3. Estimation of the dynamic PAC equation using non-linear least squares conditional on the estimated VAR-implied expectations terms.

Long-run Target Consumption

The target consumption equation is specified as:

$$c_t^* = \eta_0 + \eta_L y_t^{L,p} + \eta_T y_t^{T,p} + \eta_P y_t^{P,p} + \eta_D w_t + \gamma^T t, \quad (3)$$

where the permanent income components are constructed using VAR-based expectations. The equation is estimated country-by-country using a Bayesian linear regression model with independent priors for the coefficient vector and the precision matrix centred on ECB-BASE estimates. The estimation is implemented using a Gibbs sampler that alternates between draws of the coefficient vector conditional on the error precision and draws of the error precision conditional on the coefficients (see e.g. [Koop \(2003\)](#)). The reported estimates correspond to the posterior means.

VAR Estimation

The VAR systems underlying expectations formation described in more details in Appendix

B are estimated jointly across countries using a pooled SUR framework:

$$Y = X\alpha + \mathcal{E}, \quad \mathbb{V}(\mathcal{E}) = \Sigma \otimes I_T. \quad (4)$$

The estimation proceeds in two stages. First, pooled FGLS estimation is used to obtain common coefficients across countries:

$$\hat{\alpha}^{(j+1)} = (X^\top (\hat{\Sigma}^{(j)-1} \otimes I_T) X)^{-1} X^\top (\hat{\Sigma}^{(j)-1} \otimes I_T) Y. \quad (5)$$

The covariance matrix is iteratively updated from the residuals until convergence. The pooled estimates are subsequently used as priors in the Bayesian SUR estimation, which relaxes the full-pooling restriction and allows for limited cross-country heterogeneity. Further details are provided in Appendix D.

PAC Estimation

Conditional on the estimated VAR system, the expectations term Z_t is obtained from the companion-form representation of the VAR and can be expressed as:

$$Z_t = \sum_{j=0}^{\infty} d_j \iota H^{j+1} x_{t-1} = h' x_{t-1} \quad (6)$$

where, coefficient vector h' is a non-linear function of the VAR matrix H , discount factor β and a_k coefficients. x are the VAR variables which are a subset of the full model variable set. ι is a selection vector that picks variables in x that are considered to be relevant for the imperfect information set under which the VAR-forecast is formed.

Important to point out is that the estimated VARs are an input into the estimation of the PAC equations, whereby the expectation terms depend non-linearly on the coefficients of the underlying VAR model. Therefore, in the final step the equation (2) is estimated using non-linear least squares. Standard errors are computed from the final linear approximation conditional on the estimated expectations term.¹³

¹³Alternatively one can use iterative LS approach, as done e.g. in the FRB-US model.

3.3.2 Estimation of the financial block

The estimation of the financial block is different from the ones of the other blocks. It proceeds in four stages. First, ten-year expected values of the short-term interest rate, euro area output gap, and country output gap are computed as finite-horizon present values using VAR-based expectations constructed from a trend-component auxiliary model with a 40-quarter horizon. Second, the euro area 10-year term premium, defined as the gap between the 10-year OIS rate and the VAR-implied expected short rate, is estimated via Bayesian linear regression methods as an AR(1) process augmented with expected output gap and the first difference of the US term premium. We impose a tight prior on the AR coefficient, a moderate prior on the output gap coefficient, and a diffuse prior on the constant. Third, country-specific lending rates for household consumption loans, household mortgage loans, long-term corporate lending rates, household deposit rates, corporate bond yields, and cost of equity are all modelled as weighted averages of the country 10-year government bond yield and the euro area short-term rate, where the weight on the long rate is estimated in a Bayesian linear regression method. The country bond yield itself is the sum of the OIS rate and a country risk premium and is estimated separately in the country-premium block.¹⁴ Fourth, the credit spreads for each of the six instruments, defined as the residual between the actual lending rate and its benchmark rate mixture, are all estimated individually as AR(1) equations expanded with an expected output gap term again via Bayesian linear regressions. Detailed values of the estimated coefficient are presented in Table 10.

3.3.3 Estimation methods landscape

Other PAC blocks are estimated by an appropriate adjustment of the methodology illustrated above to the specific block variables. Non-PAC blocks (except WAPRO), rely on standard error-correction two-stage estimation techniques in combination with Bayesian estimation, or directly on single equation Bayesian methods. The Wage-Price-Output (WAPRO) gap block is estimated as a reduced form system using Bayesian techniques. Table 1 gives an overview of all the methods applied for the estimation of ECB-MC model.

¹⁴The country premium block is estimated by using the Seemingly Unrelated Regression estimation (SUR) on the fixed-effect cross-country model. The estimation sample covers the sovereign debt crisis period which facilitates the extraction of the cross-country risk differentials.

Table 1. Estimation methods by block: ECB-MC model

Block	Estimation framework	Sample
Business investment	PAC; SUR Pooled FGLS; Bayesian linear regression; NLS	2000Q1–2018Q4
Consumption	PAC; SUR Pooled FGLS; Bayesian linear regression; NLS	1999Q1–2018Q4
Country premium	Fixed-effects panel SUR	2008Q2–2013Q4
Deflators	ECM long-run / short-run; OLS	2000Q1–2018Q4
Exchange rate	Bayesian linear regression	2000Q1–2016Q4
Financial/NFA	Bayesian linear regression	1999Q1–2018Q4
Fiscal	OLS	1999Q1–2018Q4
HICP	ECM long-run / short-run; Bayesian linear regression; OLS ECMs	1996Q2–2018Q4
House prices	PAC; SUR Pooled FGLS; Bayesian linear regression; NLS	2008Q4–2018Q4
Inventories	OLS	2000Q1–2016Q4
Labour	PAC; SUR Pooled FGLS; Bayesian linear regression; NLS	1999Q1–2018Q4
NFA	OLS	1998Q3–2017Q1
Property income	PAC; SUR Pooled FGLS; Bayesian linear regression; NLS	2000Q1–2018Q4
Residential investment	PAC; SUR Pooled FGLS; Bayesian linear regression; NLS	1999Q1–2018Q4
Trade	Bayesian linear regression; OLS	1995Q3–2018Q4
Wage	PAC; SUR Pooled FGLS; Bayesian linear regression; NLS	1996Q2–2018Q4
Wapro	System-wide Bayesian estimation	1995Q4–2018Q4
Wealth	Calibrated	—

Notes. PAC = Polynomial Adjustment Cost framework. SUR Pooled FGLS = Seemingly Unrelated Regression Pooled Feasible Generalised Least Squares. NLS = Non-Linear Least Squares. Bayesian linear regression = Gibbs sampler.

We use a Dynare toolkit with features that have been specifically developed for estimation and simulation of the ECB-MC model.¹⁵ More specifically, model simulation is carried

¹⁵See the reference manual, [Adjemian, Juillard, Karamé, Mutschler, Pfeifer, Ratto, Rion & Villemot \(2024\)](#), for more details. See also [Adjemian, Boku, Darracq Pariès, Müller & Zimic \(2024\)](#) for specific aspects related to the solution methods implemented for PAC based model within Dynare. The interested reader can find

1 out using specific Dynare solution routines for backward-looking models. Given that each
2 endogenous variable in the ECB-MC model can be associated with a particular equation and
3 can be expressed as equal to some left-hand side expression, a Dynare solution algorithm is
4 implemented which efficiently exploits these characteristics.
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8 Given that the ECB-MC includes five countries, for the sake of brevity a selected sample of
9 the estimated coefficients is reported in the Appendix A.1.4. An online appendix reports
10 the full list of model equations and variables.¹⁶
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14 **4 Forecasting with the ECB-MC model**

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16 This section describes how the ECB-MC model is currently used in the quarterly macroe-
17 conomic projection exercises of the ECB. This process adopts a bottom-up methodology,
18 initiating forecasts at the national level before aggregating the results to the euro area
19 level. The ECB-MC is used to produce the forecast for the five largest euro area coun-
20 tries: Germany, France, Italy, Spain, and the Netherlands.¹⁷ We have described in other
21 papers the forecasting procedures and techniques at the core of the ECB/Eurosystem staff
22 macroeconomic projections. The interested reader can check, for instance, the Guide to
23 the ECB/Eurosystem staff macroeconomic projection exercises ([European Central Bank](#)
24 (2016)) to get a sense of the key characteristics of the projection exercises, the main actors
25 involved, and the use of models and judgement in the preparation of the final numbers and
26 scenarios.¹⁸
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37 the implementation example details therein.

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39 ¹⁶In an interactive web-page type documentation of the model we will also report a list of codes from the
40 ECB's Data Portal indicating, when available, the time series used for estimation. It should be noted that
41 the statistical time series are continuously revised. Therefore the recent data snapshot might not fully reflect
42 the same information as used during the estimation process. In addition, a subset of data is restricted and
43 therefore not accessible by general public. This interactive page is currently under construction.

44 ¹⁷The five countries are largely representative of the general and specific most important euro area
45 characteristics and account for more than 80 percent of the euro area GDP. The forecast of the remaining
46 euro-area countries is generated with an additional framework, a small scaled error correction type model.
47 The forecast follows a bottom-up approach, forecasting each country individually and then aggregating the
48 results. For details on the process, aggregation and tools see, [European Central Bank \(2016\)](#).

49 ¹⁸See also [Ciccarelli et al. \(2023\)](#), the blog on models and projections for a good understanding of the
50 main models and of the model-based forecast augmented with judgement. The ECB Occasional Paper on
51 Macroeconometric models for forecasting and policy analysis, [Ciccarelli et al. \(2024\)](#), also illustrates with
52 details the policy and forecasting use of the modelling portfolio.
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4.1 The role of models in the ECB forecasting process

Models play an important role in the ECB projection exercises: (1) They provide a clear accounting framework (identities) and a medium-term path for the economy based on estimated historical relationships (behavioural equations). (2) They help build a forecast narrative, in terms of driving forces and dynamics for the outlook. (3) They are the cornerstone for conducting risk and scenarios analysis. In the next pages we will focus on the special role of the ECB-MC in the projection process and on the various steps that are needed to build the projection narrative. Before describing the role of the ECB-MC in this process, however, we need to clarify two main aspects of the process itself.

First, it is necessary to describe the modelling portfolio that supports the policy preparation. We have already said in Section 2 that the ECB uses a suite of models for projections to provide a good balance between model diversity and specialisation, in line with international best practice. The models can be divided into three broad groups, as illustrated in Figure 1 and described in [Ciccarelli et al. \(2023\)](#):

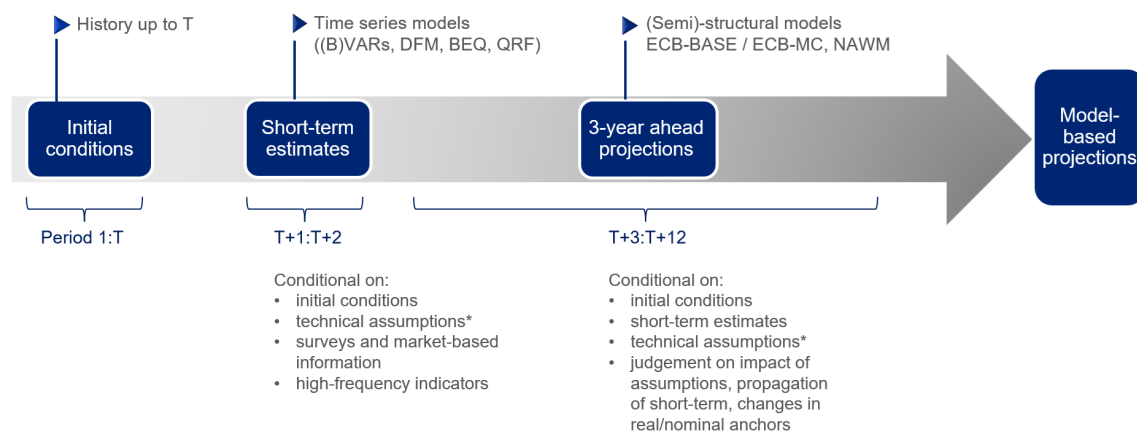
1. **The main projections models.** These are ‘semi-structural’ models, namely models that are built around a small theoretical core specifying long-run relationships for the key macroeconomic variables, while the short-run dynamics are empirically fitted with PAC error-correction type of regressions. The ECB-MC is used to produce country projections while the ECB-BASE model is the main ECB model for euro area aggregate data. The latter plays an important role in cross-checking the bottom-up aggregation of country projections, in scenario analyses and in monetary policy preparation.
2. **Dynamic stochastic general equilibrium (DSGE) models.** These are general equilibrium micro-funded models where the macroeconomic outcomes are the aggregate consequences of the simulated decisions of the various economic agents. Turns out that these models are not so straightforward to use for forecasting purposes in the given institutional setup. Therefore, they generally complement the main projection models with counterfactual scenario analyses as well as monetary policy evaluations. The New Area-Wide Model II (NAWM) is one of the main euro-area structural models for scenario forecasting and monetary policy analysis at the ECB. Beside specific NCB models, multi-country versions of the NAWM also exist which jointly models the euro

area and the world economy (see EAGLE model (Jacquinot et al. (2010)) and DREAM model (Darracq-Pariès et al. (2026))). These DSGE multi-country models have been designed for conducting quantitative policy analysis of macroeconomic development interdependences within the euro area and between euro area regions and the world economy.

3. **Empirical ‘satellite’ models.** These models refer to a wide range of econometric and statistical frameworks used for nowcasting and short-term forecasting, cross-checking projections, or modelling sectoral or financial factors not fully captured by the main models. Satellite models include, for instance, time series models such as vector autoregressive, factor and machine learning models. Recent examples include new econometric methods developed to produce short-term forecasts with higher frequency information during the pandemic or cope with exceptionally high energy price volatility during the current inflation surge (e.g. Eraslan et al. (2025), Banbura et al. (2025)).

Figure 4 broadly summarises the general projection steps and the tools involved. More details of these steps can be found in Ciccarelli et al. (2024).

Figure 4: The projections steps



Note: The figure shows the various steps to obtain the final projections. Short-term estimates and three year ahead projections share a common set of exogenous assumptions.

The various complementary models are needed to complete the narrative around the forecast. This is mainly done with three types of analysis: sensitivity, risk, and scenario analysis.

- The sensitivity analysis assesses the consequences for inflation or GDP growth of

different paths assumed for selected assumptions. This is done using the same ECB-MC model for countries, and ECB-BASE for the euro area.

- The risk analysis looks at how likely it is that economic developments will deviate from the baseline depending on events that would push the mode of the baseline distribution to the left or to the right. This risk distribution is typically estimated by resorting to satellite models and tools (e.g. [Lenza et al. \(2023\)](#)) and the Growth-at-Risk model (e.g. [Figueres & Jarociński \(2020\)](#) or [Carboni et al. \(2026\)](#)).
- The scenario analysis examines the consequences of hypothetical events or economic conditions that deviate from the baseline, in a wider and more holistic manner than in the sensitivity (where assumptions are typically changed one at a time) or risk analysis (which looks only at factors affecting the skewness of the baseline distribution). This is typically performed with DSGE models (NAMW II) or the ECB-BASE.

A second important defining element of the projection narrative is the so-called “judgement”, namely the impact of factors that are not or cannot be included in structure of one model and that require additional expert judgement or specialised models and data.

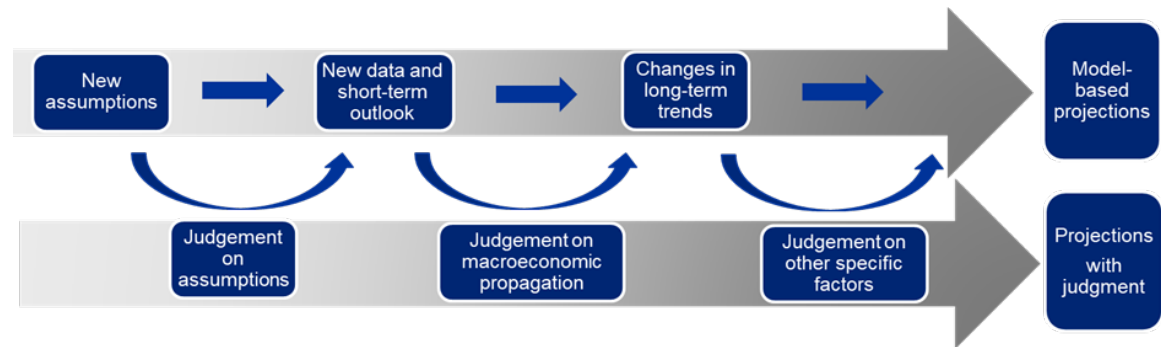
It is important to understand from the outset that, while the ECB/Eurosystem staff projection exercises are model-based, they are not purely limited to traditional statistical time-series analysis, as is common in most central banks and policy institutions. They are instead nuanced exercises, continuously calibrated via expert judgement to assimilate a plethora of information streams and inputs, mirroring the dynamic economic landscape. The process itself therefore relies less on purely mechanical usage of the economic-models, and rather uses economic models as a starting point, as an organizing and constraining device and finally as a cross-checking device. Therefore, the final projections incorporate a fair amount of judgement as a result of additional information included in the model and iterative discussions that happen during the projection exercise.¹⁹

As described in Section 2, the connection between all models in a projection exercise should be seen as an assembly line where the judgemental intervention happens at each stage of the

¹⁹[Ciccarelli et al. \(2024\)](#), [European Central Bank \(2016\)](#), ECB Blogs <https://www.ecb.europa.eu/press/blog/date/2023/html/ecb.blog230705~d16c61381c.en.html> and <https://www.ecb.europa.eu/press/blog/date/2025/html/ecb.blog.20250115~f016f263dd.en.html>

process (see figure 5).

Figure 5: Projections and judgement



Note: The figure shows the conceptual steps of the ECB forecasting process.

The forecasting process at the ECB is therefore distinct in two fundamental ways from standard statistical exercise. Firstly, it is not a purely statistical endeavour but a judicious blend of data, models, and insights from diverse sectors, ensuring a reflection of the ever-changing economic context. Secondly, the forecasting is inherently continuous, building on a rich repository of insights from past iterations, and is primarily conducted as a sequence of updates. Given these features, semi-structural models emerge as flexible tools for forecasting within central banks, adept at harnessing the intricate balance between model-based consistency and the flexibility to incorporate judgemental adjustments.

4.2 The role of the ECB-MC

The ECB-MC model can be used in two versions, a so-called ‘forecasting’ version and a ‘linked’ version. The former uses single country models to produce the country-level projections that are based on exogenous assumptions (e.g. interest rates, international outlook, fiscal variables) but excludes possible spillover effects across countries. In the ECB/Eurosystem forecasting framework, the spillovers are computed as a separate step via the Trade Consistency Exercise (TCE)²⁰. The linked version of the model, instead, links the country blocks with a trade matrix and allows to simultaneously cover the country dynamics, and cross-border spillovers. In addition, it also allows for activating the monetary policy

²⁰See [European Central Bank \(2016\)](#) for more details on the ECB/Eurosystem forecasting process

rule, which in the forecasting set-up is treated exogenously.²¹ All properties and uses of the model will be presented in this paper only for the single-country or ‘forecasting’ version. A more comprehensive analysis of the linked version is discussed in a separate forthcoming paper that will present properties and characteristics of the full model.

The ECB-MC has been, among other things, especially designed to include this systematic judgement into macroeconomic forecasts, based on practices (add-factor adjustment) that are quite common across central banks and institutions. Therefore, it is important that the model have some desirable characteristics which make it conformable to economic theory and consistent with the data. However, as we will see later in this section, its purpose in the ‘assembly line’ of the projection exercises is not to produce ultimate point forecasts, and hence a standard evaluation of its forecasting properties based e.g. on RMSE, or similar metrics, is less relevant and must be performed with accurate caveats. Instead, an evaluation of the final ECB/Eurosystem forecast is regularly performed and published, sometimes with the indication of the sources of forecast errors (see for instance [Chahad et al. \(2024\)](#)).

We can therefore summarise the role of the ECB-MC in the process into three areas: (i) provide a starting point for the derivation of the baseline; (ii) accommodate additional and judgemental information obtained outside the model, and (iii) perform sensitivity analysis and forecast decomposition to evaluate the potential impact of uncertain assumptions on the baseline. In what follows, we will describe in detail the main input of the ECB-MC in the forecast-generation process of the ECB and examine some of the main properties that make it suitable for this purpose.

The forecast-generation process with the ECB-MC model—schematically summarised in Figure 6 — progressively incorporates the relevant information arriving in a new forecasting round. It begins with integrating new data releases and assessing the impact of updated assumptions on the previous baseline. It then proceeds by updating the projections to reflect the short-term outlook and the incorporation of expert judgement.

²¹The Linked setup extends the model by accounting for intra-euro area bilateral trade flows and competitors’ prices defining countries’ competitiveness positions. These linkages are modelled following [Hubrich & Karlsson \(2010\)](#) whereby foreign demand is computed as a weighted average of import volumes of the trading partners and competitors’ prices are measured as a weighted average of trading partners’ export prices. Common monetary policy across all countries follows a modified policy rule from the Appendix [A.1.4](#), with inflation and output gaps variables corresponding to a weighted average of the ‘Big-5’ economies and the weights correspond to GDP shares in the euro area normalized to 1. The nominal exchange rate is modelled in a similar fashion.

The process results in a baseline projection that can be interpreted as a conditional forecast, combining model-based outputs with judgemental inputs. In the next subsections, we describe each step in more detail and conclude with a brief discussion on how model-based uncertainty is incorporated into the final forecast narrative.

Figure 6: The forecast-generation process with the ECB-MC model

<u>Update starting from previous baseline:</u>	(1) Impact of assumptions	(2) Projection update	(3) Conditional forecast
Technical assumptions update	Yes	Yes	Yes
Incoming data release	No	Yes	Yes
Short-term outlook update	No	Yes	Yes
Residuals	No	No	Yes

Note: ‘Assumptions’ include interest rates, external economic conditions affecting external demand and prices, energy and agricultural prices, fiscal plans. ‘Incoming data release’ refers to the release of new data or data revisions. ‘Short-term outlook update’ refers to the results of nowcasting models for GDP growth and inflation. ‘Residuals’ refers to the need to adjust the equation residuals to include judgemental input which is not directly captured in the model.

4.2.1 Impact of assumptions

Assumptions refer to projections of exogenous variables, including external conditions (such as global demand, competitors’ export and import prices, exchange rates, oil and gas prices), financial variables (such as interest rates, lending rates and asset prices), and fiscal variables (including government consumption, investment and taxation).²²

Assessing the impact of these assumptions is typically the first step in updating a forecast.²³ This exercise provides a transparent mapping from incoming information to forecast revisions. It allows policymakers and analysts to quantify the contribution of external factors to changes in the outlook, abstracting from other elements such as model adjustments or expert judgement. For example, evaluating how revisions to interest rate expectations or external demand translate into domestic inflation and activity is a critical first step in forming an informed policy assessment.

In the forecasting use of the ECB-MC model, most financial, fiscal and external variables are treated as exogenous conditioning variables. This means that, while some of these

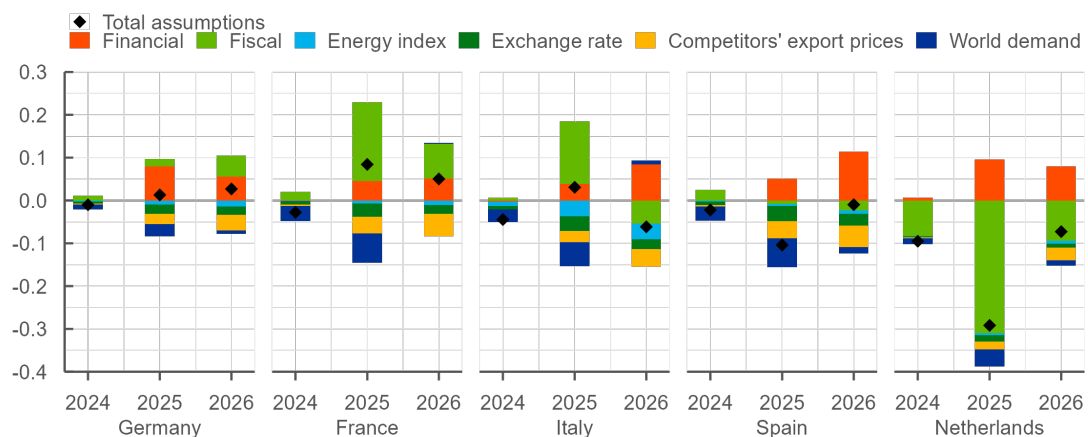
²²Detailed description of methods and tools used to produce the aforementioned assumptions can be found in [European Central Bank \(2016\)](#).

²³Note that, in practice, we assess the impact of changes in assumptions on revisions to the baseline forecast, reflecting the inherently sequential nature of the ECB/Eurosystem forecasting process.

variables might inherently have endogenous dynamics, and the model includes ‘behavioural’ or endogenous equations for them, namely dynamic equations that could be used to forecast them (think of a simultaneous equation model or e.g. a VAR), their dynamics is not used over the forecast horizon. Instead, these variables are ‘exogenized’ and the forecast of the other variables is obtained conditionally to externally projected future paths of these assumptions.

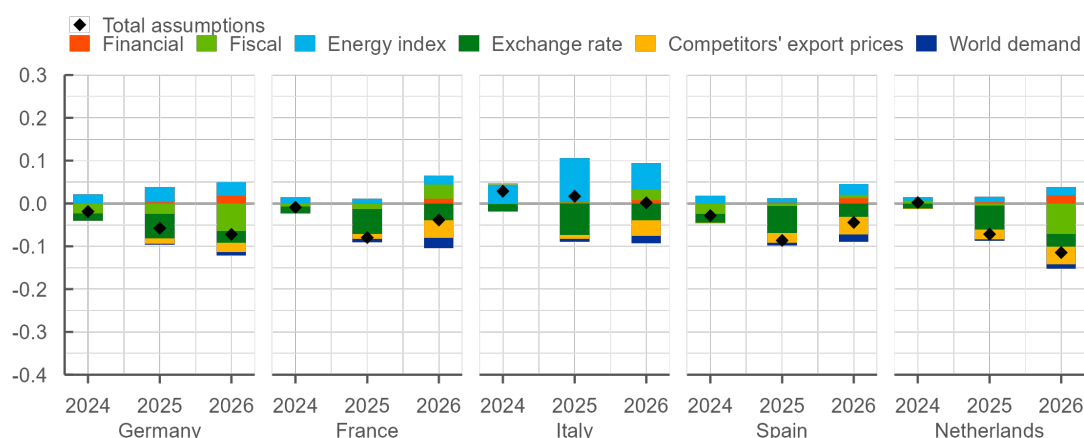
Two illustrative examples – which refer to one of the recent projection exercises – are reported in Figure 7 and Figure 8. Figure 7 highlights the contributions of changes in assumptions to GDP growth across the five largest euro area countries, while Figure 8 illustrates the corresponding impact on HICP annual inflation rate. Without entering in the economic content of the charts, the example demonstrates the depth and precision of the analysis provided to policymakers, effectively identifying the key drivers behind the projections for a given round. Naturally, the contributions of assumptions to the primary projections differ across rounds, reflecting evolving economic conditions.

Figure 7: Example of change in GDP projections due to change in the assumptions



Note: The figure depicts the impact of assumptions on real GDP growth, in percentage points. Financial assumptions include interest rates, asset prices and lending rates; the Energy index refers to Oil and gas prices which are in USD; exchange rates include USD and the euro effective exchange rate. Fiscal assumptions refer to fiscal measures, including net indirect taxes. World demand includes both intra- and extra-euro area trade.

Figure 8: Example of change in HICP projections due to change in the assumptions



Note: The figure depicts the impact of assumptions on HICP inflation, in percentage points. Financial assumptions include interest rates, asset prices and lending rates; the Energy index refers to Oil and gas prices which are in USD; exchange rates include USD and the euro effective exchange rate. Fiscal assumptions refer to fiscal measures, including net indirect taxes. World demand includes both intra- and extra-euro area trade.

Finally, it is important to emphasise that, although the forecasting application of the ECB-MC treats assumptions as exogenous within a given iteration, the ECB forecasting framework is designed such that changes in the macroeconomic outlook are ultimately reflected in the assumptions themselves. While these assumptions are produced using separate satellite models, the ECB forecasting process is inherently iterative and involves multiple rounds of discussion, assessment, and revision. During this process, the assumptions are updated sequentially to incorporate the evolving macroeconomic outlook, including insights derived from the ECB-MC in earlier rounds. These iterations are intended to ensure consistency between the final macroeconomic projections and the underlying assumptions.

4.2.2 Projection update

The impact of assumptions can be combined with new data and the short-term outlook to provide a model-based indication of the likely forecast path in the absence of additional judgement. This combined indicator—referred to as the ‘Projection update’—offers a simple benchmark for the model-consistent forecast. Despite its simplicity, it offers a useful reference point for assessing the extent of expert judgement applied beyond the main model over the medium-term horizon.

Central bank forecasting is characterised by both the use of judgement and the continuity of

1 projection rounds. While we do not explore here the underlying reasons for judgement or
2 the implications of iterative forecasting for forecasters’ behaviour, these features shape how
3 forecast revisions are interpreted.²⁴
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6 In practice, central banks rely on summary indicators to track the drivers of forecast revisions.
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8 The ‘Projection update’ is one such indicator: by capturing all information available at the
9 start of a round—updated assumptions, new data, and the short-term outlook—it provides
10 a benchmark against which the final baseline projection can be assessed.
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14 The protocol for conducting a ‘Projection update’ follows a sequence of steps. First, to
15 preserve the implicit expert judgement from the previous forecasting exercise, model residuals
16 consistent with the previous forecast are identified. In straightforward cases, these residuals
17 can be directly extracted from the earlier model simulation results. However, in cases of
18 significant historical data revisions (e.g., level shifts or benchmark updates), determining
19 consistent residuals requires a process known as ‘splicing’.
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26 Splicing, in simple terms, uses the revised historical data levels as a new starting point,
27 extending them with growth rates from the previous forecast. This reconstructed dataset
28 is then used to “invert” the model, enabling the computation of residuals that align with
29 the updated data. This approach is particularly suited to semi-structural models, such as
30 ECB-MC, while fully structural models may require alternative methods. Exploring those
31 alternatives, however, lies beyond the scope of this paper.
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38 Once the residuals consistent with the previous forecast have been identified, the model is
39 run conditional on the latest forecast assumptions and the short-term outlook derived from
40 satellite models. This conditioning is essential, as the main forecasting model—typically
41 specified at quarterly frequency—lacks the granularity needed to fully incorporate high-
42 frequency incoming information. Central banks therefore rely on complementary tools, such
43 as nowcasting and early-estimate models, to obtain a more accurate assessment of current
44 economic conditions (see e.g. [Eraslan et al. \(2025\)](#)).
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51 Integrating insights from these satellite tools into the main model allows this information
52 to feed through to the medium-term horizon. Together with updated assumptions, this
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55 ²⁴For a discussion of why ECB projections combine model-based computations with expert judgement, see
56 e.g. [Ciccarelli et al. \(2023\)](#).
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step yields the ‘Projection update’, which provides a model-based benchmark for evaluating the extent of expert judgement applied over the medium term horizon. An example of the ‘Projection update’ indicator together with the corresponding (judgemental) baseline projections is illustrated in Figure 9.

4.2.3 Add-factoring process

As reminded several times, forecasts are rarely produced purely on the basis of a mechanical application of economic models. Instead, they integrate various off-model pieces of information that forecasters have in real time but cannot be easily integrated within the model’s structure. The final step in the forecasting process therefore involves incorporating expert judgement into the model-based projections. In semi-structural models like ECB-MC, this can be done in two main ways: either by exogenising certain variables and imposing a fixed path over the forecast horizon, or by adjusting selected model residuals—a method commonly referred to as “add-factoring”. Fundamentally, both approaches achieve the same goal: modifying the model-based forecast to integrate off-model information.

This off-model information can originate from various sources. It might come from other, often smaller and more specialized models (referred to as satellite models) that focus on specific economic channels. It can also include soft indicators, such as survey results, or expert knowledge gained from previous forecasting rounds and experience with potential model misspecifications.

The “add-factoring” method is the most widely used because it is relatively straightforward to implement. It also enables the propagation of judgement throughout the forecast horizon. For instance, an adjustment to a residual in one quarter is propagated by the model’s solution to subsequent quarters, ensuring consistency across the forecast. Examples of such procedures and the relative treatment of the residuals are provided in the paper by [Angelini et al. \(2019\)](#). The add-factoring process is essentially the last step in the process of constructing the baseline projections.

Figure 9 illustrates how the indicators described above are used in the final stages of the forecasting process. The figure presents two consecutive projections for quarterly GDP growth in the five largest euro area countries, together with the two model-based indicators

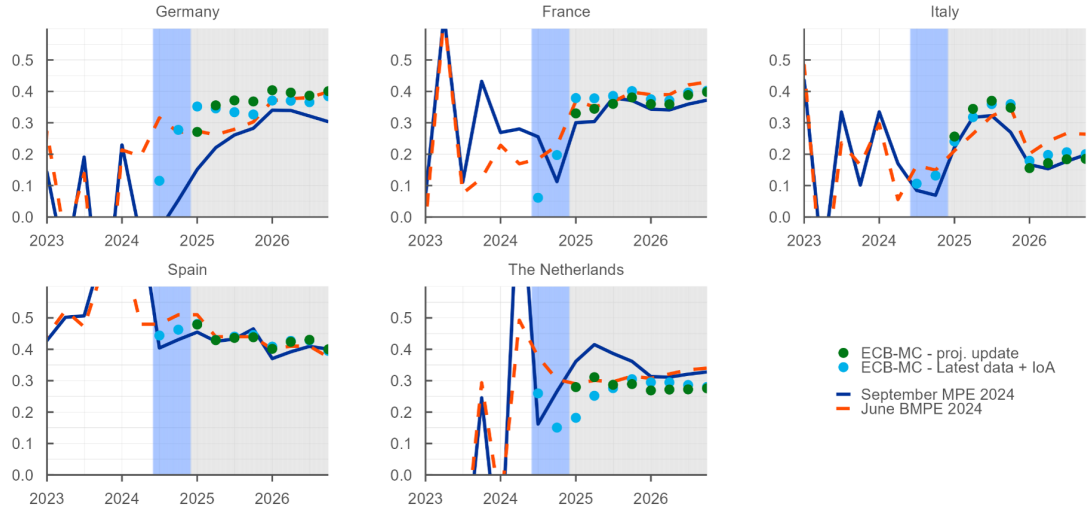
introduced earlier: the ‘Impact of assumptions’ and the ‘Projection update’.

The red dashed line and the blue solid line represent the final projections from two successive forecasting rounds, corresponding respectively to the previous and the current projection exercise. The blue dots show an intermediate step in which the previous forecast is updated solely with revised assumptions. In the short-term horizon (blue-shaded area), the gap between these blue dots and the final current projection (solid blue line) reflects the incorporation of short-term information derived from satellite models and expert assessment. This captures the fact that the near-term outlook is strongly influenced by high-frequency indicators and external information that are not fully embedded in the main model and are therefore incorporated through judgemental adjustments.

The green dots, starting after the short-term horizon (here: 2024Q3–2024Q4), represent the ‘Projection update’ indicator. This measure conditions on the short-term outlook and captures the model-based propagation of revised exogenous assumptions under the absence of additional medium-term judgement. As such, it provides a benchmark for the model-consistent forecast path.

At the medium-term horizon (here: 2025Q1–2026Q4), the difference between the final projection (solid blue line) and the ‘Projection update’ reflects the cumulative impact of expert judgement. This includes both the propagation of judgement introduced over the short-term horizon and additional direct adjustments implemented through add-factors in the medium term to account for information and considerations not explicitly captured by the model. A more formal derivation of these concepts is provided in the following section.

Figure 9: Example of Projection update for real GDP



Note: The figure depicts how the ‘Projection update’ (green dots) is used to cross check the baseline forecast (blue solid line) for GDP growth. The blue dotted line shows the latest statistical data combined with the impact of assumptions (“Latest data + IoA”). The red dashed line is the previous forecast.

Source: ECB/Eurosystem projections database and own calculations.

4.2.4 Decomposing the baseline projections

While it is challenging to quantify judgement precisely, it is instructive to examine how much of this judgement is contained in the forecast. The forecast decomposition presented here aims to do just that, leveraging the concept of the ‘Projection update’. It is essential to note that this decomposition is relative to the previous forecast, providing an analysis of changes rather than an absolute level of judgement. The decomposition of the baseline forecast into different components allows for a detailed understanding of the factors driving changes between forecast iterations. It is based on the quantities derived in the previous sections on ‘Impact of assumptions’ and ‘Projection update’ concepts and decomposes the change in forecasts into the impact of assumptions, statistical carry-over, impact of history and short-term outlook, and implied judgement. The (change in the) baseline forecast therefore can be decomposed as follows:

$$\Delta Forecast = IoA_t + SC_t + IH_t + IJ_t \quad (7)$$

where:

- IoA_t represents the impact of assumptions, calculated as the difference between updated

and previous forecasts after incorporating new external information for key economic indicators (e.g., interest rates, external demand, fiscal plans).

- SC_t denotes the statistical carry-over effect, capturing the purely statistical impact of data revisions and it is independent of the model. It's calculated to show the growth contribution from the previous year's level, assuming no change in the current year's quarterly growth rates:

$$SC_t = \frac{Q_{t-1:4}}{\bar{Q}_{t-1}} - 1 \quad (8)$$

Here, $Q_{t-1:4}$ represents the level of the variable of interest (e.g., GDP) in the fourth quarter of the previous year, and $\bar{Q}_{t-1}$ is the average level over the entire previous year. This calculation quantifies the extent to which the growth rate for the current year is influenced by the level attained at the end of the previous year, assuming no quarterly growth within the current year.

- IH_t stands for the impact of history and short-term outlook. This component reflects the adjustments made to account for revised historical data and short-term forecasts, as identified in the 'Projection update' process. It is calculated by taking the 'Projection update' and subtracting both the impact of assumptions and the statistical carry-over effect:

$$IH_t = PU_t - IoA_t - SC_t \quad (9)$$

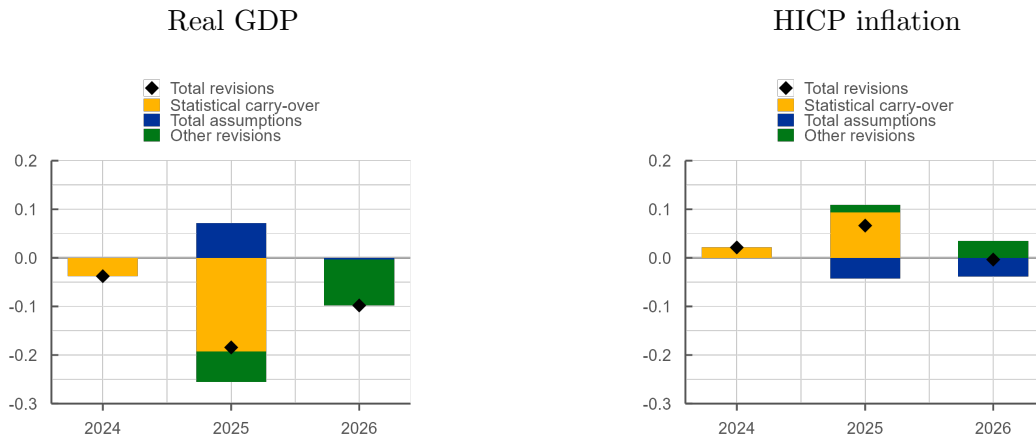
where PU_t denotes the final projection update (i.e. the update consistent with the assumptions embedded in the final baseline projections). This calculation captures the changes due to historical data revisions and the updated short-term forecasts. It contains both statistical carry-over, but also endogenous model propagation from the lagged dynamics.

- IJ_t can be interpreted as implied judgement, reflecting forecasters' subjective adjustments that go beyond the mechanical application of the model and the latest data.

These components collectively explain the transition from the previous forecast to the updated baseline, highlighting the roles of new information, methodological updates,

and experts' judgement. The 'Projection update' protocol, involving steps from inverting the previous forecast to adjusting for new assumptions, history and short-term outlook, underpins this decomposition by providing the mechanisms through which each component influences the overall forecast.

Figure 10: Example of Forecast decomposition



Note: This figure depicts the factors driving the revision of projections for real GDP growth and HICP inflation for aggregated Big-5 country projections, in percentage points. Other revisions refer to the impact of history and short-term outlook and implied judgement.

Source: ECB/Eurosystem projections database and own calculations.

4.3 Assessing the forecast performance

We conclude this section with some considerations on the forecasting properties of the model. As already said, the ECB baseline projections are not generated by the model in isolation. Rather, the model operates within a conditional framework that incorporates a rich set of assumptions determined outside the ECB-MC model, including interest rates, exchange rates, fiscal assumptions and external demand which are treated as exogenous variables. In addition, the (B)MPE forecast is anchored by near-term inflation projections (NIPE) produced by National Central Banks (see [European Central Bank \(2016\)](#)) over the first four quarters. Expert judgement is then introduced through add-factors, which completes the baseline projection. As a result, a purely model-based forecast is not observed in practice. This complicates a direct assessment of model performance while the layered process that combines model structure, conditioning assumptions and judgement makes it difficult to attribute forecast errors to their underlying sources.

Therefore, the 'evaluation metric' for the model cannot be the same one used in standard

academic time series practices and must be adapted to the main purposes of the model. Concretely, to evaluate its forecasting ability we ask the question: What would the (conditional) forecast of the model be if we fed it with the ‘right’ assumptions, namely with the financial, fiscal, and external data that occurred over the evaluation period (as opposed to the real time assumptions). The model residuals are then derived from a Bayesian estimation procedure using the unobserved components model (UCM) framework described in [Angelini et al. \(2019\)](#). The evaluation period (2018–2025) encompasses large and atypical shocks, most notably the COVID-19 pandemic and the surge in energy prices following Russia’s invasion of Ukraine. To mitigate the influence of these extreme events, we condition the model on observed energy prices. This allows the evaluation to focus on the model’s ability to capture underlying macroeconomic dynamics, rather than its response to (asymmetric) large shocks.

Table 2 reports the root mean squared forecast errors (RMSE) of the HICP conditional forecasts obtained from pseudo-real-time simulations for each country. In this setup, the model is conditioned on the ex-post realisations of assumptions and energy prices. Overall, the model-based forecasts outperform the random walk benchmark across most countries and horizons. The only exception is France at the short horizon, where the model underperforms relative to the benchmark. Table 2 also reveals that the model’s forecasting performance varies across countries. This reflects a key design feature of the ECB-MC model. While originally conceived as a country-level framework, the ECB-MC was also developed as a multi-country counterpart to the ECB-BASE model. As a result, the model structure is deliberately kept relatively homogeneous across countries, closely mirroring the architecture of ECB-BASE. Cross-country heterogeneity in model outcomes therefore arises primarily from differences in calibration and estimation, rather than from structural differences in model specification.

In addition, we conduct a counterfactual exercise based on the information set available at the time of the March 2022 Macroeconomic Projection Exercise (MPE), which coincides with the onset of the war in Ukraine and the beginning of the subsequent inflation surge.²⁵ Starting from this information set, we run the model under two alternative conditioning

²⁵Note that March 2022 MPE exercise was produced with the NMCM model, i.e. the predecessor of the ECB-MC.

Table 2: Relative forecast accuracy of conditional forecasts for HICP inflation

Country	RMSE ratio: CF relative to RW		
	$h = 4$	$h = 8$	$h = 12$
DE	0.52	0.39	0.43
FR	1.14	0.81	0.74
IT	0.40	0.32	0.27
ES	0.55	0.55	0.41
NL	0.60	0.44	0.40
Average	0.64	0.50	0.45

Notes: Entries report the ratio of root mean squared forecast errors (RMSE) from conditional forecasts (CF) relative to the random walk benchmark (RW). Values below one indicate that CF outperforms RW. The evaluation is based on a common sample across forecasting methods. The number of observations is $N = 21$ for horizons $h = 4$ and $h = 8$, and $N = 11$ for horizon $h = 12$.

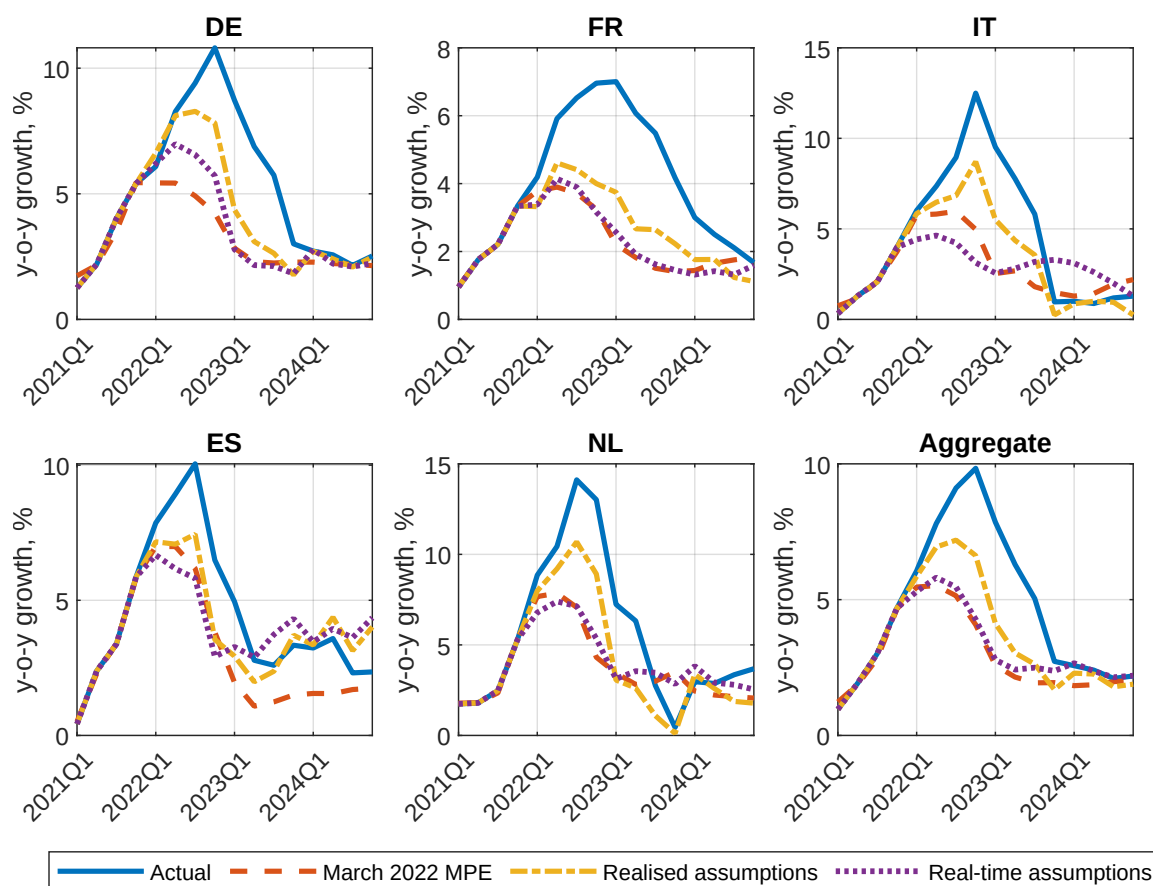
paths. First, we use real-time assumptions, i.e. the assumptions available at the time of the March 2022 exercise. Second, we re-run the model using ex-post realised assumptions, including the actual evolution of energy prices.²⁶

The comparison between these two simulations allows us to assess the extent to which forecast errors can be attributed to unexpected developments in conditioning variables, as opposed to limitations of the model itself. The results indicate that, for most countries, incorporating realised assumptions leads to a noticeable improvement in forecast accuracy. However, even under perfect foresight with respect to exogenous inputs, the model is not able to fully capture the magnitude of the inflation surge. This suggests that a significant part of the inflation dynamics during this period was driven by factors that are only imperfectly represented in the model, such as supply bottlenecks and elevated uncertainty, which imply non-linear features that the model cannot capture.

Figure 11 illustrates these results by comparing the two model-based simulations with the March 2022 MPE projections and the realised path of HICP inflation.

²⁶A similar exercise was conducted with the ECB-BASE model, for details see, [Angelini, Darracq Pariès & Zimic \(2025\)](#).

Figure 11: HICP year-on-year inflation: actual data and alternative forecast vintages



5 Conclusion

The paper introduces the ECB Multi-Country Model (ECB-MC), a key tool in the European Central Bank's efforts to analyze and forecast the complex dynamics of the Eurozone economy. As one of the primary models used in the ECB's Broad Macroeconomic Projection Exercises, the ECB-MC plays a vital role in shaping monetary policy decisions and promoting economic stability within the Eurosystem.

The ECB-MC captures the complexities of the largest five euro area economies. By incorporating detailed expenditure and sectoral structure that replicates national accounts, international trade relationships, and monetary policy transmission mechanisms, the ECB-MC provides policymakers with a comprehensive framework for forecasting and analyzing the potential impact of various economic shocks and policy interventions.

Moreover, the ECB-MC integrates seamlessly into the Eurosystem forecasting process, which

prioritizes robust modeling techniques to deliver accurate and reliable economic projections. The model's infrastructure not only facilitates the smooth incorporation of expert judgement but also provides a rigorous framework through its model-based indicators. These indicators enable forecasters to quantify and monitor the extent of judgement applied.

In summary, the ECB Multi-Country Model demonstrates the ECB's commitment to transparently improving macroeconomic modeling in support of its price stability mandate. As the Eurozone faces ongoing economic challenges, ECB tools and models will evolve accordingly. For the time being, the ECB-MC model remains a valid tool in support of policymakers for their informed decisions to strengthen and sustain the Eurozone economy.

A The model's structure

This section describes the main blocks and equations of the ECB-MC model. It aims at giving a comprehensive overview of the model. However, as the theoretical underpinnings are the same as described in [Angelini et al. \(2019\)](#), below we recall the main features only.

We use capital letters to denote the variables in levels, while small letters refer to logarithmic transformation. The estimated coefficients corresponding the the key equations presented in this section are reported in [E](#).

A.1 The demand block

A.1.1 Household consumption

We distinguish two types of households: one type are the liquidity constrained households that follow a rule-of-thumb behaviour, and the other type are optimizing households that maximise their expected lifetime utility subject to resource constraint.

The consumption (c) of liquidity constrained households moves in line with their labour and transfer income (yh^L and yh_T , respectively):

$$\Delta c_t = \Delta(yh_t^L + yh_t^T) \quad (10)$$

Empirical implementation of the consumption behaviour of optimising households is, in turn, based on the PAC approach whereby we define the consumption target and we assume that agents face some adjustment costs that delay reaching the target.

The (log) consumption target (c^*) is then expressed as the sum of the different permanent income and wealth components with different propensities to consume:

$$c_t^* = \eta_0 + \eta_T eyh_t^T + \eta_P eyh_t^P + \eta_L eyh_t^L + \eta_D hw_t^D \quad (11)$$

where, eyh^L , eyh^T , eyh^P denote expected permanent labour, transfer and property incomes, respectively, and hw^D is financial and housing wealth of households. Note that, by construction, $\eta_T + \eta_P + \eta_D + \eta_L = 1$.

The short-run consumption dynamics of the unconstrained households is assumed to follow the error-correction equation (PAC approach), which, combined with the spending of rule-of-thumb consumers, allows to derive the following dynamic behaviour of the aggregate consumption:

$$\Delta c_t = (1 - \theta) \left(a_0 (c_{t-1}^* - c_{t-1}) + \sum_{i=1}^{m-1} a_i \Delta c_{t-i} + \beta_1 x_t + \mathbb{E}_{t-1} \sum_{j=0}^{\infty} d_j \Delta c_{t+j}^* \right) + \theta \Delta (y h_t^L + y h_t^T) + \epsilon_t^C \quad (12)$$

where θ is the share of rule-of-thumb consumers, a_0 is the coefficient on the deviation of consumption from its target and a_i gives the weights on the backward looking terms. The term $\mathbb{E}_{t-1} \sum_{j=0}^{\infty} d_j \Delta c_{t+j}^*$ represents the expectations of future targets. Finally, same as in the case of ECB-BASE model, also the country version dynamic equations have been arbitrarily augmented with an additional explanatory variable, x_t , which denotes the spread between the lending rate on consumption and the risk-free rate to account for direct effects of financial factors on durable consumption.

A.1.2 Investment

Business investment: We derive business investment starting from the premise of two types of agents (firms): cash-flow constrained and optimizing agents that maximize profits subject to capital accumulation constraint.

The output of a firm is given by the Cobb-Douglas production function with constant returns to scale and two production inputs, capital K_t and labour N_t :

$$Y_t = F(N_t, K_t) = N_t^\alpha K_t^{1-\alpha} \quad (13)$$

The costs of the production inputs are the relative price of the investment good, RP_t , and wages, W_t .²⁷ The depreciation rate of capital is given by δ .

In this set-up, the solution of the firm's optimization problem yields a formula for the user

²⁷To ease the description and without loss of generality we have dropped the technology progress term from the production function.

cost of capital, UC , which can be expressed in terms of investment costs (determined by the depreciation rate, financing cost for business investments (R_{t+1}^{ib}) and net capital gains given by the relative price growth:

$$(1 - \alpha) \frac{Y_{t+1}}{K_{t+1}} = RP_t \left\{ R_{t+1}^{ib} + \delta - (1 - \delta) \left(\frac{RP_{t+1} - RP_t}{RP_t} \right) \right\} \equiv UC_{t+1} \quad (14)$$

We then further derive the target capital stock as:

$$K_t^* = \frac{S_t^K Y_t}{UC_t} \quad (15)$$

where S_t^K denotes the capital to output share. Now, taking into account also the law of motion for capital, we can then derive the target for business investment:

$$IB_t^* = \left(G_{t+1}^{K^*} + \delta \right) K_t^* \quad (16)$$

where $G_{t+1}^{K^*}$ is the growth rate of the (target) capital stock, which is approximated by the real GDP growth.

The short-run dynamics of aggregate investment are then given by the following equation:

$$\Delta ib_t = \left(1 - \theta^{ib} \right) \left(a_0^{ib} (ib_{t-1}^* - ib_{t-1}) + \sum_{k=1}^{m-1} a_k^{ib} \Delta ib_{t-k} + \mathbb{E}_{t-1} \sum_{j=0}^{\infty} d_j^{ib} \Delta ib_{t+j}^* \right) + \theta^{ib} \Delta y_{t-1} + \epsilon_t^{ib} \quad (17)$$

where ib_t is the log of business investment, a_0^{ib} is the mean reversion parameter associated with previous period deviations from the target investment, a_k^{ib} is an autoregressive coefficient associated with k quarters lagged business investment, and d_j reflects the effect of today's adjustment of investment decisions due to expected changes in the investment target by $\mathbb{E}_{t-1} \Delta ib_{t+j}$. Finally θ^{ib} represents the share of cash-flow constrained agents.

Residential investment Residential investment is modelled from the firm's side. We first derive the user cost of housing capital (UC^H)

$$(1 - \alpha) \frac{Y_{t+1}^H}{KH_{t+1}} = RP_t^H \left\{ R_{t+1}^{NFC} + \delta^H - (1 - \delta^H) \left(\frac{RP_{t+1}^H - RP_t^H}{RP_t^H} \right) \right\} \equiv UC_{t+1}^H \quad (18)$$

where Y^H represents output in the residential sector, KH is the housing capital stock, RP^H are relative prices (house prices over the residential investment deflator), δ^H is a depreciation rate of housing capital and R_{t+1}^{NFC} is the lending rate for non-financial corporations. We then assume that the target for residential investment can be expressed as a function of output, relative prices and pure user costs of housing capital (user costs excluding relative house prices):

$$IH_t^* = Y_t (UC_{-RP_t^H}^H)^{\beta_1^{ih}} (RP_t^H)^{\beta_2^{ih}} \quad (19)$$

where a clear distinction between relative house prices and user costs of housing capital is made to allow the examination of different elasticities of investment to the two respective components.

The short-run equation for residential investment is derived using the PAC approach and additionally including the de-trended real GDP growth rate as an accelerator effect:

$$\Delta ih_t = a_0^{ih} (ih_{t-1}^* - ih_{t-1}) + \sum_{i=1}^2 a_i^{ih} \Delta ih_{t-i} + \mathbb{E}_{t-1} \sum_{j=0}^{\infty} d_j^{ih} \Delta ih_{t+j}^* + \theta^{ih} (\Delta y_t - \Delta \bar{y}_t) + e_t^{ih} \quad (20)$$

Here a_0^{ih} is a mean reversion parameter for residential investment, a_i^{ih} are autoregressive coefficients, θ^{ih} measures the accelerator effect, and the coefficients d_j are loadings on expected future target dynamics.

A.1.3 Government

The government block within the ECB-MC is an advanced component derived from the fiscal framework used in the ECB-BASE model. This block provides a comprehensive representation of government revenues and expenditures and their interactions with the domestic economy. On the revenue side, the model incorporates implicit tax rates aligned with data from government finance statistics, covering taxes on income and wealth, as well as taxes on production and imports. On the expenditure side, it includes government consumption, investment, and social transfers. Government revenues are modelled using implicit rates, calculated as the ratio of revenue categories reported in the government finance statistics to the corresponding macroeconomic base (e.g., the macro base for taxes on production and imports includes final private consumption and government purchases). Expenditures are modelled as a proportion of potential output. The fiscal block encompasses

variables related to nominal, real, and implicit deflators. Detailed modelling specifications for each variable are discussed in the ECB-BASE paper and will be further elaborated in the forthcoming paper.

For forecasting purposes at the ECB, the fiscal block is treated as exogenous, with its key inputs provided by the Fiscal Policy Division. As this paper focuses on the application of the ECB-MC for forecasting, further details of the fiscal model are omitted here. It is important to note that while the fiscal block influences macroeconomic outcomes, it does not exhibit feedback effects within the current forecasting framework.

A.1.4 Foreign trade

The ECB-MC trade block follows the traditional framework of [Goldstein & Khan \(1985\)](#). However, given the multi-country setup of the model, a special attention is paid to modelling separately intra- and extra-euro area flows. The modelling of extra-euro area trade volumes and prices is essential from the perspective of analysing the impact of the external shock that materialize outside the euro area. In turn, the intra-euro area variables enable the consistency of the euro-area dynamics as the demand for exports in one euro area country is automatically reflected in imports of other euro area countries, taking into account the respective trade shares. In the context of the ECB’s Macroeconomic Projection Exercises when the “single-country version” of the model is employed, this trade consistency is ensured “ex-post” via a procedure described in [Hubrich & Karlsson \(2010\)](#).

Trade volumes For each country we model explicitly total imports and exports as well as their extra-euro area components. All trade volumes are modelled in real terms.

Starting with the total imports (MTR), we assume that in the long-run they are determined by the import demand and the differential between import and domestic prices:

$$mtr_t^* = \alpha_0^{mtr^*} + wer_t + \alpha_1^{mtr^*} (meed_t - yed_t) + \alpha_2^{mtr^*} trend_t \quad (21)$$

where mtr^* is the target (long-run) total import, $\alpha_0^{mtr^*}$ includes deterministic terms, $meed$ is the non-energy total import deflator, yed is the GDP deflator, and wer is the import demand proxied by the weighted average of import content of domestic final demand components,

and can be considered as an indicator of import absorption.

The short-run dynamics of total import is governed by the speed of the mean reversion to the long-run optimal level, change in import demand and change in nominal non-energy import prices:

$$\Delta mtr_t = \alpha_0^{mtr} + y_t^* + trend_t + \alpha_1^{mtr}(mtr_{t-1} - mtr_{t-1}^*) + \alpha_2^{mtr}(\Delta wtr_t - y_t^*) + \alpha_3^{mtr} \Delta meed_t \quad (22)$$

where y^* is an HP-filtered potential output of country, in logs.

The behavioural equations for the extra-euro area imports, MXR , mimics the specification for the total import with the only exception being the use of nominal exchange rates ($EENX$) instead of the non-energy import deflator.

$$\Delta mxr_t = y_t^* + trend_t + \alpha_1^{mtr}(mxr_{t-1} - mxr_{t-1}^*) + \alpha_2^{mtr}(\Delta wtr - y_t^*) + \alpha_3^{mtr} \Delta eenx_{t-1} \quad (23)$$

On the exports side, total exports, XTR , are modelled with the long-run behaviour determined by the euro area world demand and relative export prices:

$$xtr_t^* = \alpha_0^{xtr} + wdr_t + \alpha_1^{xtr}(xtd_t - cxd_t) + \alpha_2^{xtr} trend_t \quad (24)$$

where xtr_t^* is the target (long-run) total exports, wdr is world demand (including intra-euro area demand), xtd is the export deflator, and cxd are the competitors' export prices. Similarly as in the case of imports, the long-run elasticity of export to world demand is assumed to be one.

The short-run dynamics of total exports is then governed by the dynamic error-correction term approach:

$$\Delta xtr_t = \alpha_0^{xtr} + y_t^* + trend_t + \alpha_1^{xtr}(xtr_{t-1} - xtr_{t-1}^*) + \alpha_2^{xtr} \Delta(wdr_t - y_t^*) + \alpha_3^{xtr} \Delta(xtd_t - cxd_t) \quad (25)$$

The behavioural equations for the extra-euro area export, XXR , mimics the specification for the total exports with the only exception being the use of world demand which is computed based on non-euro area trading partners only and the change in nominal effective exchange

rates, $EENX$, expressed as EUR against a bundle of foreign currencies.

$$\Delta xrr_t = y_t^* + trend + \alpha_1^{xrr}(xrr_{t-1} - xrr_{t-1}^*) + \alpha_2^{xrr}\Delta(wdr_t - y_t^*) + \alpha_3^{xrr}\Delta eenx \quad (26)$$

The remaining trade volumes: intra-euro area imports, MNR , and intra-euro area exports, XNR are derived by accounting identities, i.e. $MNR = MTR - MXR$ and $XNR = XTR - XXR$, respectively.

Trade deflators Likewise trade volumes, we model trade deflators from a top-down approach, i.e. we explicitly model total import and export deflators and the extra-euro area deflators. The intra-euro area prices are then re-computed to match the model consistency. On the import side, the core deflator is the non-energy total import deflator ($MEED$). The prices for energy/oil and imported commodities, (MED) are taken from external models and are used as conditioning assumptions. Those prices are normally expressed in US dollars and therefore need to be converted into Euro in the model.

The equation for (log) target $MEED$:

$$meed_t^* = \alpha_0^{meed^*} + \alpha_1^{meed^*}cmd_t + (1 - \alpha_2^{meed^*})yed_t + \alpha_3^{meed^*}trend_t \quad (27)$$

where cmd is competitors' import prices and yed is GDP deflator, both expressed in logs.

Short-run dynamics of $MEED$:

$$\begin{aligned} \Delta meed_t = \pi_t^e + \alpha_0^{meed} + \alpha_1^{meed}(meed_{t-1} - meed_{t-1}^{star}) + \alpha_2^{meed}\Delta(cmd_t - \pi_t^e) \\ + \alpha_3^{meed}\Delta(yed_t - \pi_t^e) + \alpha_4^{meed}trend_t \end{aligned} \quad (28)$$

where π^e is the long-term domestic inflation expectations.

Short-run dynamics of total import deflator (MTD) are then modelled as a weighted average of:

$$mtd_t = (1 - phi^{med}) * meed_t + phi^{med} * (med_t - exr_t) \quad (29)$$

where exr is the nominal euro-US dollar exchange rate, in logs.

Derivation of target extra-euro area deflator is based on the indicator of the competitors' import prices ($CMDEX$), energy prices (MED), nominal euro-US dollar exchange rate

(*EXR*) and GDP deflator (*YED*):

$$\Delta mxd_t^* = cmdex_t + \alpha_0^{mxd^*} + \alpha_1^{mxd^*} (med_t - exr_t - cmdex_t) + \alpha_2^{mxd^*} (yed_t - cmdex_t) + \alpha_3^{mxd^*} trend_t \quad (30)$$

Short-run dynamics of extra-euro area import deflator (*MXD*) then follow:

$$\begin{aligned} \Delta mxd_t = \alpha_1^{mxd} (mtd_{t-1} - mtd_{t-1}^*) + \alpha_2^{mxd} (\Delta(med_t - exr_t) - \Delta yed_t) \\ + \alpha_3^{mxd} (\Delta cmdex_t - \Delta yed_t) + \Delta yed_t + trend_t \end{aligned} \quad (31)$$

Total target export deflator is modelled as a function competitors' export prices, domestic and import costs of export production, and energy prices expressed in euro.

$$xtd_t^* = cxd_t + \alpha_0^{xtd^*} + \alpha_1^{xtd^*} (yed_t - cxd_t) + \alpha_2^{xtd^*} (med_t - exr_t - cxd_t) + \alpha_3^{xtd^*} trend_t \quad (32)$$

Short-run dynamics:

$$\begin{aligned} \Delta xtd_t = \alpha_0^{xtd} + \alpha_1^{xtd} (xtd_{t-1} - xtd_{t-1}^*) + \alpha_2^{xtd} (\Delta cxd_t - \Delta yed_t) + \alpha_3^{xtd} (\Delta(med_t - exr_t) - \Delta yed_t) \\ + \Delta yed_t + trend_t \end{aligned} \quad (33)$$

Finally, the target for extra-euro area exports deflator is modelled as:

$$xxd_t^{star} = cxdex_t + \alpha_0^{xxd^{star}} + \alpha_1^{xxd^{star}} (yed_t - cxdex_t) + \alpha_2^{xxd^{star}} trend_t \quad (34)$$

while its short-run dynamics follow:

$$xxd_t = \alpha_1^{xxd} (xxd_{t-1} - xxd_{t-1}^{star}) + \alpha_2^{xxd} \Delta(cxdex_t - yed_t) + \Delta yed_t + trend_t \quad (35)$$

Having derived total and extra-euro area real volumens and prices, the remaining variable are obtained from identities.

Exchange Rate and Price/Interest Parity In the context of the ECB/Eurosistem Macroeconomic Projection Exercises, the exchange rates as well as a set of euro area and foreign interest rate is given as an assumptions. The paths for those variables are not determined

by ECB-MC model.²⁸

A.2 The supply block

The supply side of the model is centred around a standard Cobb-Douglas production function.

$$Y_t(i) = F^i(N_t(i), K_t(i)) = A_t (K_t)^\alpha (\zeta^t N_t)^{(1-\alpha)}$$

where K_t denotes aggregate capital stock and N_t denotes aggregate employment. A_t is total factor productivity and ζ_t is labour augmenting technology.

Firms choose optimal capital (K_t) and labour input (N_t) to minimize total input costs $R_{K,t}K_t + w_tN_t$ subject to a technology constraint given by the production function.

A.2.1 Production function

Based on the production function we can calculate the potential output, taking as input aggregated capital stock and trend employment. To derive the latter we replace aggregate employment by aggregate hours worked:

$$H_t = \frac{H_t}{N_t} * \underbrace{\frac{N_t + U_t}{WAP_t}}_{part.rate} * WAP_t * \underbrace{\frac{N_t}{LF_t}}_{empl.rate} \quad (36)$$

where N_t denotes employment headcount, WAP_t denotes working age population and LF_t is the labour force. Denoting the total hours per employee/employment headcount $\frac{H_t}{N_t}$ as HPE_t , the participation rate as $LFPR$ and the employment rate as $(1 - U_t)$ where U_t is the unemployment rate, we can rewrite the production function as:

$$Y_t = A_t K_t^\alpha (HPE_t * LFPR_t * WAP_t * (1 - U_t))^{1-\alpha} \quad (37)$$

Taking logs and denoting variables in logs with small case letter, we obtain the formula for potential output:

$$y_t^* = a_t^* + \alpha * k_t + (1 - \alpha)(hpe_t^* + lfpr_t^* + wap_t^* + \log(1 - u_t^*)) \quad (38)$$

where ‘*’ indicates the trend value of the corresponding variable.

²⁸In the linked version of the model we use the same UIP condition as defined in [Angelini et al. \(2019\)](#)

A.2.2 Labour market

The Labour market block of the ECB-MC is centred around an equation for total employees. Furthermore the block features an equation for hours worked, a measure for unemployment and a separate modelling of self-employed and the labour force participation rate.

Employees We assume that firms reach the optimal employment target (N_t^*) only gradually, therefore we use a PAC approach to derive an equation for employees:

$$\Delta n_t = a_0^n (n_{t-1}^* - n_{t-1}) + a_1^n \Delta n_{t-1} + \mathbb{E}_{t-1} \sum_{j=0}^{\infty} d_j \Delta n_{t+j}^* + e^n \quad (39)$$

and the target variable is determined by:

$$n_t^* = mc + \log(1 - \alpha) + y_t - w_t$$

where mc is the log of Lagrange multiplier on the technology constraint.

Self-employed The dynamics of employees and self-employed are not always well aligned, making a separate modelling of self-employed (n_t^{SE}) necessary, but ensuring that the ratio of self-employed to employee stays constant in steady state. Self-employed are related to the growth in the target of employees and to an error-correction term:

$$\Delta n_t^{SE} = \Delta n_{t-1}^* - \beta_{EC} (n_{t-1}^{SE} - n_{t-1}^{SE,*})$$

The target for self-employed is given by a constant fraction of the employees target.

Labour force participation rate The change in the labour force participation rate (lfp_t) is modelled as a function of an error-correction term and the change in the unemployment rate:

$$\Delta lfp_t - \Delta lfp_t^T = -0.5(lfp_{t-1} - lfp_{t-1}^T) - \beta_u \Delta u_t$$

Unemployment rate Finally, we compute the unemployment rate as

$$U_t = 100 \left(1 - \frac{Employment}{Labour\ force} \right) \quad (40)$$

where the labour force is calculated as the working age population multiplied by the labour market participation rate.

A.2.3 Prices and wages

Similar as in the ECB-BASE model, we use a small semi-structural general equilibrium model to determine jointly the output gap, inflation, wages and unemployment. The model is centred around a **Wage-Price** and **Output gap** specification hence thereafter referred to as WAPRO. Derivation and estimation technique for euro area WAPRO is described in [Angelini et al. \(2019\)](#). The only difference between WAPRO modelling applied to the counties (ECB-MC) vs the euro area (ECB-BASE) is the country coverage. In case of ECB-BASE the model is used to analyse intra- vs extra-euro area dynamics while in ECB-MC we model an euro area country vs the rest of the world. The model is estimated with Bayesian techniques.

Price inflation

The key inflation indicator in the ECB-MC is GDP deflator which is modelled via a reduced form of the New Keynesian Phillips Curve, as in [Galí et al. \(2012\)](#) combined with a variant of [Cogley & Sbordone \(2008\)](#). The equation shows that actual inflation depends both on a measure of expected inflation and on past inflation. Marginal costs are approximated via a combination of the output gap and the wage gap.

$$\pi_t = \frac{1}{1 + \beta_\pi \delta_\pi} \left\{ (1 - \delta_\pi + \delta_\pi \beta_\pi - \beta_\pi) \bar{\pi}_t + \beta_\pi (\mathbb{E}_t \pi_{t+1}) + \delta_\pi \pi_{t-1} + \beta_y^\pi \left(\hat{w}_t + \left(\frac{\alpha}{1-\alpha} \right) \hat{y}_t \right) \right\} + e_t^\pi \quad (41)$$

where α is the capital share in the model.

Note that inflation is modelled with reference to an inflation attractor $\bar{\pi}$. To accommodate medium term deviation of inflation from the inflation target, we model the time-varying inflation attractors as follows:

$$\bar{\pi}_t = (1 - \beta_\pi) \pi_{ss}^{EA} + \beta_\pi \bar{\pi}_{t-1} + \delta_\pi (\pi_{t-1} - \bar{\pi}_{t-1}) + e_{\bar{\pi},t} \quad (42)$$

HICP and demand deflators The modelling of HICP (and sub-aggregates) and the demand deflators essentially relies on two inputs, the GDP deflator and the import price deflator. The GDP deflator determined in equation (41) is the central domestic price deflator in the model. As described above it is closely linked to the output gap, unemployment and wages. The import price deflator is reflecting international factors such as foreign inflation, the exchange rate and oil prices. We follow a bottom-up approach wherein the total HICP is modelled as a weighted sum of the components.

The general long-run expression for HICP-based measures of prices takes the following form:

$$p^* = (1 - \omega)p^y + \omega p^m \quad (43)$$

where p^* represents a long-run level of a particular price measure, p^y is the GDP deflator, p^m is log of import deflator, and ω is the import content in the final use of respective sector in the economy determined by 'Input/Output' tables.

In terms of model equations, this specification translates into the following formulas for:

- long-run core inflation (HEF):

$$hef_t^* = \alpha_0^{hef^*} + (1 - \omega^{hef})yed_t + \omega^{hef}meed_t + trend_t \quad (44)$$

- long-run energy inflation (HEG):

$$heg_t^* = \alpha_0^{heg^*} + (1 - \omega^{heg})yed_t + \omega^{heg}(med_t - exr_t) + trend_t \quad (45)$$

- long-run food inflation (HIF):

$$hif_t^* = yed_t + \alpha_0^{hif^*} + \alpha_1^{hif^*}trend_t \quad (46)$$

where yed denotes the GDP deflator, $meed$ are the import prices excluding energy costs, med are the exported energy prices, and exr are the euro-dollar exchange rates, all expressed in logs.

The short-run price dynamics follow the error-correction model with the mean reversion constructed around the long-run price level and short run coefficient constrained in line with an imposed dynamic homogeneity. In practice implying the following equations:

- short-run core inflation:

$$\begin{aligned}\Delta hef_t = & \pi_t^e/400 + \alpha_0^{hef} + \alpha_1^{hef}(hef_{t-1} - hef_{t-1}^*) + \alpha_2^{hef}(\Delta yed_t - \pi_t^e/400) \\ & + \alpha_3^{hef}(\Delta meed_t - \pi_t^e/400) + \alpha_4^{hef}(\Delta hef_{t-1} - \pi_{t-1}^e/400) + \alpha_5^{hef}(\Delta hef_{t-2} - \pi_{t-2}^e/400) \\ & + \alpha_6^{hef}\Delta\tau_t^{hef} + \alpha_7^{hef}\Delta\tau_{t-1}^{hef} \quad (47)\end{aligned}$$

- short-run energy inflation:

$$\begin{aligned}\Delta heg_t = & \alpha_0^{heg} + \pi_t^e/400 + \alpha_1^{heg}(heg_{t-1} - heg_{t-1}^*) + \alpha_2^{heg}(\Delta(med_t - exr_t) - \pi_t^e/400) \\ & + \alpha_3^{heg}(\Delta(med_{t-1} - exr_{t-1}) - \pi_{t-1}^e/400) + \alpha_4^{heg}(\Delta heg_{t-1} - \pi_{t-1}^e/400) + \alpha_5^{heg}\Delta\tau_t^{heg} \\ & + \alpha_6^{heg}\Delta\tau_{t-1}^{heg} \quad (48)\end{aligned}$$

- short-run food inflation:

$$\begin{aligned}\Delta hif_t = & \pi_t^e/400 + \alpha_0^{hif} + \alpha_1^{hif}(hif_{t-1} - hif_{t-1}^*) + \alpha_2^{hif}(\Delta yed - \pi_t^e/400) \\ & + \alpha_3^{hif}(\Delta yed_{t-1} - \pi_{t-1}^e/400) + \alpha_4^{hif}(\Delta hif_{t-1} - \pi_{t-1}^e/400) + \alpha_5^{hif}\tau_t^{hif} \\ & + \alpha_6^{hif}\tau_{t-1}^{hif} \quad (49)\end{aligned}$$

where π_{t-1}^e are the long-term inflation expectations and τ^{hef} , τ^{hef} , τ^{hif} are the indirect tax rates applied on consumption, energy, and food, respectively.

We then use total HICP measure as an anchoring indicator for modelling the private consumption deflator, which in the long-run is modelled as:

$$cod_t^* = \alpha_0^{cod^*} + \alpha_1^{cod^*}trend_t + hicmp_t \quad (50)$$

while its short-term dynamics follow the equation:

$$\Delta cod = \alpha_0^{cod} + \alpha_1^{cod} \Delta hicp_t + \alpha_2^{cod} (cod_{t-1} - cod_{t-1}^*) + \alpha_3^{cod} \Delta yed_t + (1 - \alpha_1^{cod} - \alpha_2^{cod} - \alpha_3^{cod}) \Delta med \quad (51)$$

Wage Inflation

The wage inflation gap is modelled via a reduced form of the New Keynesian Phillips Curve. The equation relates actual wage inflation gap to future wage inflation, past, present and future domestic price inflation, present and future euro area price inflation, the deviation of the labour share of income from its trend, the deviation of trend output growth from its steady state and the deviation of the unemployment rate from its trend. In formula:

$$\begin{aligned} \pi_t^w + \pi_t^C = & \frac{1}{1 + \beta_w \delta_w} \{ (1 - \beta_\pi^w) (1 - \delta_\pi^w) (\bar{\pi}_t + \Delta \bar{y}_t) \\ & + \beta_\pi^w \mathbb{E}_t(\pi_{t+1}^w + \pi_{t+1}^C) + \gamma_\pi^w (\pi_{t-1}^w + \pi_{t-1}^C) - \beta_u^{\pi^w} \hat{u}_t \} \\ & + \xi_w \hat{w}_t + e_t^{\pi^w} \end{aligned} \quad (52)$$

Here we follow the approach by [Galí et al. \(2012\)](#) and relate the unemployment gap to the gap in the marginal rate of substitution and a variant of [Cogley & Sbordone \(2008\)](#).

A.3 The financial block

In the context of the individual country models, we take reference short-term rate as given and model the country-specific long-term market rates. The long-term rates are determined as the sum of the average of expected short-term rates, term premium, and country-specific risk premium. Expectations about the future short-term rates are derived from the Base VAR setting. In such setting, the VARs share a core set of macro variables: the policy rate, the GDP deflator, and the output gap. The system of the core VAR variables is augmented by the specific variable for which expectations are being formed.²⁹ The term premium is modelled separately as a function of expected macroeconomic conditions on euro area level and external market developments. The country risk premium depends on the expected macroeconomic conditions and a country's fiscal position. Finally, the short-term rate and the long-term market rate, alongside the risk premiums associated with particular debt and

²⁹For details on the Base VAR specification see Appendix A.2 page 62 [Angelini et al. \(2019\)](#).

equity instruments, provide the basis for the construction of lending rates and financing cost measures used in other parts of the ECB-MC model.³⁰

A.3.1 Lending rates

We model six particular lending rates/financing cost measures:

- the consumer loan rate, one of the determinants of consumption,
- the mortgage rate, a determinant of house prices and property income,
- the deposit rate, a determinant of property income,
- the lending rate for Non financial Corporations,
- the corporate bond yields,
- the cost of equity

We construct the lending rates/financing cost measures as a composite of the short-term reference rate and the long-term market interest rate, plus a residual risk spread corresponding to each financing instrument $i, i = 1 : 6$:

$$LR_t^i = v_s^i R_t(1) + v_l^i R_t(40) + RPR_t^i \quad (53)$$

where v_s^i and v_l^i are weights assigned to the short-term rate (3-month) and the long-term rate (10-year) and RPR denotes a country-specific risk spread associated with each particular financing instrument. The latter are modelled in the following way:

$$RPR_t^i = \alpha^{RPR^i} + \rho^{RPR^i} RPR_{t-1}^i + \beta^{RPR^i} \frac{1}{m} \mathbb{E}_t \sum_{i=0}^m \hat{y}_{t+i} + e_t^{RPR^i} \quad (54)$$

where $\hat{y}_{t+i}$ correspond to expected output gap for a given country and $e_t^{RPR^i}$ captures the idiosyncratic shocks.

Most of the corresponding data is directly observable within the MFI interest rate statistics (MIR) dataset published regularly by the ECB. The unobservable cost of equity is inferred from the Hasia-Fuller Dividends Discount Model.

³⁰See figure 3 in section 3.1.

A.3.2 Interest-bearing income and wealth

The interest-bearing income and wealth components play an important role in the ECB-MC as they are one of the key transmission channels of financial shocks in the model. Property income and wealth enter into the determination of target consumption. In addition, property income and several other interest-bearing instruments contribute to households' nominal disposable income. This explicit modelling of determinants of wealth in the ECB-MC results in a rich set of transmission channels from changes in prices of different asset classes on consumption and therefore on the whole economy.

Households wealth

The households' wealth is defined as a sum of gross housing wealth GHW_t and the net financial wealth NFW_t . The stock of nominal housing depends on the house prices and nominal residential investment IH_t , the latter being driven by mortgage-related lending rates. In turn, the net financial wealth (NFW) is a function of the nominal net disposable income and a revaluation term, which depends on the households' financial portfolio. Empirically, we can write:

Households' wealth:

$$HHW_t = GHW_t + NFW_t \quad (55)$$

where GHW_t is the gross housing wealth and NFW_t is the net financial wealth, where both series are observed within the sector accounts statistics.³¹

The stock of nominal housing wealth :

$$GHW_t = IH_t + (1 - \delta^h)GHW_{t-1}(RPP_t/RPP_{t-1}) \quad (56)$$

where IH_t are nominal residential investments, where RPP_t is a house prices index (new and existing dwellings), and δ^h is the depreciation rate associated with the housing stock.

Household net financial wealth:

$$NFW_t = (Y_t - C_t - IH_t) + NFW_{t-1}REV_t \quad (57)$$

³¹Other parts of the household wealth consist of inventories, other investments, non produced assets, capital transfers and adjustments for pensions. These components are assumed to be exogenous and are not modelled explicitly.

where Y_t is the Nominal Disposable Income, C_t is the Nominal (total) Consumption, IH_t is the Nominal Residential Investment (gross fixed capital formation, dwellings) and REV_t is the Revaluation Term³².

Property income

The property income ($DINP$) is the most interest-sensitive component of the nominal disposable income. In the ECB-MC it is expressed by the following identity:³³

$$DINP_t = GOS_t + IRN_t + DDN_t + RIN_t + OIN_t + RTN_t + OCT_t - WTP_t - SCP_t \quad (58)$$

where we explicitly model household gross operating surplus (GOS_t), the net interest income (IRN_t), and the dividend income (DDR_t)³⁴, reinvested earnings (RIN_t), other investment income (OIN_t), non-residential property rents (RTN_t), and other capital transfers (OCT_t). Taxes (WTP_t) and social contributions (SCP_t) associated with the property income are modelled in the fiscal block. As implied above, all modelled variables, whose specification is provided in following subsections, are directly observable within the sector accounts statistics.

Gross operating surplus of households

The gross operating surplus of households is modelled as a share of nominal GDP (Y^N) and is assumed to be primarily determined by income from the residential housing capital:

$$\frac{GOS_t}{Y_t^N} = \alpha^{GS} + \beta_1^{GS} \frac{KH_t * IHD_t}{Y_t^N} + \beta_2^{GS} \frac{HPI_t}{P_t^c} + e_t^{GS} \quad (59)$$

where the first term represent the nominal housing stock, whereas the second term reflects house prices relative to the private consumption deflator. Real housing capital, KH , is specified in the Residential Investment block in Section A.1.2, while the residential investment deflator (IHD), the private consumption deflator (P^c) and the residential property price index ($RPPI$) are directly observed variables.

³²For more details on the revaluation term, please see [Angelini et al. \(2019\)](#)

³³All variables are expressed in net terms (credit - debit) and relate to the household sector.

³⁴Household gross operating surplus is not including mixed income, which we assume to be a part of the labour income.

Net interest income

The net interest income of households is modelled as a share of GDP and depends on its own autoregressive term, previous period's net foreign assets (NFA) position relative to GDP, the general level of interest rates captured by STN , and a spread between mortgage rate (R^m) and deposit rate (R^d) that captures the net concept given by interest payable and receivable:

$$\begin{aligned} \frac{IRN_t}{Y_{t-1}^N} = & \alpha^{IRN} + \rho^{IRN} \frac{IRN_{HH,t-1}}{Y_{t-2}^N} + \beta_1^{IRN} \frac{NFA_{t-1}}{Y_{t-1}^N} \\ & + \beta_2^{IRN} STN_t + \beta_3^{IRN} (R_t^m - R_t^d) + e^{IRN} \end{aligned} \quad (60)$$

In the estimation, the NFA series is observed in the sector accounts statistics, while mortgage and deposit rates are constructed as suggested by equation (53).

Dividends

The dividends are modelled within the PAC framework, implying a dependency of the real dividends growth on: (i) the degree to which dividends were out of equilibrium in the previous period, (ii) lagged dividend growth, and (iii) expected growth of desired dividend income. The desired (log-) level of dividend income is assumed to be a constant fraction of the (real) gross operating surplus and mixed income:

$$ddr_t^* = \alpha^D + gosmir_t + t \quad (61)$$

where we also include a linear trend to improve the empirical fit. The corresponding short-run PAC equation is then:

$$\Delta ddr_t = a_0^d (ddr_{t-1}^* - ddr_{t-1}) + \sum_{i=1}^3 a_i^d \Delta ddr_{t-i} + \mathbb{E}_t \sum d_i^d \Delta ddr_{t+i}^* + e_t^{ddr} \quad (62)$$

House prices

The target of house prices is derived from an inverted demand curve resulting from the optimization problem of the consumer who chooses between different consumption subcom-

ponents. The resulting equilibrium condition is the following:

$$RP_t^{H,*} = C_t^* \frac{1}{KH_t} \frac{1}{UC_t^{HP}} \chi \quad (63)$$

where RP are relative prices (House Prices over Consumption Deflator), C^* is the target consumption and UC^{HP} is user cost of housing ownership.³⁵ The condition in (63) is the basis for the following empirical specification:

$$rp_t^{H*} = \alpha^{H*} + \beta_1^{H*} y h_t - \beta_1^{H*} k h_t + \beta_3^{H*} uc_t^{HP} \quad (64)$$

where the user costs are defined as:

$$uc_t^{HP} = \alpha^{UC_{hp}} + \delta_t^H + r_t^{mt} - \pi_t^e + \tau_t - 0.4 \Delta^e rp_t^H \quad (65)$$

with r_t^{mt} being the mortgage rate, π^e the inflation expectations, τ the taxes on housing and $\Delta^e rp^H$ the expectations on relative price growth.

The short-run equation for relative prices is the usual PAC equation:

$$\Delta rp_t^H = a_0^{rph} (rp_{t-1}^{H*} - rp_{t-1}^H) + \sum_{i=1}^1 a_i^{rph} \Delta rp_{t-i}^H + \mathbb{E}_{t-1} \sum_{j=0}^{\infty} d_j^{rph} \Delta rp_{t+j}^{H*} + e_t^{rph} \quad (66)$$

Net foreign assets

Accumulation of the net foreign assets (NFA) is assumed to depend on the trade balance and a residual component capturing the net interest income and a revaluation term:

$$NFA_t - NFA_{t-1} = TB_t + REST_t \quad (67)$$

The residual term, $REST$, is modelled relative to nominal GDP and is assumed to be driven by changes in the spread between interest rates on foreign liabilities (IR^{FL}) and interest

³⁵Note that the user costs of housing ownership differ from the user costs of housing capital defined in the residential investment section.

rates on foreign assets (IR^{FA}), and by changes in domestic and world prices (P^y and P^W):

$$\frac{REST_t}{Y_{t-1}^N} = \alpha^{nfa} + \beta_1^{nfa} \Delta(IR^{FL} - IR^{FA}) + \beta_2^{nfa} \Delta(P_t^y) + \beta_3^{nfa} \Delta(P_t^W) + e_t^{nfa} \quad (68)$$

The empirical interest rates used in (68) are observed implicitly as a ratio between property income of foreign assets/liabilities and total foreign assets/liabilities:

$$IR_t^{FA} = \frac{IPN_t^{FA}}{FA_t} \cdot 400; R_t^{FL} = \frac{IPN_t^{FL}}{FL_t} \cdot 400.$$

For modelling purposes the interest rate on foreign assets is expressed in terms of a foreign long-term rate ($R_t^{US_{10Y}}$), whereas foreign liabilities are determined by the domestic long-term rate ($R_t^{CC_{10Y}}$):

$$IR_t^{FA} = c_t^{FA} + \beta^{ifa} R_t^{US_{10Y}} + e_t^{ifa} \quad (69)$$

$$IR_t^{FL} = c_t^{FL} + \beta^{ifl} R_t^{CC_{10Y}} + e_t^{ifl} \quad (70)$$

All interest rate variables are demeaned and c is set to follow an autoregressive process, implying the interest rate parity in the long run.³⁶

B VAR expectations

The ECB-MC model assumes that economic agents form expectations using a reduced-form structure based on VARs. In particular, expectations are formed using country-specific VARs that incorporate a common monetary policy channel through a shared “Base VAR” block. This setup allows agents to use limited information and preserves the modularity of the model.

B.1 General VAR Structure:

Each expectation-relevant equation features a target variable (e.g. target consumption) and the Base VAR, which includes inflation, interest rate, and output gap. The Base VAR has two components, the country specific one and the euro area wide. The latter is common across countries and is extended to reflect the euro area’s common monetary policy by

³⁶Namely: $c_t^{FA} = 0.99c_{t-1}^{FA}$; $c_t^{FL} = 0.99c_{t-1}^{FL}$

incorporating EA-wide aggregates.

The full VAR system includes both country-specific and EA-level variables. The structure of the system is represented below by a block-inclusion matrix (1 = variable included, 0 = not included):

	y_t^*	π_t^{EA}	gap_t^{EA}	i_t^{EA}	π_t^c	gap_t^c	$\pi_t^{EA,*}$	$\pi_t^{c,*}$	i_t^*	g_t^*
y_t^*	1	0	0	1	1	1	0	1	1	1
π_t^{EA}	0	1	1	1	0	0	1	0	1	0
gap_t^{EA}	0	1	1	1	0	0	1	0	1	0
i_t^{EA}	0	1	1	1	0	0	1	0	1	0
π_t^c	0	0	0	1	1	1	0	1	1	0
gap_t^c	0	0	0	1	1	1	0	1	1	0
$\pi_t^{EA,*}$	0	0	0	0	0	0	1	0	0	0
$\pi_t^{c,*}$	0	0	0	0	0	0	0	1	0	0
i_t^*	0	0	0	0	0	0	0	0	1	0
g_t^*	0	0	0	0	0	0	0	0	0	1

Rows denote the dependent variable in each VAR equation, while columns indicate the variables included in the corresponding information set. Relative to the ECB-BASE specification in [Angelini et al. \(2019\)](#), the ECB-MC extends the VAR structure by augmenting the country-specific information set with euro area aggregates and by imposing block exogeneity restrictions.

The variable y_t^* denotes the target variable entering the corresponding PAC equation, while π_t^{EA} , gap_t^{EA} and i_t^{EA} denote euro area inflation, the euro area output gap and the common euro area short-term interest rate, respectively. Country-specific variables are denoted by superscript c , with π_t^c and gap_t^c representing domestic inflation and the domestic output gap. Variables with superscript $*$ denote long-term targets entering the VAR system, namely the euro area inflation target $\pi_t^{EA,*}$, the country-specific inflation target $\pi_t^{c,*}$, the long-term interest rate target i_t^* , and the long-term growth rate of the target variable g_t^* .

We assume that euro area aggregates do not respond to country-specific variables. In

particular, euro area inflation and the euro area output gap evolve independently of domestic country conditions within the VAR system. The short-term interest rate depends only on euro area aggregates, reflecting the common monetary policy environment of the monetary union. Country-specific inflation and output gap variables affect the interest rate only indirectly through their contribution to euro area aggregates.

B.2 VAR form

The exact setup of the Base VAR is:

$$\Delta z_t = \Lambda_0(z_{t-1} - z_{t-1}^*) + \sum_{k=1}^K \Lambda_k \Delta z_{t-k} + \varepsilon_t \quad (71)$$

where z_t is a $n \times 1$ vector of variables normally containing target variable, inflation, interest rate, and output gap as explained above. Λ_0 is $n \times n$ matrix containing coefficients that control how variables respond to the deviation of the lagged level from the long-term targets. Λ_k are $n \times n$ matrices that collect auto-regressive coefficients for K lags. The Base VAR does not contain any constant terms. Assuming the VAR is stable, one can show that in the long-term, VAR-based forecasts will converge to the long-term expectations $z_{t+h} \xrightarrow{h \rightarrow \infty} z_{t-1+h}^*$.

To close the model, we assume that the long-term targets follow a random walk process, $z_t^* = z_{t-1}^* + \eta_t$, while the long-term target for the output gap is fixed at zero. Long-term expectations for inflation and interest rates are sourced from Consensus Forecasts and interest rate swaps, respectively. Target variables are constrained to grow at their assumed long term steady state growth rate (e.g. target consumption with long-term potential output growth rate), which also follows a random walk.

C ECB-MC SUR Pooled Model: System-Weighted Pooled FGLS

C.1 Setting

C.1.1 Panel structure

Consider a panel of N countries indexed $c \in \{1, \dots, N\}$ observed over T time periods $t = 1, \dots, T$. Each country is associated with one endogenous variable of interest. The model

structure can be thought of as a variant of the Error-Correction (ECM) specification since it contains both *short-run* terms (first or second differences capturing dynamic adjustment) and *error-correction* terms (ensuring convergence to the long-run equilibrium).

A generic panel equation specific per country contains with K regressors. The regressors are partitioned into:

- Country-specific regressors $\mathbf{x}_{t-1}^c \in \mathbb{R}^{K_c}$: variables that differ across countries (own lags, country GDP gap, country specific inflation ...).
- Common regressors $\mathbf{w}_{t-1} \in \mathbb{R}^{K_w}$: variables shared by all countries (e.g. a common interest rate).

This implies that the total regressors per equation: $K = K_c + K_w$.

C.2 The pool restriction

The **full pool restriction** imposes that the coefficient are *identical* across all N countries. This reduces the number of free parameters from $N \times K$ to K , which improves estimation efficiency given that the parameter homogeneity across countries is a reasonable assumption which in our setting seems reasonable. Therefore, under this restriction the model for country c is:

$$y_t^c = \boldsymbol{\alpha}_c^\top \mathbf{x}_{t-1}^c + \boldsymbol{\alpha}_w^\top \mathbf{w}_{t-1} + \varepsilon_t^c, \quad (72)$$

where $\mathbf{y}$ denotes the variable being modelled here (see Sections C.4.1 and C.5 for the two cases in ECB-MC), $\boldsymbol{\alpha}_c \in \mathbb{R}^{K_c}$ multiplies the country-specific regressors and $\boldsymbol{\alpha}_w \in \mathbb{R}^{K_w}$ multiplies the common regressors. The full parameter vector is $\boldsymbol{\alpha} = (\boldsymbol{\alpha}_c^\top, \boldsymbol{\alpha}_w^\top)^\top \in \mathbb{R}^K$.

subsectionSingle-period stacked formThe SUR System Under Full Pooling

C.2.1 Single-period stacked form

Collect all N country equations at time t into a system. Define:

$$\mathbf{y}_t = \begin{pmatrix} y_t^1 \\ \vdots \\ y_t^N \end{pmatrix} \in \mathbb{R}^N, \quad \boldsymbol{\varepsilon}_t = \begin{pmatrix} \varepsilon_t^1 \\ \vdots \\ \varepsilon_t^N \end{pmatrix} \in \mathbb{R}^N. \quad (73)$$

The matrix of regressors at time t must reflect the fact that country-specific regressors differ across equations while common regressors are identical for every country:

$$\mathbf{X}_{t-1} = \begin{pmatrix} \mathbf{x}_{t-1}^{1\top} & \mathbf{w}_{t-1}^\top \\ \mathbf{x}_{t-1}^{2\top} & \mathbf{w}_{t-1}^\top \\ \vdots & \vdots \\ \mathbf{x}_{t-1}^{N\top} & \mathbf{w}_{t-1}^\top \end{pmatrix} \in \mathbb{R}^{N \times K}, \quad (74)$$

$$= [\mathbf{X}_{c,t-1} \mid \mathbf{1}_N \mathbf{w}_{t-1}^\top]$$

where $\mathbf{X}_{c,t-1} \in \mathbb{R}^{N \times K_c}$ stacks the country-specific regressors row by row, and $\mathbf{1}_N \mathbf{w}_{t-1}^\top \in \mathbb{R}^{N \times K_w}$ repeats the common regressor vector $\mathbf{w}_{t-1}$ in every row. Then the system at time t is given by:

$$\mathbf{y}_t = \mathbf{X}_{t-1} \boldsymbol{\alpha} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\alpha} = \begin{pmatrix} \boldsymbol{\alpha}_c \\ \boldsymbol{\alpha}_w \end{pmatrix} \in \mathbb{R}^K. \quad (75)$$

The variance-covariance (VCV) matrix of the error terms allows cross-country error correlations (Seemingly Unrelated Regressions structure):

$$\boldsymbol{\Sigma} \equiv \mathbb{E}[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^\top] = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_{22} & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_{NN} \end{pmatrix}. \quad (76)$$

By stacking over all T observations and denoting $\mathbf{Y} = (\mathbf{y}_1^\top, \dots, \mathbf{y}_T^\top)^\top \in \mathbb{R}^{NT}$, $\boldsymbol{\varepsilon} \in \mathbb{R}^{NT}$ and $\mathbb{X} \in \mathbb{R}^{NT \times K}$ is obtained by stacking $\mathbf{X}_{t-1}$ over $t = 1, \dots, T$:

$$\mathbb{X} = \begin{pmatrix} \mathbf{X}_0 \\ \mathbf{X}_1 \\ \vdots \\ \mathbf{X}_{T-1} \end{pmatrix} = \begin{pmatrix} \mathbf{X}_{c,0} & \mathbf{1}_N \mathbf{w}_0^\top \\ \mathbf{X}_{c,1} & \mathbf{1}_N \mathbf{w}_1^\top \\ \vdots & \vdots \\ \mathbf{X}_{c,T-1} & \mathbf{1}_N \mathbf{w}_{T-1}^\top \end{pmatrix} \in \mathbb{R}^{NT \times K}. \quad (77)$$

the full model and its VCV matrix can be written as:

$$\mathbf{Y} = \mathbb{X} \boldsymbol{\alpha} + \boldsymbol{\varepsilon}, \quad \mathbb{E}[\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^\top] = \boldsymbol{\Sigma} \otimes \mathbf{I}_T. \quad (78)$$

C.3 The Pooled FGLS Estimator

C.3.1 Feasible GLS (FGLS) by iteration

Given that Σ is unknown we resort to using FGLS as in Zellner (1962). In this case the algorithm implemented is standard and relies on the iterative procedure to estimate Σ from the residuals of the current α iterate. More precisely the algorithm proceeds as follows:

1. **Initialise** with the pooled OLS estimate: $\hat{\alpha}^{(0)} \leftarrow (\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top \mathbf{Y}$.
2. **Compute residuals**: $\hat{\mathbf{E}}^{(j)} = \mathbf{Y} - \mathbb{X} \hat{\alpha}^{(j)}$, reshape into an $T \times N$ matrix $\hat{\mathcal{E}}^{(j)}$.
3. **Estimate covariance**:

$$\hat{\Sigma}^{(j)} = \frac{1}{T} \hat{\mathcal{E}}^{(j)\top} \hat{\mathcal{E}}^{(j)}. \quad (79)$$

4. **GLS update**:

$$\hat{\alpha}^{(j+1)} = \left(\mathbb{X}^\top ((\hat{\Sigma}^{(j)})^{-1} \otimes \mathbf{I}_T) \mathbb{X} \right)^{-1} \mathbb{X}^\top ((\hat{\Sigma}^{(j)})^{-1} \otimes \mathbf{I}_T) \mathbf{Y}. \quad (80)$$

5. **Convergence check**: if $\max_k |\hat{\alpha}_k^{(j+1)} - \hat{\alpha}_k^{(j)}| < \tau$, stop; otherwise set $j \leftarrow j + 1$ and return to step 2.

Convergence and efficiency. As shown by Zellner (1962), under the assumption of Gaussian error terms, iterating to convergence yields the maximum likelihood SUR estimator. This variant of this algorithm is implemented in the Dynare routine `pooled_fgls` using the Cholesky factorisation of $\hat{\Sigma}$ for numerical stability: the GLS step (80) is computed as a QR decomposition of $\hat{\mathbf{L}}^{-1} \mathbb{X}$, where $\hat{\mathbf{L}} = \text{chol}(\hat{\Sigma}) \otimes \mathbf{I}_T$.

C.4 Two cases for the dependent variable in ECB-MC

C.4.1 Case 1: The Dependent Variable is defined as a gap variable

The endogenous variable y_t^c is a **percentage gap** between a (constructed) series and its trend or equilibrium. The dependent variable entering the model specification in this case is $\Delta y_t^c = y_t^c - y_{t-1}^c$, the *change in the gap*.

C.4.2 Variable construction

Let s_t^c denote the underlying series of interest (e.g. a ratio or share) and h_t^c its trend or equilibrium benchmark. The gap variable is:

$$y_t^c = 100 \left(\frac{s_t^c}{h_t^c} - 1 \right). \quad (81)$$

The trend h_t^c may be obtained in a various way, such as an externally constructed equilibrium/target serie or plain HP-filtered (or any other statistically trend extracted) series.

C.4.3 Order of integration

Because y_t^c is already expressed as a *deviation from trend*, it is typically stationary or near-stationary (I(0) or weakly I(1)). The first difference Δy_t^c is then stationary, consistent with an **I(1)-ECM** specification.

C.4.4 The equation

The generic country equation is:

$$\begin{aligned} \Delta y_t^c = & \underbrace{\alpha_1 \Delta y_{t-1}^c}_{\text{short-run AR: gap momentum}} + \underbrace{\alpha_2 y_{t-1}^c}_{\text{ECM: gap mean reversion}} + \underbrace{\sum_{k \in \mathcal{S}_c} \alpha_k \Delta x_{k,t-1}^c}_{\text{short-run country terms}} + \underbrace{\sum_{k \in \mathcal{L}_c} \alpha_k x_{k,t-1}^c}_{\text{ECM country terms}} \\ & + \underbrace{\sum_{k \in \mathcal{S}_w} \alpha_k \Delta w_{k,t-1}}_{\text{short-run common terms}} + \underbrace{\sum_{k \in \mathcal{L}_w} \alpha_k \tilde{w}_{k,t-1}}_{\text{ECM common terms}} + \varepsilon_t^c. \end{aligned} \quad (82)$$

In this case, the level y_{t-1}^c is the **own error-correction term**. It directly measures the distance

of the gap from its long-run equilibrium (zero). The equilibrium relationship is simply $y^c = 0$; no cointegrating vector needs to be separately estimated. This is an implicit normalisation: the trend h_t^c is defined such that the long-run level of y_t^c is zero by construction.

C.4.5 Matrix form (Case 1)

Define country-stacked vectors $\mathbf{y}_t$, $\mathbf{x}_t^{(k)}$, $\mathbf{1}_N$ as before. The full system under full pooling is:

$$\begin{aligned} \Delta \mathbf{y}_t = & \alpha_1 \Delta \mathbf{y}_{t-1} + \alpha_2 \mathbf{y}_{t-1} + \sum_{k \in \mathcal{S}_c} \alpha_k \Delta \mathbf{x}_{t-1}^{(k)} + \sum_{k \in \mathcal{L}_c} \alpha_k \mathbf{x}_{t-1}^{(k)} \\ & + \sum_{k \in \mathcal{S}_w} \alpha_k \mathbf{1}_N \Delta w_{k,t-1} + \sum_{k \in \mathcal{L}_w} \alpha_k \mathbf{1}_N \tilde{w}_{k,t-1} + \boldsymbol{\varepsilon}_t. \end{aligned} \quad (83)$$

The key feature is that the ECM term is simply $\alpha_2 \mathbf{y}_{t-1}$ — a scalar times the lagged *endogenous* variable itself.

C.5 Case 2: The Dependent Variable is a Growth Rate

The endogenous variable is the **growth rate** of some underlying level variable B_t^c , defined as $b_t^c = \Delta \ln B_t^c$ (or an annualised/scaled version). The model is written in second differences: $\Delta^2 b_t^c = \Delta b_t^c - \Delta b_{t-1}^c$, i.e. the *change in the growth rate* (acceleration).

C.5.1 Variable construction

Let B_t^c be the underlying level variable (in ECB target variables as private consumption, business investment, residential investment targets). The **growth rate** is:

$$b_t^c = \Delta \ln B_t^c = \ln B_t^c - \ln B_{t-1}^c. \quad (84)$$

The model's dependent variable is $\Delta^2 b_t^c = b_t^c - 2b_{t-1}^c + b_{t-2}^c$, the **acceleration** of the growth rate.

C.5.2 Order of integration

This specification is consistent with b_t^c (the growth rate) following an I(1) process, which implies that the level B_t^c is I(2). This is an I(2)-ECM: the second difference of the log-level

is modelled with error-correction terms in the first-difference (growth-trend gap).

C.5.3 The equation

Let τ_t^c denote the long-run equilibrium growth rate (trend). The generic country equation is:

$$\begin{aligned} \Delta^2 b_t^c = & \underbrace{\alpha_1 \Delta^2 b_{t-1}^c}_{\text{short-run AR: acceleration inertia}} + \underbrace{\alpha_2 (\Delta b_{t-1}^c - \tau_{t-1}^c)}_{\text{ECM: growth rate gap from trend}} + \underbrace{\sum_{k \in \mathcal{S}_c} \alpha_k \Delta x_{k,t-1}^c}_{\text{short-run country terms}} + \underbrace{\sum_{k \in \mathcal{L}_c} \alpha_k x_{k,t-1}^c}_{\text{ECM country terms}} \\ & + \underbrace{\sum_{k \in \mathcal{S}_w} \alpha_k \Delta w_{k,t-1}}_{\text{short-run common terms}} + \underbrace{\sum_{k \in \mathcal{L}_w} \alpha_k \tilde{w}_{k,t-1}}_{\text{ECM common terms}} + \varepsilon_t^c. \end{aligned} \quad (85)$$

In this case, the ECM term $\Delta b_{t-1}^c - \tau_{t-1}^c$ measures the **deviation of the growth rate from its long-run trend**. The trend τ_t^c is typically an exogenous estimate of the equilibrium growth rate of a particular variable (e.g. potential business investment growth).

Important to notice is that in Case 1 the ECM term is the level of the gap itself (y_{t-1}^c), and the dependent variable is its first difference. In Case 2 the ECM term is the gap of the *growth rate* from its trend ($\Delta b_{t-1}^c - \tau_{t-1}^c$), and the dependent variable is the *second difference* of the log-level. Obviously, both are error-correction mechanisms, but differing in an integration order.

C.5.4 Matrix form (Case 2)

Define country-stacked vectors as before, and $\boldsymbol{\tau}_t = (\tau_t^1, \dots, \tau_t^N)^\top$. The full system under full pooling is:

$$\begin{aligned} \Delta^2 \mathbf{b}_t = & \alpha_1 \Delta^2 \mathbf{b}_{t-1} + \alpha_2 (\Delta \mathbf{b}_{t-1} - \boldsymbol{\tau}_{t-1}) + \sum_{k \in \mathcal{S}_c} \alpha_k \Delta \mathbf{x}_{t-1}^{(k)} + \sum_{k \in \mathcal{L}_c} \alpha_k \mathbf{x}_{t-1}^{(k)} \\ & + \sum_{k \in \mathcal{S}_w} \alpha_k \mathbf{1}_N \Delta w_{k,t-1} + \sum_{k \in \mathcal{L}_w} \alpha_k \mathbf{1}_N \tilde{w}_{k,t-1} + \boldsymbol{\varepsilon}_t. \end{aligned} \quad (86)$$

The key difference from Case 1 is the ECM term $\alpha_2 (\Delta \mathbf{b}_{t-1} - \boldsymbol{\tau}_{t-1})$: it involves a first difference of the endogenous variable *minus* an exogenous trend, rather than the level of the endogenous variable itself.

C.6 Side-by-Side Comparison

Table 3 provides the summary comparison of the cases previously described.

Table 3: Structural comparison of the two cases

Feature	Case 1: Gap variable	Case 2: Growth rate
Endogenous var. z_t^c	Gap: $y_t^c = 100(s_t^c/h_t^c - 1)$	Growth rate: $b_t^c = \Delta \ln B_t^c$
Dependent var.	Δy_t^c (first diff. of gap)	$\Delta^2 b_t^c$ (second diff. of log-level)
Integration order	I(1)-ECM for y_t^c	I(2)-ECM for $\ln B_t^c$
Own AR term	$\alpha_1 \Delta y_{t-1}^c$	$\alpha_1 \Delta^2 b_{t-1}^c$
Own ECM term	$\alpha_2 y_{t-1}^c$	$\alpha_2 (\Delta b_{t-1}^c - \tau_{t-1}^c)$
Equilibrium	$y^c = 0$ (built into definition)	$\Delta b^c = \tau^c$ (exogenous trend)
Stability cond.	$\alpha_2 < 0$	$\alpha_2 < 0$
Estimation	Pooled FGLS (80)	Pooled FGLS (80)
Pool restriction	$\alpha^c = \alpha$ for all c	$\alpha^c = \alpha$ for all c

D Bayesian estimation of the VAR/ SUR model

The estimation of the VAR systems in ECB-MC is conducted using a Bayesian Seemingly Unrelated Regressions (SUR) framework as described in Ando and Zellner (2010). This approach allows for efficient estimation of the system while incorporating informative priors to address limited time-series length, especially at the country level.

Closely following Ando & Zellner (2010) for each equation (country) $j = 1, \dots, N$, let $\mathbf{y}_j \in \mathbb{R}^T$ collect the observations, $\mathbf{X}_j \in \mathbb{R}^{T \times k_j}$ the regressors, and $\alpha_j \in \mathbb{R}^{k_j}$ the coefficient vector. Stack across equations:

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_N \end{pmatrix}, \quad \mathbf{X} = \text{diag}(\mathbf{X}_1, \dots, \mathbf{X}_N), \quad \alpha = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_N \end{pmatrix}. \quad (87)$$

The SUR model in a matrix form reads as:

$$\mathbf{y} = \mathbf{X}\alpha + \mathbf{u}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \Sigma \otimes \mathbf{I}_T), \quad (88)$$

where $\Sigma \in \mathbb{R}^{N \times N}$ is an $N \times N$ symmetric matrix with diagonal elements $\{\sigma_1^2, \dots, \sigma_N^2\}$ and off-diagonal elements σ_{ij} . Let $\mathbf{R}$ denote the $N \times N$ matrix with (i, j) -th element:

$$R_{ij} = (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\alpha}_i)^\top (\mathbf{y}_j - \mathbf{X}_j \boldsymbol{\alpha}_j). \quad (89)$$

The normal likelihood function can be written as:

$$L(\mathbf{y}|\boldsymbol{\alpha}, \Sigma) = \frac{1}{(2\pi)^{TN/2} |\Sigma|^{T/2}} \exp\left\{-\frac{1}{2} \text{tr}[\mathbf{R} \Sigma^{-1}]\right\}, \quad (90)$$

Notice that $\mathbf{R} = \mathbf{E}^\top \mathbf{E}$, with $\mathbf{E}$ the $T \times N$ matrix whose j -th column is $\mathbf{y}_j - \mathbf{X}_j \boldsymbol{\alpha}_j$.

D.1 Prior distributions

In the absence of prior information, [Ando & Zellner \(2010\)](#) use Jeffreys' invariant prior:

$$\pi_1(\boldsymbol{\alpha}, \Sigma) = \pi_1(\boldsymbol{\alpha}) \pi_1(\Sigma) \propto |\Sigma|^{-(N+1)/2}. \quad (91)$$

The resulting joint posterior is:

$$g(\boldsymbol{\alpha}, \Sigma|\mathbf{y}) \propto |\Sigma|^{-(T+N+1)/2} \exp\left\{-\frac{1}{2} \text{tr}[\mathbf{R} \Sigma^{-1}]\right\}. \quad (92)$$

On the other hand there are cases when prior information on the coefficients is available, In this case [Ando & Zellner \(2010\)](#) adopt a normal prior for $\boldsymbol{\alpha}$ combined with the same diffuse prior for Σ :

$$\pi_2(\boldsymbol{\alpha}, \Sigma) = \pi_2(\boldsymbol{\alpha}) \pi_2(\Sigma), \quad \pi_2(\boldsymbol{\alpha}) = \mathcal{N}(\boldsymbol{\alpha}_0, \mathbf{A}_\beta^{-1}), \quad \pi_2(\Sigma) \propto |\Sigma|^{-(N+1)/2}, \quad (93)$$

where $\boldsymbol{\alpha}_0 \in \mathbb{R}^K$ is the prior mean and $\mathbf{A}_\beta \in \mathbb{R}^{K \times K}$ (positive definite) is the prior precision matrix.

D.2 Posterior distributions

[Ando & Zellner \(2010\)](#) show that under Jeffreys' prior conditional posteriors read as

$$g(\boldsymbol{\alpha}|\Sigma, \mathbf{y}) = \mathcal{N}(\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\Omega}}_\alpha), \quad g(\Sigma|\boldsymbol{\alpha}, \mathbf{y}) = \mathcal{IW}(\mathbf{R}, T), \quad (94)$$

with:

$$\hat{\alpha} = [\mathbf{X}^\top (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T) \mathbf{X}]^{-1} \mathbf{X}^\top (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T) \mathbf{y}, \quad (95)$$

$$\hat{\Omega}_\alpha = [\mathbf{X}^\top (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T) \mathbf{X}]^{-1}. \quad (96)$$

Under informative normal prior posteriors are given by

$$g(\alpha | \boldsymbol{\Sigma}, \mathbf{y}) = \mathcal{N}(\bar{\alpha}, \bar{\Omega}_\alpha), \quad g(\boldsymbol{\Sigma} | \alpha, \mathbf{y}) = \mathcal{IW}(\mathbf{R}, T), \quad (97)$$

with updated mean and covariance:

$$\bar{\alpha} = \bar{\Omega}_\alpha [\mathbf{X}^\top (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T) \mathbf{X} \hat{\alpha} + \mathbf{A}_\beta \alpha_0], \quad (98)$$

$$\bar{\Omega}_\alpha = [\mathbf{X}^\top (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T) \mathbf{X} + \mathbf{A}_\beta]^{-1}. \quad (99)$$

D.3 Gibbs sampler

Since the full conditional posteriors (97) are standard distributions, [Ando & Zellner \(2010\)](#) point out that a two-block Gibbs sampler can be directly used:

1. **Initialise:** set $\alpha^{(0)}$ (e.g. to the equation-by-equation OLS estimates).
2. For $s = 1, 2, \dots, S$:
 - (a) **Draw $\boldsymbol{\Sigma}^{(s)}$:** given $\alpha^{(s-1)}$, form the $T \times N$ residual matrix $\mathbf{E}^{(s-1)}$ with j -th column $\mathbf{y}_j - \mathbf{X}_j \alpha_j^{(s-1)}$, set $\mathbf{R}^{(s)} = \mathbf{E}^{(s-1)\top} \mathbf{E}^{(s-1)}$, then sample:

$$\boldsymbol{\Sigma}^{(s)} \sim \mathcal{IW}(\mathbf{R}^{(s)}/T, T). \quad (100)$$

- (b) **Draw $\alpha^{(s)}$:** given $\boldsymbol{\Sigma}^{(s)}$, compute $\hat{\alpha}$ from (95), then $\bar{\Omega}_\alpha$ and $\bar{\alpha}$ from (98)–(99), and sample:

$$\alpha^{(s)} \sim \mathcal{N}(\bar{\alpha}, \bar{\Omega}_\alpha). \quad (101)$$

3. **Discard** the first S_{burn} draws as burn-in; retain $S_{\text{keep}} = S - S_{\text{burn}}$ draws.

The posterior mean of α and credible intervals are computed from the retained draws $\{\alpha^{(s)}\}$.

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E Estimation results of selected blocks

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Table 4: Consumption

Parameter	Description	Target									
		DE		FR		IT		ES		NL	
		Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD
η_0	Constant of target consumption	0.422	0.0618	0.491	0.0650	0.014	0.5293	0.568	0.0890	0.44	0.0786
η_T	Weight of permanent transfer income	0.286	0.0010	0.214	0.0010	0.206	0.0097	0.285	0.0010	0.286	0.0010
η_P	Weight of permanent property income	0.098	0.0010	0.134	0.0010	0.106	0.0096	0.097	0.0010	0.097	0.0010
η_L	Weight of permanent labour income	0.537	0.0010	0.619	0.0010	0.505	0.0098	0.537	0.0010	0.537	0.0010
γ^T	Coeff. of time trend	-0.001	0.0000	-0.000	0.0000	-0.001	0.0000	-0.002	0.0000	-0.002	0.0000
Dynamic PAC Equation											
Parameter	Description	DE		FR		IT		ES		NL	
		Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
		Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
a_0	Coeff. of past deviation from target consumption	0.140	0.0788	0.252	0.1362	0.150	0.0963	0.033	0.0360	0.100	0.0703
a_1	Coeff. of lagged consumption growth	0.176	0.2302	0.055	0.1323	0.200*		0.667	0.0963	0.013	0.1561
β_x	Weight of spread on the lending rate on consumption	-0.002*		-0.000	0.0024	-0.002*		-0.001	0.0043	-0.002*	
θ	Share of liquidity constrained households	0.100*		0.073	0.0830	0.237	0.0943	0.100*		0.05	0.0678
R^2		0.108		0.162		0.509		0.518		0.103	

Note: Note that the estimation of the dynamic part is done using non-linear least squares. We report the standard errors computed using [White & Domowitz \(1984\)](#) approach. This table contains estimation results for the Big-5 countries with the following abbreviations: DE for Germany, FR for France, IT for Italy, ES for Spain, and NL for the Netherlands. Furthermore, Est. stands for estimate, SE for standard error and PSD for posterior standard deviation of the estimation results. Estimates with a * are either calibrated or follow by construction identities. Thus, other estimates are conditional on these calibrations.

Table 5: Business Investment

Dynamic PAC Equation											
Parameter	Description	DE		FR		IT		ES		NL	
		Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
a_0^{ib}	Coeff. of past deviation from target investment	0.283	0.0870	0.118	0.0533	0.115	0.0337	0.138	0.0486	0.405	0.0992
a_1^{ib}	Coeff. of one-period lagged investment growth	-0.127	0.1905	0.674	0.2008	0.156	0.2405	0.573	0.3734	-0.393	0.1705
a_2^{ib}	Coeff. of two-period lagged investment growth	0.426	0.2337	0.345	0.1912	0.385	0.1453	-0.039	0.1449	0.012	0.1340
R^2		0.440		0.602		0.494		0.341		0.251	

Note: Note that the estimation of the dynamic part is done using non-linear least squares. We report the standard errors computed using [White & Domowitz \(1984\)](#) approach. This table contains estimation results for the Big-5 countries with the following abbreviations: DE for Germany, FR for France, IT for Italy, ES for Spain, and NL for the Netherlands. Furthermore, Est. stands for estimate, SE for standard error and PSD for posterior standard deviation of the estimation results. Estimates with a * are either calibrated or follow by construction identities. Thus, other estimates are conditional on these calibrations.

Table 6: Residential Investment

Dynamic PAC Equation											
Parameter	Description	DE		FR		IT		ES		NL	
		Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
a_0^{ih}	Coeff. of past deviation from target residential investment	0.057	0.0248	0.035	0.0149	0.211	0.0543	0.084	0.0462	0.006	0.0280
a_1^{ih}	Coeff. of one-period lagged investment growth	-0.014	0.1011	0.770	0.1051	-0.000	0.0809	0.315	0.1133	0.054	0.0995
a_2^{ih}	Coeff. of two-period lagged investment growth	-0.084	0.0897	0.086	0.1005	0.143	0.0857	0.275	0.1053	0.239	0.0800
a_3^{ih}	Coeff. of three-period lagged investment growth	0.175	0.1546	-0.195	0.0841	0.064	0.0928	0.157	0.1176	0.215	0.1084
R^2		0.089		0.742		0.325		0.416		0.144	

Note: Note that the estimation of the dynamic part is done using non-linear least squares. We report the standard errors computed using [White & Domowitz \(1984\)](#) approach. This table contains estimation results for the Big-5 countries with the following abbreviations: DE for Germany, FR for France, IT for Italy, ES for Spain, and NL for the Netherlands. Furthermore, Est. stands for estimate, SE for standard error and PSD for posterior standard deviation of the estimation results. Estimates with a * are either calibrated or follow by construction identities. Thus, other estimates are conditional on these calibrations.

Table 7: Total Import Volumes

Parameter	Description	Target											
		DE		FR		IT		ES		NL			
		Est.	PSD	Est.	SE	Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD
α_0^{mtr*}	Constant of target import	-0.272	0.0061	-0.382	0.0069	-0.409	0.0050	-0.135	0.0048	0.264	0.0066		
α_1^{mtr*}	Coeff. of relative import prices	-0.296	0.0246	-0.301	0.0240	-0.305	0.0243	-0.343	0.0215	-0.283	0.0227		
α_2^{mtr*}	Coeff. of import trend	0.004	0.0000	0.003	0.0000	0.003	0.0000	0.001	0.0000	0.002	0.0000		
Dynamic error-correction Equation													
Parameter	Description	DE		FR		IT		ES		NL			
		Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD
α_0^{mtr}	Constant of import growth	-0.004	0.0079	-0.005	0.0079	-0.004	0.0079	-0.003	0.0079	-0.008	0.0079		
α_1^{mtr}	EC Coeff. related to target deviations	-0.094	0.1982	-0.094	0.1976	-0.097	0.1944	-0.096	0.1945	-0.094	0.1971		
α_2^{mtr}	Coeff. related to import demand changes	1.496	0.2049	1.496	0.2050	1.497	0.2048	1.497	0.2048	1.498	0.1988		
α_3^{mtr}	Coeff. related to non-energy prices changes	0.002	0.1004	0.002	0.1004	0.001	0.1004	0.001	0.1003	0.002	0.1004		
R^2		-0.767		-0.415		-0.263		-0.226		0.887			

Note: Note that the estimation of the dynamic part is done using non-linear least squares. We report the standard errors computed using [White & Domowitz \(1984\)](#) approach.

This table contains estimation results for the Big-5 countries with the following abbreviations: DE for Germany, FR for France, IT for Italy, ES for Spain, and NL for the Netherlands. Furthermore, Est. stands for estimate, SE for standard error and PSD for posterior standard deviation of the estimation results. Estimates with a * are either calibrated or follow by construction identities. Thus, other estimates are conditional on these calibrations.

Table 8: Total Export Volumes

Parameter	Description	Target							
		DE		FR		IT		ES	
		Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD
α_0^{xtr*}	Constant of target export	1.073	0.0048	0.638	0.0039	0.425	0.0052	-0.366	0.0075
α_1^{xtr*}	Coeff. of relative export prices	-1.028	0.1604	-1.151	0.1656	-1.148	0.1520	-1.135	0.1575
α_2^{xtr*}	Coeff. of export trend	0.001	0.0000	-0.005	0.0000	-0.006	0.0000	0.001	0.0000
Dynamic error-correction Equation									
Parameter	Description	DE		FR		IT		ES	
		Est.	SE	Est.	SE	Est.	SE	Est.	SE
		Est.	SE	Est.	SE	Est.	SE	Est.	SE
α_0^{xtr}	Constant of export growth	-0.011	0.0013	-0.009	0.0013	-0.010	0.0013	-0.008	0.0013
α_1^{xtr}	EC Coeff. related to target deviations	-0.104	0.0988	-0.100	0.0917	-0.098	0.0894	-0.098	0.0932
α_2^{xtr}	Coeff. related to export demand changes	0.909	0.4364	0.886	0.4403	0.918	0.4373	0.898	0.4457
α_3^{xtr}	Coeff. related to export prices changes	-0.164	0.1940	-0.159	0.1950	-0.164	0.1943	-0.160	0.1959
R^2		0.272		-0.492		0.073		-0.175	

Note: Note that the estimation of the dynamic part is done using non-linear least squares. We report the standard errors computed using [White & Domowitz \(1984\)](#) approach.
This table contains estimation results for the Big-5 countries with the following abbreviations: DE for Germany, FR for France, IT for Italy, ES for Spain, and NL for the Netherlands. Furthermore, Est. stands for estimate, SE for standard error and PSD for posterior standard deviation of the estimation results.
Estimates with a * are either calibrated or follow by construction identities. Thus, other estimates are conditional on these calibrations.

Table 9: Estimated coefficients of the financial block

Equation	Parameter	Pass-through weights									
		DE		FR		IT		ES		NL	
		Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD	Est.	PSD
Consumer loans	10Y yield weight	0.685	0.0148	0.919	0.0054	0.428	0.0028	0.651	0.0062	0.867	0.0153
Housing loans	10Y yield weight	0.696	0.0036	0.828	0.0057	0.580	0.0009	0.566	0.0002	0.773	0.0099
Corporate bonds	10Y yield weight	0.522	0.0035	0.670	0.0043	0.591	0.0022	0.493	0.0017	0.647	0.0030
Cost of equity	10Y yield weight	0.666	0.0535	1.035	0.0310	0.994	0.0264	0.985	0.0169	0.874	0.0416
Deposits	10Y yield weight	0.169	0.0071	0.393	0.0164	0.737	0.0064	0.463	0.0021	0.349	0.0250
Long term loans	10Y yield weight	0.291	0.0037	0.351	0.0080	0.420	0.0031	0.402	0.0025	0.217	0.0070

Note: Estimated coefficients of the financial block for the five largest euro area countries. Pass-through equations relate retail interest rates to sovereign long-term yields and short-term policy rates. Est. stands for estimate and PSD for posterior standard deviation of the estimation results.

Table 10: Estimated coefficients of the financial block

Spread Equations											
Equation	Parameter	DE		FR		IT		ES		NL	
		Est.	Posterior std.	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Consumer loan spread	Persistence	0.684	0.0056	0.677	0.0095	0.868	0.0052	0.843	0.0034	0.550	0.0045
	Expected growth	-0.157	0.0009	-0.219	0.0088	-0.155	0.0051	-0.135	0.0037	-0.169	0.0083
	Constant	1.788	0.1409	1.250	0.1448	0.602	0.1481	0.853	0.0999	2.121	0.1054
Housing loan spread	Persistence	0.273	0.0056	0.763	0.0062	0.750	0.0065	0.877	0.0052	0.576	0.0047
	Expected growth	-0.165	0.0009	-0.222	0.0084	-0.137	0.0041	-0.093	0.0027	-0.149	0.0077
	Constant	1.135	0.0131	0.224	0.0059	0.149	0.0045	0.086	0.0035	0.666	0.0121
Deposit spread	Persistence	0.420	0.0095	0.951	0.0310	0.786	0.0071	0.669	0.0260	0.748	0.0036
	Expected growth	-0.139	0.0010	-0.164	0.0083	-0.166	0.0057	-0.076	0.0029	-0.161	0.0074
	Constant	0.062	0.0025	0.032	0.0020	-0.336	0.0203	-0.142	0.0074	0.133	0.0031
Corporate bond spread	Persistence	0.315	0.0102	0.806	0.0148	0.866	0.0108	0.783	0.0140	0.260	0.0102
	Expected growth	-0.178	0.0010	-0.209	0.0105	-0.043	0.0045	-0.024	0.0031	-0.143	0.0081
	Constant	0.575	0.0076	0.151	0.0095	0.065	0.0068	0.144	0.0083	0.457	0.0050
Cost of equity spread	Persistence	0.766	0.0025	0.805	0.0038	0.739	0.0051	0.802	0.0041	0.583	0.0032
	Expected growth	-0.168	0.0009	-0.185	0.0093	-0.188	0.0092	-0.144	0.0067	-0.171	0.0009
	Constant	1.261	0.0791	1.142	0.1311	1.452	0.2004	1.209	0.1633	2.247	0.1115
Long-term lending spread	Persistence	0.280	0.0081	0.927	0.0058	0.795	0.0115	0.866	0.0006	0.474	0.0064
	Expected growth	-0.142	0.0009	-0.170	0.0090	-0.115	0.0052	-0.112	0.0026	-0.107	0.0072
	Constant	0.960	0.0158	0.079	0.0054	0.122	0.0067	0.139	0.0067	0.612	0.0113

Note: Estimated coefficients of the financial block for the five largest euro area countries. Spread equations are estimated separately for each country. Est. stands for estimate and PSD for posterior standard deviation of the estimation results.

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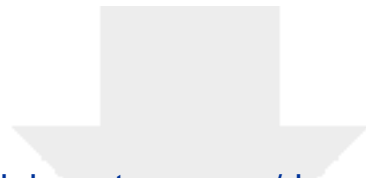
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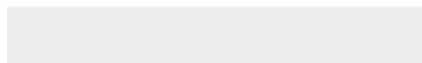
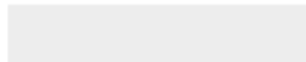
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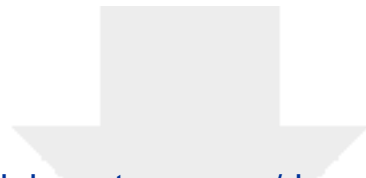


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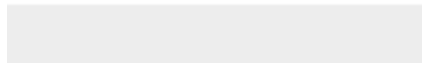
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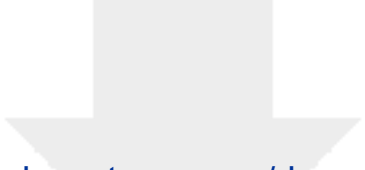




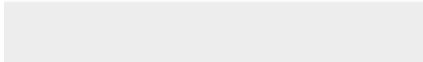
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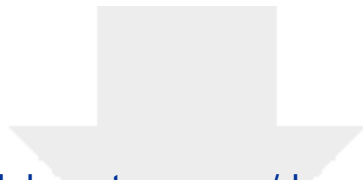
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